

The Volatility of Information Rents in Dynamic Mechanism Design

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Abstract

This paper brings the martingale methods of continuous-time contracting to Myersonian dynamic mechanism design. I show in a canonical setting that implementability is characterized by a core object, the volatility of the agent's information rents—the rent's sensitivity to innovations in his type. Under the usual envelope conditions, truthtelling after the initial report is locally incentive compatible if and only if the volatility satisfies a supermartingale condition; when the volatility has finite variation, the same condition is necessary and sufficient for global incentive compatibility, and an integral constraint ensures truthful initial reporting. Using this characterization, I cast the design problem as optimal control, which applies even when the first-order approach fails: one program solves for the optimal mechanism in a regular finite-variation class, and another for the optimal locally incentive-compatible mechanism. The two bracket the designer's true payoff. I apply the results to financial market design (a monopolistic dealer, then an exchange) and show how a smoothed dynamic bid–ask spread balances rent extraction today against surplus extraction tomorrow.

Keywords: dynamic mechanism design, contracts, persistent private information, optimal control, market design, market microstructure.

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1 Introduction

Dynamic mechanism design in the Myersonian tradition has developed largely in discrete time (Courty and Li, 2000; Battaglini, 2005; Eső and Szentes, 2007; Pavan et al., 2014).¹ By contrast, the broader dynamic contracting literature has benefited greatly from continuous-time analysis using martingale methods (DeMarzo and Sannikov, 2006; Sannikov, 2008; Williams, 2011). Although the two literatures study closely related problems—the first with transferable utility, the second without—they have done so with surprisingly distinct methods. This paper extends martingale methods to the Myersonian setting and investigates how the two literatures’ approaches relate, in search of a unified viewpoint. In the process, it obtains new results for each literature.

The main result characterizes implementability in terms of a core object, the volatility of the agent’s information rent—the rent’s sensitivity to innovations in his type. Under the usual envelope conditions, truthtelling after the initial report is locally incentive compatible if and only if the volatility is non-negative and, suitably discounted, a supermartingale under truthful reporting. When the volatility has finite variation, the same conditions are necessary and sufficient for global incentive compatibility, and an integral constraint ensures truthful initial reporting. The characterization also applies without transfers and is thus new to both literatures, for reasons discussed below.² Because the conditions after the initial report are local in time, the design problem admits techniques from optimal control. Taken together, the results offer a *complete* characterization of the locally incentive-compatible mechanisms,³ identify a natural class on which the characterization is *global*, and provide a method for optimizing over each. I use these tools to analyze a benchmark problem in financial market design where the first-order approach—the standard method—fails.

¹The notable exception is Bergemann and Strack (2015), who solve dynamic revenue-maximization problems in continuous time. Their analysis proceeds within the first-order approach, obtaining first-order conditions by, in their words, “precisely the approach taken in discrete time”; the separation of methods persists even where the clock is continuous. I discuss the relation below.

²In brief, the canonical reference, Williams (2011), restricts the agent to positive drift distortions, whereas I allow distortions of either sign.

³Definition 1 rules out a payoff gain at quadratic order from an arbitrarily small movement away from truthtelling toward any smooth admissible reporting strategy. Technically, when that quadratic effect is zero, higher-order terms may still make a small deviation profitable. Under the conditions of Appendix C, however, the mechanism can be perturbed at arbitrarily small cost to the designer so that the quadratic effect is strictly negative in every nonzero direction.

In the baseline model, a principal allocates a good or asset to a single agent and charges transfers. The agent’s marginal value for the allocation—his type—follows a Markov diffusion, either a mean-reverting Ornstein–Uhlenbeck process or a geometric Brownian motion. The agent alone observes his type, so the principal’s mechanism responds only to his reports. The agent can misreport his initial type and thereafter distort the drift of his reports. The agent’s information rent is the marginal value of his private information: by the envelope theorem, the expected discounted allocation he will receive, weighted by the impulse response of the type process.

By the martingale representation theorem, the agent’s rent is an Itô process with a volatility. I show that implementability after the initial report rests entirely on this volatility (Theorem 1a), with truthful initial reporting ensured separately by the integral constraint at date zero (Theorem 1b). The supermartingale property reflects an economic tradeoff: reducing the exposure of the rent to changes in the type today limits how quickly that exposure can be raised in the future.

The characterization makes the design problem amenable to optimal control, with the rent and its volatility as state variables. The formulation does not rely on the first-order approach, whose validity requires stringent assumptions (Pavan et al., 2014; Battaglini and Lamba, 2019). Two programs result: one solves for the optimal mechanism among finite-variation mechanisms satisfying the date-zero integral condition, and the other for the optimal locally incentive-compatible mechanism, under the conditions of Theorems 2a and 2b. The finite-variation class is natural on two counts: local incentive compatibility implies global after $t = 0$, and in discrete time its mechanisms become *linear contracts*, in which the rent—a discounted measure of the future allocation—is linear, at each date, in the agent’s current type.

Together the two programs bracket the designer’s payoff. They are most useful when the first-order approach is invalid, and both preserve the central dynamic tradeoff the designer must then balance—extracting surplus today against extracting it tomorrow. In the environments covered by the irrelevance theorem of Eső and Szentes (2017), this tradeoff disappears when the first-order approach is valid: the designer earns as much as if the agent’s future types were observed directly. Analysis of the invalid case is thus economically important, and this paper provides one route into it.

I also show how the analysis connects to the discrete-time theory. There, incentive compatibility amounts to the usual envelope condition together with *integral monotonicity* (Pavan et al., 2014): truthful reporting must dominate every one-shot misreport in an integrated sense (reviewed in Section 2). Integral monotonicity, however, is non-local and well known to be intractable to

impose, both computationally and analytically. For the linear contracts described above, the loadings—the date-by-date coefficients, which are functions of past reports—are the discrete counterpart of the volatility. Integral monotonicity then collapses by elementary algebra to the finite-difference form of the continuous-time drift condition (Proposition 3).

The volatility characterization also applies without quasilinear transfers to contracts satisfying the usual envelope condition, as in dynamic insurance (Proposition 6). Most of that literature follows the first-order approach. Within this class, when the agent can over- or underreport the increments of his type, I complete the characterization of local incentive compatibility. Here the connection to the discrete-time theory is quite useful: from integral monotonicity, one can conjecture, without any stochastic calculus, the correct conditions for local incentive compatibility and sufficient conditions for global.

I apply the machinery to a classic setting in finance: a market maker trading with a single trader whose hedging needs are private and persistent—the dynamic counterpart of the monopolistic dealer’s problem (Glosten, 1989). I show that the first-order approach fails here outright because the trader’s side of the market is endogenous: whether he buys or sells depends on the terms of trade. I solve the two control programs, whose solutions both involve a smoothed dynamic bid–ask spread. The spread balances rent extraction today against surplus extraction tomorrow. Using this result, I then show that, under the conditions stated in the application, a related design problem—a revenue-maximizing exchange in which many traders participate and the market must clear—reduces to those single-trader problems. The setting belongs to a benchmark class of models in financial market microstructure, in which traders bear quadratic holding costs (e.g., Du and Zhu, 2017; Malamud and Rostek, 2017; Chen and Duffie, 2021; Antill and Duffie, 2021).

Related literature. The paper sits at the junction of two literatures: Myersonian dynamic mechanism design and dynamic contracting more broadly. On the Myersonian side, where revenue matters to the designer,⁴ the field goes back to Baron and Besanko (1984) and Besanko (1985). Contributions include Courty and Li (2000), Battaglini (2005), Esó and Szentes (2007), Krähmer and Strausz (2011), and Boleslavsky and Said (2013). Pavan et al. (2014) give the general treatment. Pavan (2017) and Bergemann and Välimäki (2019) survey this literature, which has primarily developed in discrete time. I analyze this setting in continuous time, arriving at a distinct characterization of imple-

⁴As opposed to the parallel strand on implementing efficient allocations in dynamic environments; see Bergemann and Välimäki (2019) for a survey of both.

mentability, and exploit tools from optimal control. The closest paper is by [Bergemann and Strack \(2015\)](#), who derive revenue-maximizing mechanisms in continuous time within the first-order approach, by the envelope methods of the discrete theory; the present paper instead confronts second-order incentives with martingale methods.

Within the Myersonian tradition, a recent line of work analyzes the failure of the first-order approach directly. [Battaglini and Lamba \(2019\)](#) show that with persistent types the failure is generic. [Krasikov and Lamba \(2021, 2026\)](#) solve for optimal contracts in specific environments where the approach fails; [Garrett and Pavan \(2015\)](#) and [Garrett et al. \(2018\)](#) develop variational methods that deliver optimality conditions and robust predictions without assuming validity of the first-order approach; and [Li and Shi \(2025\)](#) retain the local incentive constraints in a two-period screening problem where the standard relaxation breaks down, showing that stochastic mechanisms can then be essential. The control programs of Section 4 take a similar approach in an infinite-horizon setting: the local program optimizes subject to all local incentive constraints, while the finite-variation program imposes global continuation incentive compatibility together with a sufficient condition for truthful reporting at date zero. In static settings, local first- and second-order conditions are often sufficient for global incentive compatibility ([Carroll, 2012](#)), but in dynamic settings no comparable result is available.

On the contracting side, the paper extends the martingale methods of continuous-time contract theory ([Sannikov, 2008](#); [Williams, 2011](#)) to screening with transfers and a nondegenerate initial type. The first-order approach, however, is popular in contract theory, including the dynamic insurance and public finance literature (e.g., [Zhang, 2009](#); [Kapička, 2013](#); [Farhi and Werning, 2013](#)). Most closely, [Williams \(2011\)](#) obtains sufficient conditions for incentive compatibility in a transfer-free model of dynamic insurance, with deviations restricted to positive drift distortions; his conditions constrain the loading of the agent’s promised marginal utility—the counterpart, in his setting, of the rent volatility here. The restriction prevents the agent from ever correcting a lie, and its economic motivation is perhaps unclear ([Bloedel et al., 2023](#)). In contrast, I consider the full set of drift distortions and obtain a complete characterization of local incentive compatibility, which is fundamentally second order: with both signs admitted, truthful reporting is interior rather than a corner of the deviation set—a strict first-order condition certifies a corner, but an interior maximum is a second-order matter. Williams’s first-order approach has nonetheless been influential in continuous-time contracting (e.g., [Prat and Jovanovic, 2014](#); [DeMarzo and Sannikov, 2017](#); [Cisternas, 2018](#)).

This paper is partly inspired by [Bloedel et al. \(2023\)](#), who revisit Williams’s

model and identify issues in its analysis of incentive compatibility and optimality. They go on to analyze a class of *self-insurance contracts*—contracts implementable as consumption–saving problems for the agent—and show that the optimal one generically strictly dominates the contract Williams had identified as optimal in his leading application. They also show that the analysis has close discrete-time analogues. Acciaio et al. (2025) likewise revisit Williams’s model, characterizing incentive compatibility within parametric families of contracts linear in the report.⁵

Outline. Section 2 reviews the discrete-time characterization of incentive compatibility and the first-order approach. Section 3 sets out the model. Section 4 presents the volatility characterization and the two control programs that bracket the designer’s payoff. Section 5 develops the discrete-time analogy and the linear-contracts interpretation. Section 6 applies the results to financial market design—first the dealer, then the exchange. Section 7 outlines the logic of the main proofs through the master deviation formula, which expresses the agent’s gain from any deviation in terms of the rent volatility. Section 8 extends the results to contracts without transfers, and Section 9 concludes.

2 Background

We recall, informally, the characterization of incentive compatibility for Markov environments due to Pavan et al. (2014). Time is discrete, $t \in \{0, 1, \dots\}$. The agent privately observes his type θ_t , which follows a Markov process, and reports a type each period. A direct mechanism specifies an allocation process $X = \{X_t\}$ and a transfer process $T = \{T_t\}$, each X_t and T_t measurable with respect to the history of reports through period t . The agent’s flow utility is $u(\theta_t, X_t) - T_t$, discounted by a factor $\beta \in (0, 1)$ per period.

Because the type process is Markov, θ_s can be written as a function of θ_t and the shocks arriving between t and s . The impulse response $I_{s,t} = \partial\theta_s/\partial\theta_t$ is the marginal effect of the time- t type on the time- s type, holding those intermediate shocks fixed. Consider a one-shot misreport: the agent’s current type is θ_t , he reports $\hat{\theta}_t$, and he reports truthfully thereafter. Let $X^{\hat{\theta}_t}$ be the

⁵Those contracts—linear in the *report*, with deterministic coefficients—are distinct from the linear contracts of Section 5, which are linear in the *innovations*, with predictable coefficients.

allocation stream that follows. The information rent is

$$D_t(\hat{\theta}_t, \theta_t) = \mathbb{E} \left[\sum_{s=t}^{\infty} \beta^{s-t} I_{s,t} u_{\theta}(\theta_s, X_s^{\hat{\theta}_t}) \mid \mathcal{F}_t \right],$$

the expected discounted stream of marginal utilities with respect to the type, weighted by the impulse response. $D_t(\theta_t, \theta_t)$ is the on-path rent. Write $W_t(\theta_t)$ for the agent's expected continuation payoff at a truthful history under truthful reporting.

Proposition 1 (Pavan et al. 2014). *A mechanism (X, T) is incentive compatible if and only if, at every truthful history:*

(i) (Envelope / ICFOC.) *The agent's continuation payoff satisfies*

$$W'_t(\theta_t) = D_t(\theta_t, \theta_t).$$

(ii) (Integral monotonicity.) *For every type θ_t and every alternative report $\hat{\theta}_t$,*

$$\int_{\hat{\theta}_t}^{\theta_t} [D_t(l, l) - D_t(\hat{\theta}_t, l)] dl \geq 0.$$

Condition (i) is the first-order (envelope) condition: it pins down transfers given the allocation, and is the dynamic analogue of the Myersonian payoff formula. It is, by itself, only *necessary* for incentive compatibility. Condition (ii)—integral monotonicity—is what upgrades the local condition to a global one: truthful reporting must dominate every one-shot misreport in the integrated sense. It requires that, integrated against the impulse response, raising the report weakly raises the (marginal-utility-relevant) allocation—a single dimension of monotonicity, but one that must hold in the appropriate *integrated* sense over the whole continuation path rather than period by period.

The difficulty with (ii) is twofold. In general, it is difficult to *verify* and it is computationally intractable to *impose* (except in the case of a few periods). The first-order approach is the standard response: impose only the envelope condition (i), optimize the relaxed problem, and hope the solution satisfies (ii).

3 Benchmark Model

We fix a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carrying a standard Brownian motion $B = \{B_t\}_{t \geq 0}$ and an independent random variable $\theta_0 \sim F$. Let $\{\mathcal{F}_t\}_{t \geq 0}$

be the $\mathbb{P}$ -augmentation of the initial enlargement of the Brownian filtration by θ_0 . The filtration satisfies the usual conditions, and since $\theta_0 \perp\!\!\!\perp B$ the process B remains a standard Brownian motion with respect to it, with $\mathcal{F}_0 = \overline{\sigma(\theta_0)}$ non-degenerate. All processes are taken to be adapted to $\{\mathcal{F}_t\}$.

Types. A principal allocates to an agent whose marginal value for the allocation—his type θ_t —is privately known and evolves according to a Markov diffusion with initial condition θ_0 . I treat two cases throughout:

$$\text{(OU)} \quad d\theta_t = \lambda(\bar{\theta} - \theta_t) dt + \gamma dB_t, \quad (1)$$

$$\text{(GBM)} \quad d\theta_t = \mu \theta_t dt + \gamma \theta_t dB_t. \quad (2)$$

In (1), $\lambda \geq 0$ is the mean-reversion rate and $\bar{\theta}$ the long-run mean. In (2), μ is the growth rate. In both, $\gamma > 0$ is the volatility parameter. I assume the initial type's law F admits a density f strictly positive on the interior of its support Θ , an interval in $\mathbb{R}$; under case (2), $\Theta \subseteq (0, \infty)$ and $r > \mu$. The latter is the usual discounted-growth condition.

Mechanisms and reports. The principal does not observe the agent's type. She commits to a mechanism that responds to his *reported* type. The agent reports a process $\hat{\theta} = \{\hat{\theta}_t\}$, choosing the initial value $\hat{\theta}_0 \in \Theta$ freely and thereafter perturbing the report's drift—replacing the Brownian motion B that drives the type by $\hat{B} := B + \int_0^\cdot a_s ds$ for an adapted process a :

$$d\hat{\theta}_t = \lambda(\bar{\theta} - \hat{\theta}_t) dt + \gamma d\hat{B}_t \quad (\text{OU}), \quad d\hat{\theta}_t = \mu \hat{\theta}_t dt + \gamma \hat{\theta}_t d\hat{B}_t \quad (\text{GBM}).$$

Truthful reporting is $\hat{\theta}_0 = \theta_0$ and $a \equiv 0$, so that $\hat{\theta} = \theta$. Every report law is required to be absolutely continuous with respect to the truthful report law over any finite horizon, so that all payoffs below are computed under the single measure $\mathbb{P}$.⁶⁷

A *mechanism* is a pair (X, T) : an allocation $X = \{X_t\}$, valued in a convex subset of $\mathbb{R}$ (possibly $\mathbb{R}$ itself), and a flow transfer $T = \{T_t\}$, valued in $\mathbb{R}$ and satisfying $\mathbb{E}[\int_0^\infty e^{-rt} |T_t| dt] < \infty$, so that discounted transfer streams are well defined. Both are adapted to the reported process. A distortion

⁶Condition 1(C1) states this requirement precisely. Standard Girsanov conditions, such as Novikov's condition on each finite horizon, are sufficient for it.

⁷I adopt the *strong* formulation, fixing $(\Omega, \mathcal{F}, \mathbb{P})$ and shifting the report pathwise, so all expectations are under $\mathbb{P}$. Much of the continuous-time contracting literature instead adopts the *weak* formulation (Williams, 2011; Bloedel et al., 2023), in which the probability space is itself part of the solution concept; the analysis can be adapted to this formulation.

a is a well-posed strategy if it is progressively measurable with $\int_0^T a_s^2 ds < \infty$ almost surely for every $T < \infty$, which together with absolute continuity makes the report, and the payoffs below, well defined. *Admissibility* asks more: integrability and transversality requirements on the report–truth gap $\delta_t := \hat{\theta}_t - \theta_t$ that involve the mechanism’s rent process, and are therefore stated once that object is available (Condition 1, Section 4.2). What matters now is that the admissible class $\mathcal{A}$ contains every bounded distortion with compact support and, with each a , the scalings εa for small ε ; incentive statements are relative to $\mathcal{A}$.

Agent’s problem. The agent’s flow utility from an allocation X at type θ is θX . Writing $\hat{X}_t$ and $\hat{T}_t$ for the allocation and transfer along the report path induced by $(\hat{\theta}_0, a)$, he solves

$$\sup_{\hat{\theta}_0 \in \Theta, a \in \mathcal{A}} U(\hat{\theta}_0, a), \quad U(\hat{\theta}_0, a) := \mathbb{E} \left[\int_0^\infty e^{-rt} (\theta_t \hat{X}_t - \hat{T}_t) dt \mid \theta_0 \right].$$

The mechanism is *incentive compatible* if truthtelling is optimal,

$$U(\hat{\theta}_0, a) \leq U(\theta_0, 0) \quad \text{for all } \hat{\theta}_0 \in \Theta, a \in \mathcal{A},$$

so that the agent neither misreports his initial type nor distorts thereafter.

The mechanism is *individually rational* if, having observed θ_0 , the agent prefers participation to autarky: $U(\theta_0, 0) \geq 0$ for $\mathbb{P}$ -a.e. θ_0 . Following the literature on dynamic mechanism design (Pavan et al., 2014; Bergemann and Strack, 2015), the participation constraint is imposed at $t = 0$ only.

Principal’s problem. The principal values the discounted stream of transfers and solves

$$\sup_{X, T} \Pi(X, T), \quad \Pi(X, T) := \mathbb{E} \left[\int_0^\infty e^{-rt} T_t dt \right]$$

over admissible mechanisms, subject to incentive compatibility and individual rationality. An admissible mechanism (formally defined in Section 4.2) satisfies

$$\mathbb{E} \left[\left(\int_0^\infty e^{-(r+\lambda)t} X_t dt \right)^2 \right] < \infty \quad (\text{OU}),$$

$$\mathbb{E} \left[\left(\int_0^\infty e^{-rt} \theta_t X_t dt \right)^2 \right] < \infty \text{ and } \mathbb{E} \left[\left(\frac{1}{\theta_0} \int_0^\infty e^{-rt} \theta_t X_t dt \right)^2 \right] < \infty \quad (\text{GBM}).$$

Here X_t, T_t (without hats) are outcomes under truthful reporting, an abuse of notation I maintain throughout.

Discussion. For simplicity, the benchmark combines bilinear utility with OU and GBM dynamics, canonical specifications that encompass many applications. The framework and characterization, however, extend along several dimensions; I highlight three.

- *Agent’s utility.* The analysis extends to any flow utility of the form

$$u(\theta, X) = \theta g(X) - c(X) + h(\theta),$$

where g is increasing. Since g is invertible, redefining the allocation as $g(X)$ restores the bilinear term $\theta g(X)$, so the rent characterization carries over with $g(X)$ replacing X . As in static mechanism design, terms that do not interact jointly with θ and X do not affect implementability. Appropriate regularity conditions on c ensure that the principal’s problem has a unique solution. The further relaxation of linearity in θ —a general utility $u(\theta, X)$ separately concave in θ and X —is discussed in Section 9.

- *Principal’s objective.* The principal may instead face a convex cost of supply (Mussa and Rosen, 1978), maximize any weighted combination of revenue and surplus, $\Pi + \eta U$ with $\eta \in [0, 1]$, or more generally optimize any objective obtained by adding a discounted surplus flow

$$\int_0^\infty e^{-rt} S(\theta_t, X_t) dt,$$

where, for example, $S = -c(X)$ in the case of convex production costs. Since the implementability results are independent of the principal’s objective, extending the analysis simply amounts to modifying the associated stochastic control problem.

- *Horizon.* The analysis extends directly to a finite horizon with the usual modifications, namely terminal conditions and a time-dependent value function.

4 Main Results

The main results are twofold. First, I characterize incentive compatibility via the volatility of the agent’s information rent—the sensitivity of his continuation utility to his type (Section 2). As in discrete time, subject to the integrability requirements of Proposition 2, transfers can be chosen so that the

first-order envelope conditions hold. Incentives therefore hinge entirely on this volatility. After $t = 0$, local incentive compatibility holds if and only if the volatility satisfies a supermartingale condition; when the volatility is of finite variation, the same condition is necessary and sufficient for global incentive compatibility (Theorem 1a). At $t = 0$, local incentive compatibility holds if and only if the initial volatility is bounded by the slope of the rent in the initial report; in the finite-variation class, an integral form of this bound ensures global incentive compatibility (Theorem 1b).

Second, I use this characterization to cast the principal's problem as optimal control. One control problem restricts attention to finite-variation mechanisms and produces an implementable lower bound on the principal's value. As I will show, this class is readily interpretable and rich enough to preserve the central tradeoff—extracting rent from today's report limits how responsive the rent can be to future movements of the type. A second control problem relaxes global to local incentive compatibility and produces an upper bound. When the controls that maximize its HJB define an admissible mechanism and its date-zero solution satisfies the conditions stated below, the program also delivers the optimal locally incentive-compatible mechanism in the corresponding class. Together, the bounds bracket the value of the optimal mechanism and are most useful precisely when the first-order approach fails.

4.1 The volatility of information rents

Let

$$Z := \begin{cases} \int_0^\infty e^{-(r+\lambda)t} X_t dt & (\text{OU}), \\ \frac{1}{\theta_0} \int_0^\infty e^{-rt} \theta_t X_t dt & (\text{GBM}). \end{cases}$$

Under an admissible mechanism, $Z \in L^2$, so by the Itô representation theorem

$$Z = D(\theta_0) + \int_0^\infty \sigma_t dB_t, \quad D(\theta_0) = \mathbb{E}[Z \mid \theta_0]$$

for some predictable process σ . The process σ will determine the volatility of information rents, the core object of the analysis. We write $\sigma_0(\theta_0)$ for the time-zero value of σ as a function of the initial type.

Letting $\mathbb{E}_t[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_t]$, the time- t information rent is

$$D_t := \begin{cases} \mathbb{E}_t \left[\int_t^\infty e^{-(r+\lambda)(s-t)} X_s ds \right] & (\text{OU}), \\ \frac{1}{\theta_t} \mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} \theta_s X_s ds \right] & (\text{GBM}), \end{cases}$$

i.e., the expected discounted allocation, weighted by the impulse response. At $t = 0$ it coincides with $D(\theta_0)$. The rent D_t is itself an Itô process, and by Itô's formula its volatility is

$$\text{vol}(D_t) = \begin{cases} e^{(r+\lambda)t} \sigma_t & (\text{OU}), \\ e^{rt} A_t / \theta_t & (\text{GBM}), \end{cases}$$

where, in the GBM case,

$$A_t := \gamma \theta_t (Y_t - e^{-rt} D_t)$$

and $Y_t := \theta_0 \sigma_t / (\gamma \theta_t)$ (i.e., σ in proportional terms). We write $Y_0(\theta_0)$ for the time-zero value of Y as a function of the initial type. Throughout, A plays under GBM the role that σ plays under OU.

4.2 Admissible Mechanisms and Deviations

Section 3 deferred the definition of admissibility: its requirements involve the rent process and its volatility, which are now in hand. Condition 1 completes it. The clauses serve the standard roles—enough integrability for martingale verification, enough richness in the deviation class for the necessity results to have bite—and the reader may skip ahead with little loss.

Condition 1 (Admissibility). *Write $\phi_t := \delta_t / \hat{\theta}_t$ and $\Psi_t := -\phi_t - \log(1 - \phi_t)$. For a fixed mechanism, a report $(\hat{\theta}_0, a)$ is admissible against the mechanism if it satisfies (C1)–(C4) below. For each mechanism and initial report, $\mathcal{A}$ denotes the distortions whose resulting reports are admissible against the mechanism; this dependence is suppressed in the notation. Clause (C5) governs their small scalings, and (C6) gives the additional requirements for joint deviations at date zero:*

(C1) *a is progressively measurable with $\int_0^T a_s^2 ds < \infty$ a.s. for all $T < \infty$. On every finite horizon, the law of the induced report is absolutely continuous with respect to the law under truthful continuation from the same initial report;*

(C2) $\mathbb{E} \int_0^\infty e^{-rt} (|\theta_t \hat{X}_t| + |\hat{T}_t|) dt < \infty;$

(C3) *along the truthful path, the equivalence class $S_t := e^{-\lambda t} \sigma_t$ (OU), respectively A_t (GBM), admits a predictable representative, still denoted by S , respectively A , such that, for every $T < \infty$, it lies in $L^2([0, T] \times \Omega)$ and*

$R_t \rightarrow R_0$ in L^1 as $t \downarrow 0$ for some finite $\mathcal{F}_0$ -measurable R_0 , where $R = S$, respectively $R = A$. Under truthful continuation from every fixed initial report $y \in \Theta$, the same representative satisfies these local L^2 and initial-limit requirements, with a finite deterministic limit R_0^y . Along the report induced by the deviation under consideration, the dB-integrands used in the proofs—the diffusion coefficient of the gains process and, whenever S , respectively A , is a semimartingale, the process $e^{\lambda t} \delta_t^2 \text{vol}(\sigma_t)$ (OU), respectively $\Psi_t \text{vol}(A_t)$ (GBM)—lie in $L^2([0, T] \times \Omega)$ for every $T < \infty$;

(C4) as $t \rightarrow \infty$, $e^{-rt} \mathbb{E}[|W_t|] \rightarrow 0$ and $\mathbb{E}[e^{-rt} |D_t \delta_t|] \rightarrow 0$; moreover,

$$\mathbb{E} \int_0^\infty e^{\lambda t} |\sigma_t \delta_t a_t| dt < \infty \quad (\text{OU}),$$

$$\mathbb{E} \int_0^\infty |A_t \phi_t a_t| dt < \infty \quad (\text{GBM}).$$

Whenever S , respectively A , is a semimartingale, also

$$\mathbb{E} \int_0^\infty e^{\lambda t} \delta_t^2 |\text{vol}(\sigma_t) a_t| dt < \infty \quad (\text{OU}),$$

$$\mathbb{E} \int_0^\infty \Psi_t |\text{vol}(A_t) a_t| dt < \infty \quad (\text{GBM}).$$

(C5) for every continuation direction $a \in \mathcal{A}$, the report generated from a truthful initial report by εa satisfies (C1)–(C4) against the same mechanism for all sufficiently small $\varepsilon > 0$. Start from a truthful initial report (conditioning on its fixed initial type, as in the definition of U) and use a superscript (ε, a) for evaluation along the report induced by εa . Define

$$\tilde{\delta}_t^a := \gamma \int_0^t e^{-\lambda(t-s)} a_s ds \quad (\text{OU}).$$

Whenever S (OU), respectively A (GBM), admits a semimartingale representative, the feedback terms along this scaled family satisfy the following limits, with the volatility of that representative evaluated along the scaled report:

$$\varepsilon \mathbb{E} \int_0^\infty e^{\lambda t} (\tilde{\delta}_t^a)^2 |\text{vol}(\sigma_t^{(\varepsilon, a)}) a_t| dt \longrightarrow 0 \quad (\text{OU})$$

and

$$\mathbb{E} \int_0^\infty \Psi_t^{(\varepsilon, a)} |\text{vol}(A_t^{(\varepsilon, a)}) a_t| dt = o(\varepsilon) \quad (\text{GBM})$$

as $\varepsilon \downarrow 0$.

(C6) *for the date-zero characterization, couple truthful reports from different initial values by using the same Brownian innovations, and let R^y denote S (OU), respectively A (GBM), when the initial report is y . For every $x \in \Theta$ and $T < \infty$,*

$$R_0^y \longrightarrow R_0^x, \quad \mathbb{E} \int_0^T |R_t^y - R_t^x|^2 dt \longrightarrow 0 \quad \text{as } y \rightarrow x.$$

For a feasible joint direction (h, a) , set $y_\varepsilon = x + \varepsilon h$. For all sufficiently small $\varepsilon > 0$, the report that starts from y_ε and has distortion εa satisfies (C1)–(C4) against the same mechanism. Use a superscript (ε, h, a) for evaluation along this joint report, and define

$$\Delta_t^{h,a} := h e^{-\lambda t} + \tilde{\delta}_t^a \quad (\text{OU}).$$

Whenever S (OU), respectively A (GBM), admits a semimartingale representative, the corresponding joint feedback terms satisfy

$$\varepsilon \mathbb{E} \int_0^\infty e^{\lambda t} (\Delta_t^{h,a})^2 |\text{vol}(\sigma_t^{(\varepsilon, h, a)})_{a_t}| dt \longrightarrow 0 \quad (\text{OU})$$

and

$$\mathbb{E} \int_0^\infty \Psi_t^{(\varepsilon, h, a)} |\text{vol}(A_t^{(\varepsilon, h, a)})_{a_t}| dt = o(\varepsilon) \quad (\text{GBM})$$

as $\varepsilon \downarrow 0$.

Conditions (C1)–(C4) define admissible reports relative to a mechanism and hence determine $\mathcal{A}$. A mechanism is *admissible* if it satisfies the square-integrability requirements of Section 3, the truthful-path requirements of (C3) under every fixed initial report, and (C5). In addition, every bounded progressively measurable distortion with compact support must be admissible against it from every truthful initial report. Results involving the initial report also require (C6).

The feedback limits in (C5) and (C6) are imposed direction by direction, rather than uniformly across directions; (C6) is used only at date zero.

4.3 Implementability

To motivate the results, I begin with a first-order analysis of local incentives. The point is that the agent's problem is one of singular control: first-order conditions cannot determine whether truthtelling is locally optimal, so a second-order analysis is fundamental.

To see this, suppose θ_t evolves as a Brownian motion after $t = 0$, and let $\hat{W}_t$ denote the agent's continuation utility (defined formally ahead of Proposition 2) were his type equal to his report $\hat{\theta}_t$. Payoffs are linear in the type, and, absent further distortion, the gap $\delta_t = \hat{\theta}_t - \theta_t$ persists ($\theta_s - \hat{\theta}_s = -\delta_t$ for all $s \geq t$). His true continuation utility is therefore

$$\hat{W}_t + \mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} (\theta_s - \hat{\theta}_s) \hat{X}_s ds \right] = \hat{W}_t - \delta_t D_t.$$

Suppose the agent has not distorted so far, so $\delta_t = 0$. Truth-telling passes the first-order test if $a_t = 0$ maximizes the drift of his continuation utility. A distortion a_t adds drift through two channels: $\hat{W}_t$ picks up $a_t \text{vol}_{\hat{W}_t}$ through its exposure to the Brownian motion driving the report, and the gap rises at rate γa_t ($d\delta_t = \gamma a_t dt$), for a net contribution of

$$a_t (\text{vol}_{\hat{W}_t} - \gamma D_t).$$

Because positive and negative lies are both feasible, the first-order condition requires $\text{vol}_{\hat{W}_t} = \gamma D_t$. When only positive lies are feasible, as in Williams (2011), the inequality $\text{vol}_{\hat{W}_t} \leq \gamma D_t$ suffices; when it holds strictly, every lie incurs a first-order loss and truth-telling is a local maximum. If the principal's objective is smooth in the mechanism, the strict inequality costs her only marginally, and first-order conditions are then the relevant characterization of local incentives.

In our setting, two-sided lies force the equality, and any perturbation of the volatility, at whatever cost, makes some lie strictly profitable. The singular control structure ensures that local optimality is decided at second order. The argument concerns $t > 0$ but the same logic applies to the initial report. This discussion motivates the following notion of local incentive compatibility.

Definition 1 (Local incentive compatibility). *Fix $\theta_0 \in \Theta$. A deviation direction is a pair $(h, a) \in \mathbb{R} \times \mathcal{A}$: an initial misreport $\hat{\theta}_0 = \theta_0 + \varepsilon h$ together with the drift distortion εa . It is feasible at θ_0 if $\theta_0 + \varepsilon h \in \Theta$ for every sufficiently small $\varepsilon > 0$. Truthful reporting is locally incentive compatible at θ_0 if, for every feasible direction,*

$$\limsup_{\varepsilon \downarrow 0} \frac{U(\theta_0 + \varepsilon h, \varepsilon a) - U(\theta_0, 0)}{\varepsilon^2} \leq 0,$$

and locally incentive compatible for $t > 0$ at θ_0 if this holds for every feasible direction with $h = 0$. A mechanism is locally incentive compatible, respectively locally incentive compatible for $t > 0$, if the corresponding condition holds at every $\theta_0 \in \Theta$.

Informally, when the inequality is strict, a deviation $\epsilon(h, a)$ with (h, a) smooth and bounded reduces the agent's payoff for every sufficiently small ϵ : truthtelling is a local maximum.

As in discrete time, transfers absorb the first-order effects of deviations (Proposition 2 below). The second-order conditions—the counterpart of the integral-monotonicity conditions of discrete time—reduce to local conditions on the rent volatility, and are thus amenable to optimal control.

Theorem 1a (Incentives at $t > 0$). *Fix a truthful initial report $\hat{\theta}_0 = \theta_0$, and suppose the transfers satisfy the volatility condition of Proposition 2(i). Then local incentive compatibility for $t > 0$ at this initial report (Definition 1) holds if and only if the predictable equivalence class of the rent volatility σ admits a càdlàg semimartingale representative with the following three properties:*

- (i) nonnegativity— $\sigma_t \geq 0$ (OU), respectively $A_t \geq 0$ (GBM), $dt \otimes d\mathbb{P}$ -a.e.;
- (ii) a continuous finite-variation bound—the continuous finite-variation part of $e^{-\lambda t} \sigma_t$ (OU), respectively A_t (GBM), is nonincreasing. When this part is absolutely continuous, the condition reads, $dt \otimes d\mathbb{P}$ -a.e.,

$$\text{(OU)} \quad \text{drift}(\sigma_t) \leq \lambda \sigma_t, \quad \text{(GBM)} \quad \text{drift}(A_t) \leq 0;$$

- (iii) downward jumps—almost surely, σ (OU), respectively A (GBM), has no upward jumps.

Equivalently: $e^{-\lambda t} \sigma_t$ (OU), respectively A_t (GBM), is a nonnegative supermartingale. These conditions are necessary and sufficient for global incentive compatibility at $t > 0$ when σ (OU), respectively A (GBM), is of finite variation (equivalently, Y); in that case σ , respectively Y , has no diffusion component, and the GBM condition reduces to

$$dY_t + \mu Y_t dt \leq -e^{-rt} X_t dt.$$

The supermartingale condition captures the key dynamic tradeoff of implementation: the designer may reduce the exposure to the agent's report at any time, but cannot raise it back, in expectation, along the equilibrium path. I defer further intuition to Section 5, where I also interpret the finite-variation contracts and relate them to certain “linear contracts.”

The envelope conditions eliminate first-order gains, so local effects begin at quadratic order. Higher-order terms may matter at a boundary, but a strict rent-volatility drift makes every fixed nonzero direction unprofitable when sufficiently small. Appendix C shows that weak mechanisms admit such

strict approximations at arbitrarily small cost under mild conditions. Thus the boundary issue leaves design value unchanged; in finite variation, the conditions already imply global incentive compatibility.

The conditions at $t > 0$ govern the agent's incentive to distort the *increments* of his report, holding the initial report fixed. It remains to discipline the *initial* report—the agent's one static choice, made having observed θ_0 .

Theorem 1b (Incentives at $t = 0$). *Suppose the transfers satisfy the envelope condition of Proposition 2(ii), truthful continuation is locally incentive compatible after every truthful initial report, and D is differentiable, with the corresponding one-sided derivative at any endpoint of Θ . Then truthful reporting is locally incentive compatible (Definition 1) if and only if, for all $\hat{\theta}_0 \in \Theta$,*

$$\begin{aligned}\sigma_0(\hat{\theta}_0) &\leq \gamma D'(\hat{\theta}_0) & (\text{OU}), \\ Y_0(\hat{\theta}_0) - D(\hat{\theta}_0) &\leq \hat{\theta}_0 D'(\hat{\theta}_0) & (\text{GBM}).\end{aligned}\tag{3}$$

The following global sufficient condition does not require differentiability of D . Suppose the same transfer and truthful-continuation conditions hold and, in addition, σ (OU), respectively Y (GBM), is of finite variation. Then truthful reporting is incentive compatible whenever, for all $\hat{\theta}_0, \theta_0 \in \Theta$,

$$\int_{\hat{\theta}_0}^{\theta_0} \left[D(l) - D(\hat{\theta}_0) - \frac{\sigma_0(\hat{\theta}_0)}{\gamma} (l - \hat{\theta}_0) \right] dl \geq 0 \quad (\text{OU}),\tag{4}$$

$$\int_{\hat{\theta}_0}^{\theta_0} \left[D(l) - D(\hat{\theta}_0) - (Y_0(\hat{\theta}_0) - D(\hat{\theta}_0)) \cdot \frac{l - \hat{\theta}_0}{l} \right] dl \geq 0 \quad (\text{GBM}).\tag{5}$$

The integral condition is also necessary whenever exposure cannot change appreciably during a vanishingly fast correction of a fixed initial lie. This includes smooth Markov finite-variation mechanisms with locally bounded, absolutely continuous exposure dynamics.⁸

For θ_0 near $\hat{\theta}_0$, the integrands of (4) and (5) reduce, at leading order, to positive multiples of the slack in (3): the integral conditions are the global form of the local slope bound. Figure 1 illustrates both conditions. In some settings—when the law of the initial type is not regular, for example, or in

⁸To see this, consider a report that starts at $\hat{\theta}_0$ and eliminates a fixed initial report gap over $[0, \varepsilon]$. Provided these corrective reports are admissible, if exposure changes by $o(1)$ along the correction as $\varepsilon \downarrow 0$, the deviation attains the upper bound used in the proof of the theorem; incentive compatibility then yields (4), respectively (5). The stated Markov regularity ensures this stability because finite-variation exposure has no diffusion loading and its remaining drift accumulates only $O(\varepsilon)$ over the correction interval.

the application of Section 6—the designer seeks to flatten the rent to extract revenue. There the tradeoff at $t = 0$ is especially pronounced: the local condition caps the initial volatility by the slope of the rent, so flattening the rent to extract revenue today limits how sensitive future rents can be to the agent’s type—the margin through which surplus is extracted later.

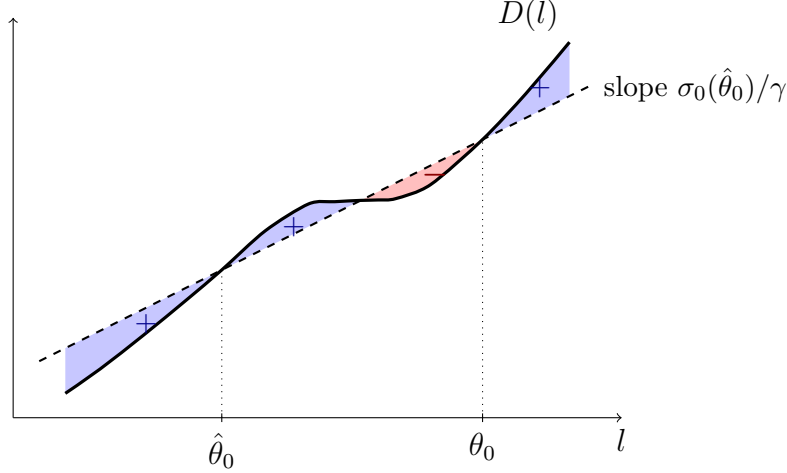


Figure 1: The time-zero condition (OU case). The rent D (solid) against the affine benchmark $D(\hat{\theta}_0) + (\sigma_0(\hat{\theta}_0)/\gamma)(l - \hat{\theta}_0)$ (dashed). The rent is nondecreasing, as required by incentive compatibility. The local condition (3) says D crosses the benchmark from below at $\hat{\theta}_0$: the rent lies above the line just to the right and below it just to the left. The integral condition (4) requires the signed area between D and the affine benchmark over the interval $\hat{\theta}_0$ to θ_0 , to be nonnegative. To the right of $\hat{\theta}_0$, regions where D exceeds the line count positively (blue) and regions where it falls short count negatively (red). To the left the roles reverse: the direction of the lie determines the sign.

We now record what the theorems require of the transfers. For a given mechanism (X, T) , define the continuation values

$$\hat{W}_t^X := \mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} \hat{\theta}_s X_s ds \right], \quad \hat{W}_t := \mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} (\hat{\theta}_s X_s - T_s) ds \right],$$

with $\text{vol}_{\hat{W}}$, $\text{vol}_{\hat{W}^X}$ their diffusion coefficients; $\hat{W}^X$ is determined by the allocation alone, while $\hat{W}$ also reflects the transfers. The notation $\hat{W}_0^X(y)$ denotes the time-zero value induced by the initial report y . The necessity statements below apply to any admissible allocation. The proposition also gives an explicit transfer construction, subject to the stated integrability requirements.

Proposition 2 (Implementing transfers). *Let an allocation X be given.*

- (i) *Local incentive compatibility at $t > 0$ requires that the transfers give the continuation value $\hat{W}$ the volatility*

$$\text{vol}_{\hat{W}_t} = \begin{cases} \gamma D_t & (\text{OU}), \\ \gamma \hat{\theta}_t D_t & (\text{GBM}). \end{cases}$$

- (ii) *Incentive compatibility at $t = 0$ requires that the transfers leave the agent a truthful value satisfying the envelope representation*

$$U(\theta_0, 0) - U(c, 0) = \int_c^{\theta_0} D(l) \, dl \quad \text{for all } c, \theta_0 \in \Theta;$$

equivalently, the expected payment is pinned down up to the constant $U(c, 0)$:

$$\mathbb{E} \left[\int_0^\infty e^{-rt} T_t \, dt \mid \theta_0 \right] = \hat{W}_0^X(\theta_0) - \int_c^{\theta_0} D(l) \, dl - U(c, 0).$$

Let g_t^ denote the volatility prescribed in (i), and set*

$$H_t := g_t^* - \text{vol}_{\hat{W}_t^X}.$$

Suppose $H \in L^2([0, T] \times \Omega)$ for every $T < \infty$ along every admissible report, and suppose the transfer flow below satisfies the admissibility conditions of Section 3; in particular, $\mathbb{E} \int_0^\infty e^{-rt} |T_t| \, dt < \infty$ under every such report. Then, for any base type $c \in \Theta$ and any value of $U(c, 0) \in \mathbb{R}$, transfers satisfying (i) and (ii) exist. One choice is

$$T_t = r \left[\hat{W}_0^X(\hat{\theta}_0) - \int_c^{\hat{\theta}_0} D(l) \, dl - U(c, 0) \right] - r \int_0^t H_s \, d\hat{B}_s.$$

The constant flow delivers the expected payment in (ii), and the martingale term gives the continuation value the volatility in (i).

The volatility requirement (i) is the continuous-time counterpart of the dynamic envelope representation from discrete-time dynamic mechanism design (Pavan et al., 2014): the agent's information rent D_t here appears as the volatility of the continuation value.⁹

⁹The Clark–Ocone formula makes the correspondence exact. When $\hat{W}$ is Malliavin differentiable, $\text{vol}_{\hat{W}_t} = \mathbb{E}_t[\mathfrak{D}_t \hat{W}]$, where $\mathfrak{D}_t$ differentiates with respect to the driving noise $\hat{B}$ at time t —exactly the channel through which the agent's drift distortions move the report. A unit displacement of $\hat{B}_t$ moves the current report by γ (OU), respectively $\gamma \hat{\theta}_t$ (GBM), so condition (i) says the continuation value responds to the current report at rate D_t : the dynamic envelope formula.

4.4 Optimal control

The characterization yields two stochastic control programs. The states are the report $\hat{\theta}_t$, the promised rent D_t , and the rent's volatility in current-value units, $\psi_t := e^{(r+\lambda)t}\sigma_t = \text{vol}(D_t)$. The first two appear already in Williams (2011); the third is new. We shall also see that continuation utility, the canonical state of dynamic contracting (Spear and Srivastava, 1987; Sannikov, 2008), is *not* a state here. I develop the OU case below; Appendix J states and proves the GBM counterparts.¹⁰

The first step is standard (Pavan et al., 2014; Bergemann and Strack, 2015): solve out the transfers. Fix a base type $c \in \Theta$, as in Proposition 2, and write $U_c := U(c, 0)$ for the base value the transfers leave the agent—a date-zero choice of the principal. Substituting the expected transfer of Proposition 2(ii) and integrating by parts against F ,

$$\Pi = \mathbb{E}\left[\int_0^\infty e^{-rt}\theta_t X_t dt\right] - \mathbb{E}[h_c(\theta_0) D(\theta_0)] - U_c, \quad (6)$$

where $h_c(\theta_0) := (\mathbb{1}\{\theta_0 \geq c\} - F(\theta_0))/f(\theta_0)$ generalizes the inverse hazard rate and reduces to $(1 - F)/f$ when c is the lowest type. Revenue is gross surplus net of information rents and the base value; the rents are priced entirely at date zero, measured from the base type (cf. the irrelevance theorem of Eső and Szentes, 2017). Individual rationality reads

$$U_c + \int_c^{\theta_0} D(l) dl \geq 0 \quad \text{for all } \theta_0 \in \Theta, \quad (7)$$

and the choice of base type is immaterial. Note what has dropped out: continuation utility is not among the states. Proposition 2 already pins down its payoff-relevant features—its volatility is the rent, and its level is priced once, at date zero—so the promise the principal carries forward is the rent D_t itself.

The problem therefore splits. Once the time-zero promise is made, the rent term and the base value in (6) are sunk, and the principal's objective coincides with surplus: for $t > 0$, maximize continuation surplus, $\mathbb{E}_t \int_t^\infty e^{-r(s-t)} \hat{\theta}_s X_s ds$, subject to delivering the promise. At $t = 0$, choose the promised rent $D(\cdot)$, the initial volatility $\sigma_0(\cdot)$, and the base value U_c . The promise evolves as

$$dD_t = [(r + \lambda)D_t - X_t]dt + \psi_t d\hat{B}_t, \quad \psi_0 = \sigma_0, \quad (8)$$

¹⁰In the costless benchmark, GBM homogeneity makes gross continuation surplus equal to $\hat{\theta}D$, so the HJB is indifferent among allocations that deliver the same rent. A strictly convex allocation cost breaks this indifference and makes the HJB select the allocation, without changing the rent characterization.

and in these units the conditions of Theorem 1a read: $\psi_t \geq 0$, $\text{drift}(\psi_t) \leq (r+2\lambda)\psi_t$, and no upward jumps; in the finite-variation class they combine into the pathwise bound $d\psi_t \leq (r+2\lambda)\psi_t dt$. Throughout the control programs, an admissible continuation must deliver the rent represented by the controlled state:

$$D_t = \mathbb{E}_t \left[\int_t^\infty e^{-(r+\lambda)(s-t)} X_s ds \right],$$

with ψ equal to its diffusion coefficient. For verification, it is useful to write the exposure dynamics in both programs as

$$d\psi_t = (r+2\lambda)\psi_t dt + v_t d\hat{B}_t - dK_t, \quad \psi_t \geq 0, \quad (9)$$

where K is adapted, càdlàg, and nondecreasing, $K_0 = 0$, and K^c denotes its continuous part. In the finite-variation program $v = 0$; in the local program v is an additional control. We write $\Xi_t := (\hat{\theta}_t, D_t, \psi_t)$. Because $K_0 = 0$, $\psi_0 = \sigma_0$ is the exposure actually delivered at date zero; the continuation policy does not make a second instantaneous adjustment there. For a smooth candidate value $V(\hat{\theta}, D, \psi)$, define

$$\begin{aligned} \mathcal{L}V := \sup_x & x(\hat{\theta} - V_D) + \lambda(\bar{\theta} - \hat{\theta})V_{\hat{\theta}} + (r+\lambda)DV_D + (r+2\lambda)\psi V_{\psi} \\ & + \frac{1}{2}\gamma^2 V_{\hat{\theta}\hat{\theta}} + \gamma\psi V_{\hat{\theta}D} + \frac{1}{2}\psi^2 V_{DD}, \end{aligned} \quad (10)$$

the supremum taken over the allocation's range. For the OU model, the reachable current-report state space is $\mathbb{R}$, even when the support Θ of the initial type is a proper interval. Accordingly, the HJB equations below are posed on $\mathbb{R} \times \mathbb{R} \times (0, \infty)$. Candidate functions extend continuously to $\psi = 0$, and their boundary traces are C^2 in $(\hat{\theta}, D)$. This boundary is absorbing. Writing $\mathcal{L}_0$ for $\mathcal{L}$ with $\psi = 0$, the state-constraint condition for either candidate $W \in \{V, \bar{V}\}$ is

$$rW(\hat{\theta}, D, 0) - \mathcal{L}_0 W(\hat{\theta}, D, 0) \geq 0.$$

For attainment, this inequality holds with equality whenever the proposed state process is at $\psi = 0$. In the local program the only admissible boundary choice is $v = 0$.

Condition 2 (Transversality). *For a class of mechanisms and each initial continuation state under consideration, a candidate value $W(\hat{\theta}, D, \psi)$ satisfies transversality if, writing $W_t := W(\hat{\theta}_t, D_t, \psi_t)$ along the states of a mechanism:*

(T1) *Along every mechanism of the class, $e^{-rt}W_t$ is of class D on each finite horizon—that is, $\{e^{-r\tau}W_\tau : \tau \leq T\}$ is uniformly integrable for every $T < \infty$ —and*

$$\liminf_{T \rightarrow \infty} e^{-rT} \mathbb{E}[W_T] \geq 0.$$

Moreover, on every finite horizon, the verification integrands

$$e^{-rt}(\gamma W_{\hat{\theta}} + \psi_{t-} W_D), \quad e^{-rt} v_t W_\psi$$

lie in $L^2([0, T] \times \Omega)$; the second integrand is present only when the volatility carries a diffusion v ;

(T2) Along the mechanism proposed to attain the bound, $e^{-rT} \mathbb{E}[W_T] \rightarrow 0$.

In both date-zero programs below, $U_c \in \mathbb{R}$, D and σ_0 are finite-valued and measurable, D is locally integrable and belongs to $L^2(F)$, and $\sigma_0 \in L^1(F)$. The functions $V(\theta_0, D(\theta_0), \sigma_0(\theta_0))$, respectively $\bar{V}(\theta_0, D(\theta_0), \sigma_0(\theta_0))$, and $h_c D$ are integrable under F .

The first program restricts attention to the finite-variation class. Within that class, the conditions of Theorems 1a and 1b guarantee global incentive compatibility: every admissible policy delivered by the program is implementable.

Theorem 2a (The finite-variation program). (OU) *Let V , nondecreasing in ψ , be a classical solution—twice continuously differentiable in $(\hat{\theta}, D)$ and once in ψ —of*

$$\min \{ rV - \mathcal{L}V, V_\psi \} = 0 \quad \text{on } \mathbb{R} \times \mathbb{R} \times (0, \infty), \quad (11)$$

with the state-constraint interpretation at $\psi = 0$, and suppose V satisfies (T1) for the finite-variation class. Then $V(\hat{\theta}, D, \psi)$ is an upper bound on the continuation surplus of every admissible finite-variation continuation satisfying the exposure constraints above and delivering rent D with volatility ψ .

The bound is attained from an initial state ξ if there are a predictable allocation X and a nondecreasing process K in (9), with $v = 0$ and $\Xi_0 = \xi$, such that:

- (a) *the state equations, completed with transfers satisfying Proposition 2(i), produce an admissible finite-variation continuation delivering rent D with volatility ψ , and (T2) holds;*
- (b) *for $dt \otimes d\mathbb{P}$ -almost every t , X_t attains the supremum over x in (10) and $rV(\Xi_{t-}) = \mathcal{L}V(\Xi_{t-})$; when $\psi_{t-} = 0$, the latter equality is interpreted as $rV = \mathcal{L}_0 V$;*
- (c) *continuous adjustment through K occurs only where $V_\psi = 0$, and, almost surely, every downward jump after date zero leaves the value unchanged:*

$$\begin{aligned} V_\psi(\Xi_{t-}) dK_t^c &= 0, \\ V(\Xi_t) &= V(\Xi_{t-}) \quad \text{at every jump of } K \text{ after date zero.} \end{aligned}$$

Under these conditions,

$$V(\xi) = \mathbb{E} \int_0^\infty e^{-rt} \hat{\theta}_t X_t dt.$$

Thus V is the maximal continuation surplus from ξ within the finite-variation class, and the stated controls attain it.

For the date-zero stage, consider the program

$$\sup_{D(\cdot), \sigma_0(\cdot), U_c} \int_{\Theta} [V(\theta_0, D(\theta_0), \sigma_0(\theta_0)) - h_c(\theta_0) D(\theta_0)] dF(\theta_0) - U_c \quad (12)$$

subject to (4), $\sigma_0 \geq 0$, and (7); the constraints are linear in $(D(\cdot), \sigma_0(\cdot), U_c)$. Assume that, for each feasible choice, there is a measurable $0 \leq \tilde{\sigma}_0 \leq \sigma_0$ such that, at every type,

$$V(\theta_0, D(\theta_0), \tilde{\sigma}_0(\theta_0)) = V(\theta_0, D(\theta_0), \sigma_0(\theta_0)),$$

and that the controls satisfying (a)–(c) can be chosen measurably with respect to θ_0 from the resulting initial states. Suppose that these controls, together with transfers satisfying Proposition 2(i)–(ii), form an admissible mechanism. The direct case is $\tilde{\sigma}_0 = \sigma_0$. Otherwise, using $\tilde{\sigma}_0$ as the actual initial exposure leaves the objective unchanged and weakly relaxes (4). Each feasible objective value is therefore implementable by a globally incentive-compatible, individually rational mechanism. Hence (12) is an implementable lower bound on the principal's value; whenever its supremum is attained, a mechanism attaining the program's value exists.

The form of (11) recurs in the local program below, and its logic is worth recording once. The finite-variation component of the volatility can be adjusted downward freely, while its upward drift is capped. Free downward adjustment makes the value nondecreasing in ψ —hence the hypothesis—and splits the state space in two. Where the volatility retains strict marginal value ($V_\psi > 0$), lowering it would destroy value, so none occurs: its drift reaches the cap—the term $(r + 2\lambda)\psi V_\psi$ in $\mathcal{L}$ —and the standard equation $rV = \mathcal{L}V$ holds. Where the marginal value is exhausted ($V_\psi = 0$), downward adjustment is costless and keeps the volatility from growing past the boundary; $V_\psi = 0$ at the junction of the two regions is the smooth-pasting condition of singular control. The second program drops the finite-variation restriction and imposes only local incentive compatibility. The volatility may then carry its own diffusion—a free control v —which contributes the additional second-order terms in (13).

Theorem 2b (The local program). (OU) Let $\bar{V}$, nondecreasing in ψ , be a classical C^2 solution of

$$\min \left\{ r\bar{V} - \mathcal{L}\bar{V} - \sup_{v \in \mathbb{R}} \left[\frac{1}{2}v^2 \bar{V}_{\psi\psi} + v(\gamma \bar{V}_{\hat{\theta}\psi} + \psi \bar{V}_{D\psi}) \right], \bar{V}_{\psi} \right\} = 0, \quad (13)$$

satisfying the state-constraint interpretation at $\psi = 0$ and (T1) for the locally incentive-compatible class, where v is the diffusion coefficient of ψ in (9). Then $\bar{V}(\hat{\theta}, D, \psi)$ is an upper bound on the continuation surplus of every admissible locally incentive-compatible continuation starting from that state. Consider the relaxed date-zero program

$$\sup_{D(\cdot), \sigma_0(\cdot), U_c} \int_{\Theta} [\bar{V}(\theta_0, D(\theta_0), \sigma_0(\theta_0)) - h_c(\theta_0) D(\theta_0)] dF(\theta_0) - U_c \quad (14)$$

subject to (7), D nondecreasing, and the following inequalities F -almost everywhere:

$$\sigma_0 \geq 0, \quad \sigma_0(\theta_0) \leq \gamma D'(\theta_0).$$

The ordinary derivative of the monotone function D exists F -almost everywhere. Every incentive-compatible, individually rational mechanism is feasible for this relaxation, so the program's value is an upper bound on the unrestricted principal's value.

The continuation bound is attained from an initial state ξ if there are predictable controls X and v , and a nondecreasing process K in (9), with $\Xi_0 = \xi$, such that:

- (a) the state equations, completed with transfers satisfying Proposition 2(i), produce an admissible continuation delivering rent D with volatility ψ ; this continuation is locally incentive compatible, and (T2) holds;
- (b) for $dt \otimes d\mathbb{P}$ -almost every t , X_t attains the supremum over x in (10); when $\psi_{t-} > 0$, v_t attains the supremum over v in (13) and the first expression there equals zero; when $\psi_{t-} = 0$, $v_t = 0$ and $r\bar{V} = \mathcal{L}_0\bar{V}$;
- (c) continuous adjustment through K occurs only where $\bar{V}_{\psi} = 0$, and, almost surely, every downward jump after date zero leaves the value unchanged:

$$\begin{aligned} \bar{V}_{\psi}(\Xi_{t-}) dK_t^c &= 0, \\ \bar{V}(\Xi_t) &= \bar{V}(\Xi_{t-}) \quad \text{at every jump of } K \text{ after date zero.} \end{aligned}$$

Under these conditions,

$$\bar{V}(\xi) = \mathbb{E} \int_0^{\infty} e^{-rt} \hat{\theta}_t X_t dt,$$

so $\bar{V}$ is the maximal continuation surplus from ξ among admissible locally incentive-compatible continuations, and the stated controls attain it.

Finally, suppose (14) has a maximizer for which D is differentiable on Θ , the two displayed inequalities hold at every type, and the controls satisfying (a)–(c) can be chosen measurably with respect to θ_0 from the initial states selected by the maximizer. Suppose these controls, together with transfers satisfying Proposition 2(i)–(ii), form an admissible mechanism. Then this mechanism is locally incentive compatible and individually rational and attains the local program value. It is optimal among admissible, locally incentive-compatible, individually rational mechanisms whose date-zero rent is nondecreasing and differentiable on Θ , whose initial exposure satisfies the two displayed inequalities at every type, and whose utilities satisfy the envelope condition of Proposition 2(ii).

Under the attainment condition in Theorem 2a, the two programs are the bracket announced at the start of the section: (12) is an implementable lower bound, and (14) dominates every incentive-compatible mechanism. Their feasible sets are convex. Whenever the candidate functions are attained continuation values—as under the attainment clauses of the theorems—the objectives are concave in the delivered pair (D, σ_0) , so the programs are convex.¹¹ If the solution of the local program happens to have finite-variation volatility and to satisfy (4), the two values coincide and the mechanism is optimal outright.

5 The Discrete-Time Analogy

This section uses a discrete-time analogue of the model to provide intuition for, and an interpretation of, the continuous-time results. I make three points. First, for *linear contracts*—mechanisms whose rent moves linearly in each period’s innovation—integral monotonicity (Pavan et al., 2014) collapses to the finite-difference form of the drift bounds of Theorem 1a. Second, these bounds must hold almost surely—path by path, not merely in expectation—and this is why finite variation emerges in the continuous theory (a process that is monotone up to an ordinary time integral is of finite variation). Third, discrete-time reasoning cuts through the stochastic calculus: an elementary argument recovers the form of the necessary and sufficient conditions for local optimality.

In continuous time, the martingale representation theorem implies that every square-integrable rent is an Itô process. In discrete time there is no

¹¹The dynamics and the surplus are linear and the allocation set is convex, so mixing two feasible mechanisms delivers the mixed rent and volatility at the mixed surplus. Concavity does not follow for an arbitrary strict supersolution used only to certify an upper bound.

such representation in general: a rent may depend on the innovations in an arbitrary, nonlinear way. Linear contracts are the class for which the representation holds by construction. The linear contracts satisfying integral monotonicity are the discrete shadow of the finite-variation mechanisms satisfying the supermartingale bound.

The discrete-time model. Fix a step $\Delta t > 0$ and a grid $\{0, \Delta t, 2\Delta t, \dots\}$, and let $\{\varepsilon_{t+1}\}_{t \geq 0}$ be innovations with $\mathbb{E}[\varepsilon_{t+1} \mid \mathcal{F}_t] = 0$ and variance Δt , each standing for a Brownian increment $B_{(t+1)\Delta t} - B_{t\Delta t}$. The type follows the Euler scheme of its diffusion,

$$\theta_{t+1} = \begin{cases} (1 - \lambda\Delta t) \theta_t + \lambda\Delta t \bar{\theta} + \gamma \varepsilon_{t+1} & \text{(OU)}, \\ \theta_t + \mu \theta_t \Delta t + \gamma \varepsilon_{t+1} & \text{(GBM)}. \end{cases}$$

Write $\beta := e^{-r\Delta t}$ for the per-period discount factor. With the bilinear utility $u(\theta, X) = \theta X$, whose marginal value is $u_\theta = X$, the discounted marginal information rent, weighted by the impulse response $I_{t,0} := \partial\theta_t/\partial\theta_0$, is $Z = \sum_{t=0}^{\infty} \beta^t X_t I_{t,0} \Delta t$. The impulse response is deterministic under OU, $I_{t,0} = (1 - \lambda\Delta t)^t$, and stochastic under GBM, $I_{t,0} = \theta_t/\theta_0$, so, writing $\beta_\lambda := \beta(1 - \lambda\Delta t)$,

$$Z = \begin{cases} \sum_{t=0}^{\infty} \beta_\lambda^t X_t \Delta t & \text{(OU)}, \\ \frac{1}{\theta_0} \sum_{t=0}^{\infty} \beta^t \theta_t X_t \Delta t & \text{(GBM)}, \end{cases}$$

the discrete counterpart of the normalized rent Z of Section 4.

Definition 2. A discrete-time mechanism is a linear contract if its discounted marginal rent is affine in the innovations: there is an $\mathcal{F}_0$ -measurable $D(\theta_0)$ and a predictable process $\{\sigma_t\}$ (each σ_t being $\mathcal{F}_t$ -measurable) with

$$Z = D(\theta_0) + \sum_{t=0}^{\infty} \sigma_t \varepsilon_{t+1}, \quad D(\theta_0) = \mathbb{E}[Z \mid \theta_0].$$

The loadings $\{\sigma_t\}$ are the discrete analogue of the rent volatility.

Proposition 3. Let (X, T) be a linear contract in the sense of Definition 2, and fix a truthful initial report. Truthful reporting satisfies integral monotonicity at every $t > 0$ if and only if, for all $t \geq 1$,

$$\text{(OU)} \quad \Delta\sigma_t \leq \lambda\sigma_t \Delta t, \quad \text{(GBM)} \quad \Delta Y_t + \mu Y_t \Delta t \leq -e^{-rt} X_t \Delta t,$$

where $\Delta\sigma_t := \sigma_t - \sigma_{t-1}$, $\Delta Y_t := Y_t - Y_{t-1}$, and $Y_t := \theta_0 \sigma_t / (\gamma \theta_t)$ is the rescaled rent loading of Section 4 (on the truthful path, $\hat{\theta} = \theta$).

These are the drift bounds of Theorem 1a, with differences in place of differentials. The inequalities must hold at every truthful history and hence, restricting attention to strongly truthful equilibria as in Pavan et al. (2014), almost surely.¹² Each period, the inequalities allow σ_t (Y_t , in the geometric case) to rise by at most order Δt . If the volatility itself had a volatility—if σ_t carried its own innovation—then σ_t would rise by order $\sqrt{\Delta t}$ with positive probability, violating the bounds for small Δt . This is the sense in which, as the period shrinks, the inequalities are compatible only with a volatility of finite variation.

We prove the Ornstein–Uhlenbeck case. The geometric case is analogous and is carried out in Appendix G.

Proof of Proposition 3, OU case. Under the Euler scheme the innovation is $\theta_{t+1} - \theta_t - \lambda(\bar{\theta} - \theta_t)\Delta t = \gamma \varepsilon_{t+1}$, so $\varepsilon_{t+1} = \frac{1}{\gamma}(\theta_{t+1} - \theta_t - \lambda(\bar{\theta} - \theta_t)\Delta t)$ and the representation of Definition 2 reads

$$Z = D(\theta_0) + \sum_{t=0}^{\infty} \frac{\sigma_t}{\gamma} (\theta_{t+1} - \theta_t - \lambda(\bar{\theta} - \theta_t)\Delta t).$$

Let $D_t(\hat{\theta}_t, \theta_t)$ denote the current-value continuation rent when the true current type is θ_t and the agent misreports $\hat{\theta}_t$ in the current period, reporting truthfully otherwise. Conditioning the representation on $\mathcal{F}_t$ under truthful reporting gives, with $D_0(\theta_0, \theta_0) = D(\theta_0)$,

$$\beta_{\lambda}^t D_t(\theta_t, \theta_t) = \mathbb{E}_t \left[D(\theta_0) + \sum_{s=0}^{\infty} \sigma_s \varepsilon_{s+1} \right] - \sum_{s=0}^{t-1} \beta_{\lambda}^s X_s \Delta t.$$

Evaluating the continuation under the one-shot misreport $\hat{\theta}_t$, and using that the flow difference vanishes for bilinear utility, $[u_{\theta}(\theta_t, X_t) - u_{\theta}(\hat{\theta}_t, X_t)]\Delta t = [X_t - X_t]\Delta t = 0$,

$$\begin{aligned} \beta_{\lambda}^t D_t(\hat{\theta}_t, \theta_t) &= D(\theta_0) + \sum_{k=0}^{t-2} \sigma_k \varepsilon_{k+1} + \frac{\sigma_{t-1}}{\gamma} (\hat{\theta}_t - \theta_{t-1} - \lambda(\bar{\theta} - \theta_{t-1})\Delta t) \\ &\quad + \frac{1}{\gamma} \mathbb{E}_t [\sigma_t (\theta_{t+1} - \hat{\theta}_t - \lambda(\bar{\theta} - \hat{\theta}_t)\Delta t)] - \sum_{s=0}^{t-1} \beta_{\lambda}^s X_s \Delta t. \end{aligned}$$

¹²Alternatively, when the innovations have full support, truthful histories carry almost every path.

Writing $\hat{\theta}_t = \theta_t + (\hat{\theta}_t - \theta_t)$ in the two innovation terms that flank period t ,

$$\hat{\theta}_t - \theta_{t-1} - \lambda(\bar{\theta} - \theta_{t-1})\Delta t = [\theta_t - \theta_{t-1} - \lambda(\bar{\theta} - \theta_{t-1})\Delta t] + (\hat{\theta}_t - \theta_t),$$

$$\theta_{t+1} - \hat{\theta}_t - \lambda(\bar{\theta} - \hat{\theta}_t)\Delta t = [\theta_{t+1} - \theta_t - \lambda(\bar{\theta} - \theta_t)\Delta t] - (\hat{\theta}_t - \theta_t)(1 - \lambda\Delta t).$$

Subtracting the truthful continuation $D_t(l, l)$, with l in the role of the true current type, the common martingale and history terms cancel and only the perturbation survives:

$$\beta_\lambda^t [D_t(l, l) - D_t(\hat{\theta}_t, l)] = \frac{1}{\gamma} (\sigma_{t-1} - \sigma_t(1 - \lambda\Delta t))(l - \hat{\theta}_t).$$

Integral monotonicity requires that truthful reporting dominate every one-shot misreport in the integrated sense

$$\int_{\hat{\theta}_t}^{\theta_t} [D_t(l, l) - D_t(\hat{\theta}_t, l)] dl \geq 0 \quad \text{for all } \theta_t, \hat{\theta}_t.$$

Because the integrand is linear in $l - \hat{\theta}_t$, the integral equals $\frac{1}{2\gamma} (\sigma_{t-1} - \sigma_t(1 - \lambda\Delta t))(\theta_t - \hat{\theta}_t)^2$, which is nonnegative for every pair $(\theta_t, \hat{\theta}_t)$ if and only if $\sigma_{t-1} \geq \sigma_t(1 - \lambda\Delta t)$ —rearranged, $\Delta\sigma_t \leq \lambda\sigma_t\Delta t$. $\square$

6 Application: Designing Financial Markets

This section applies the machinery to a benchmark problem in market microstructure: a market maker extracting revenue from a single trader whose hedging needs are private and persistent—the dynamic counterpart of the monopolistic dealer’s problem (Glosten, 1989). The first-order approach fails here outright. Because the trader may decline to trade whatever the sign of his hedging need—the participation constraint of Section 3—the first-order solution flattens the rent schedule across the middle of the type space, and the flat is incompatible with the exposure it grants. I first solve the relaxation in closed form for any symmetric regular F (Proposition 4): its rent is flat on an interval of types, its exposure is first best for every type, and the date-zero condition of Theorem 1b fails not only on the flat but at *every* interior type. The control problems of Section 4 take over. They admit closed-form continuation values, the date-zero stage reduces to a convex program, and its solution is a smoothed dynamic bid–ask spread—the classic dealer object, generated here endogenously by two-sided participation rather than by adverse selection on a single trade—with the two programs bracketing the unrestricted optimum

within one to two percent in the specifications computed below. The tight bracket does not arise because dynamics have little value. The static benchmark reported in Table 1 and derived in Appendix H.4 earns less than one third as much as the finite-variation mechanism; most revenue therefore comes from responding to later reports, and the finite-variation mechanism captures essentially all of those gains. The dealer’s problem is also the building block for the corresponding exchange: under the conditions of Corollary 2.1, the multitrader problem decomposes into single-trader problems (Section 6.6).

6.1 The dealer’s problem

A trader’s marginal value for the asset—his type—is the deviation of his endowment from his bliss point, struck by privately observed shocks that decay as the trader rebalances elsewhere:

$$d\theta_t = -\lambda \theta_t dt + dB_t, \quad \theta_0 \sim F.$$

This is the OU specification (1) with long-run mean $\bar{\theta} = 0$ and the volatility normalized to $\gamma = 1$; the mean-reversion rate $\lambda \geq 0$ is free, with $\lambda = 0$ the permanent-shock case. Throughout the application, F is symmetric about its mean, types are centered so that the mean is zero, and f is continuous and strictly positive on the interior of the support, with finite variance; the leading cases are the Gaussian and the uniform. The trader receives a flow allocation X_t (a position in the asset) and pays a transfer flow T_t ; his flow utility is

$$u(\theta_t, X_t) = \theta_t X_t - \frac{1}{2} X_t^2,$$

the bilinear benchmark with quadratic holding cost, an instance of the extension $u = \theta g(X) - c(X)$ discussed in Section 3. The type-independent holding cost can be absorbed into transfers: setting $\tilde{T}_t := T_t + X_t^2/2$ gives the trader flow payoff $\theta_t X_t - \tilde{T}_t$ and the market maker flow revenue $\tilde{T}_t - X_t^2/2$. This reduction entails no additional restriction: every individually rational mechanism with an admissible transfer has $\mathbb{E} \int_0^\infty e^{-rt} X_t^2 dt < \infty$.¹³ The market maker commits to the mechanism—allocation and transfers adapted to the reported type—and maximizes revenue $\mathbb{E} \int_0^\infty e^{-rt} T_t dt$, subject to incentive compatibility and individual rationality for every type.¹⁴

¹³Indeed, $\theta X - X^2/2 = \theta^2/2 - (X - \theta)^2/2$, and θ is discounted square-integrable, so otherwise the trader’s utility is $-\infty$. The same convention assigns utility $-\infty$ to an otherwise admissible off-path report with infinite discounted holding cost; such a report cannot be profitable, and only finite-payoff paths enter the stochastic identities.

¹⁴A weighted objective—revenue plus η times trader surplus—rescales the information-rent price by $1 - \eta$ and changes nothing structural in what follows.

In this specification the objects of Section 4 take a simple form, recorded here once and used throughout: the discounted allocation, the rent, and the current-value volatility are

$$Z = \int_0^\infty e^{-(r+\lambda)t} X_t dt, \quad D_t = \mathbb{E}_t \left[\int_t^\infty e^{-(r+\lambda)(s-t)} X_s ds \right], \quad \psi_t = e^{(r+\lambda)t} \sigma_t.$$

The conditions of Theorem 1a read: $\sigma_t \geq 0$, $\text{drift}(\sigma_t) \leq \lambda \sigma_t$, and no upward jumps; on the finite-variation class, where they are necessary and sufficient for global incentive compatibility after $t = 0$, they combine into the pathwise bound $d\psi_t \leq (r + 2\lambda)\psi_t dt$ —equivalently, $e^{-\lambda t}\sigma_t$ is nonincreasing pathwise. The date-zero condition (4) becomes

$$\mathcal{B}(\theta_0, \hat{\theta}_0) := \int_{\hat{\theta}_0}^{\theta_0} [D(l) - D(\hat{\theta}_0)] dl \geq \frac{1}{2} \sigma_0(\hat{\theta}_0) (\theta_0 - \hat{\theta}_0)^2 \quad \text{for all } \hat{\theta}_0, \theta_0 \in \Theta, \quad (15)$$

equivalently $\sigma_0(\hat{\theta}_0) \leq \sigma_0^{\max}(\hat{\theta}_0) := \inf_{m \neq \hat{\theta}_0} 2\mathcal{B}(m, \hat{\theta}_0)/(m - \hat{\theta}_0)^2$: the initial exposure is capped by the convexity of the rent in the report.

6.2 The dealer's problem as tracking

Efficiency is $X_t = \theta_t$: trade each shock to the bliss point. Write $\zeta_t := X_t - \theta_t$ for the tracking error and

$$\tilde{D}_t := D_t - \frac{\hat{\theta}_t}{r + 2\lambda} = \mathbb{E}_t \left[\int_t^\infty e^{-(r+\lambda)(s-t)} \zeta_s ds \right]$$

for the promised rent in excess of its first-best level (the second equality uses $\mathbb{E}_t[\theta_s] = e^{-\lambda(s-t)}\hat{\theta}_t$; the constant $1/(r + 2\lambda)$ is the discounted impulse response, $\int_0^\infty e^{-(r+\lambda)s} e^{-\lambda s} ds$, the rent slope of the efficient allocation). Using $\theta X - \frac{1}{2}X^2 = \frac{1}{2}\theta^2 - \frac{1}{2}\zeta^2$ and applying Proposition 2(ii) to the effective transfer $\tilde{T}$, revenue is first best, less a quadratic tracking cost, less information rents:

$$\Pi = \mathbb{E} \left[\int_0^\infty e^{-rt} \frac{1}{2} \theta_t^2 dt \right] - \frac{1}{2} \mathbb{E} \left[\int_0^\infty e^{-rt} \zeta_t^2 dt \right] - \mathbb{E}[U(\theta_0)], \quad (16)$$

where $U(\theta_0) := U(\theta_0, 0)$ is the trader's truthful value. The rent term is controlled by an envelope fact that holds in the full class, with no regularity beyond admissibility:

Lemma 1 (Envelope by convexity). *For any admissible incentive-compatible mechanism, U is convex, $U' = D$ almost everywhere, and D is nondecreasing.*

Proof. Because the type enters its own dynamics linearly, the type path is affine in θ_0 with deterministic coefficient $e^{-\lambda t}$, and every reporting strategy is a *rule*—an initial report and a drift distortion adapted to the noise—available identically to every type. The payoff of a fixed rule is therefore affine in the true type with slope $\mathbb{E} \int_0^\infty e^{-(r+\lambda)t} X_t dt$ evaluated along the rule’s report path. So U is a supremum of affine functions, hence convex; incentive compatibility says the truthful rule attains the supremum at θ_0 , so its slope $D(\theta_0)$ is a subgradient, giving $U' = D$ a.e. and monotonicity of D . $\square$

Convexity locates the binding participation constraint: U is minimized at the type θ^* with $D(\theta^*) = 0$ —the trader expecting zero net trade—and at the optimum $U(\theta^*) = 0$. Anchoring rents there, $\mathbb{E}[U] = \mathbb{E}[\Lambda(\theta_0)D(\theta_0)]$ with the *two-sided handicap*

$$\Lambda(\theta) = \begin{cases} (1 - F(\theta))/f(\theta), & \theta > \theta^*, \\ -F(\theta)/f(\theta), & \theta < \theta^* : \end{cases} \quad (17)$$

buyers’ rents are priced from above, sellers’ from below. Equivalently, the rent functional $\text{Rent}(D) = \mathbb{E}[\int_c^{\theta_0} D(l) dl] - \min_{c'} \int_c^{c'} D(l) dl$ is anchor-free and convex in D , so the date-zero program remains convex with the anchor determined endogenously; with F symmetric the optimum is symmetric and the anchor sits at the mean, $\theta^* = 0$.

The tracking form has a useful geometry. Let $\mathcal{H}$ be the Hilbert space of discounted square-integrable processes, with

$$\langle Y, Y' \rangle := \mathbb{E} \int_0^\infty e^{-rt} Y_t Y'_t dt, \quad \|Y\|^2 := \langle Y, Y \rangle.$$

Completing the square in (16) bounds revenue by

$$\frac{1}{2} \|h\|^2 - \frac{1}{2} \|X - h\|^2,$$

where the virtual-value target h is built from θ_t and $\Lambda(\theta_0)$. The minimization is over the set $\mathcal{V} \subset \mathcal{H}$ of allocations implementable by incentive-compatible, individually rational mechanisms. When the minimum is attained, symmetry makes its unique minimizer bind participation at the mean and attain this bound, as shown in Appendix H.6.

6.3 The first-order approach

The first-order approach maximizes revenue subject only to the envelope conditions and participation—ignoring the volatility conditions entirely. I solve

this relaxation in closed form: seeing exactly how it fails shows what the control problems must deliver.

The relaxation rests on an exact decomposition of the tracking cost. By (8), the excess promise evolves as

$$d\tilde{D}_t = [(r + \lambda)\tilde{D}_t - \zeta_t]dt + \nu_t d\hat{B}_t, \quad \nu_t := \psi_t - \frac{1}{r + 2\lambda}, \quad (18)$$

so ν measures the shortfall of the current-value volatility from the first-best level $1/(r + 2\lambda)$, at which the promise would carry no residual risk.

Lemma 2 (Tracking identity). *Fix an initial type and an admissible mechanism whose conditional tracking cost is finite. Then*

$$\begin{aligned} \frac{1}{2} \mathbb{E} \int_0^\infty e^{-rs} \zeta_s^2 ds &= \frac{r+2\lambda}{2} \tilde{D}_0^2 \\ &+ \frac{1}{2} \mathbb{E} \int_0^\infty e^{-rs} (\zeta_s - (r + 2\lambda)\tilde{D}_s)^2 ds + \frac{r+2\lambda}{2} \mathbb{E} \int_0^\infty e^{-rs} \nu_s^2 ds, \end{aligned} \quad (19)$$

all expectations conditional on θ_0 .

Proof. Apply Itô's formula to $e^{-rt} \frac{r+2\lambda}{2} \tilde{D}_t^2$ using (18). Completing the square gives

$$\frac{(r + 2\lambda)^2}{2} \tilde{D}^2 - (r + 2\lambda) \tilde{D} \zeta = \frac{1}{2} (\zeta - (r + 2\lambda) \tilde{D})^2 - \frac{1}{2} \zeta^2.$$

Taking expectations and letting $T \rightarrow \infty$ gives the result.¹⁵ $\square$

The identity makes the relaxation a static problem. Delivering a promised rent $D(\theta_0)$ costs at least $\frac{r+2\lambda}{2} \tilde{D}_0^2$, with equality exactly when the two nonnegative terms vanish: the exposure is first best ($\nu \equiv 0$) and the tracking error amortizes the promise ($\zeta = (r + 2\lambda)\tilde{D}$)—both of which the relaxation permits. What remains is a choice of the schedule $D(\cdot)$, trading the quadratic cost of the promise against the priced rents $\mathbb{E}[\Lambda(\theta_0)D(\theta_0)]$. Call F *regular* if the one-sided virtual value $\theta - (1 - F(\theta))/f(\theta)$ is nondecreasing—by symmetry this covers both sides, and log-concavity of f suffices—and assume it is positive

¹⁵Apply Itô's formula after the usual localization. The square-integrable promise representation and finite tracking cost justify taking expectations and removing the localization. For the terminal term, write $\rho = r + 2\lambda$; conditional Jensen and Cauchy–Schwarz give

$$e^{-rT} \mathbb{E}[\tilde{D}_T^2 \mid \theta_0] \leq \frac{1}{\rho} \mathbb{E} \left[\int_T^\infty e^{-rs} \zeta_s^2 ds \mid \theta_0 \right] \rightarrow 0.$$

for some θ , so that it has a zero $x^* > 0$; for the standard Gaussian, x^* solves $x = (1 - \Phi(x))/\phi(x)$, so $x^* \approx 0.75$, and for the uniform on $[-1, 1]$, $x^* = \frac{1}{2}$. The handicap jumps *up* by $1/f(0)$ at the anchor, so the pointwise virtual value $\theta_0 - \Lambda(\theta_0)$ jumps *down* there: a bid–ask wedge, intrinsic to two-sided participation rather than to any hazard-rate irregularity. The downward jump is what forces the ironing in the following solution.

Proposition 4 (The first-order approach, solved). *Let F be symmetric and regular as above. The relaxation is solved, uniquely up to null sets, by the rent*

$$D^{\text{FOA}}(\theta_0) = \begin{cases} \frac{\theta_0 - \Lambda(\theta_0)}{r + 2\lambda}, & |\theta_0| > x^*, \\ 0, & |\theta_0| \leq x^*, \end{cases} \quad (20)$$

the trader value $U(\theta_0) = \int_{x^*}^{|\theta_0|} D^{\text{FOA}}(l) dl$, zero precisely on the flat, and the allocation

$$X_t = \theta_t - \bar{\Lambda}(\theta_0) e^{-\lambda t}, \quad \bar{\Lambda}(\theta_0) := \theta_0 - (r + 2\lambda) D^{\text{FOA}}(\theta_0), \quad (21)$$

whose induced exposure is first best for every type: $\psi_t \equiv 1/(r + 2\lambda)$. The relaxed value is

$$\Pi^{\text{FOA}} = \frac{1}{2(r + 2\lambda)} \left(\frac{1}{r} + \mathbb{E}[(\theta_0 - \bar{\Lambda}(\theta_0))^2] \right). \quad (22)$$

The proof, in Appendix H, is a Lagrangian in the style of static mechanism design with type-dependent outside options (Jullien, 2000): participation is priced by an explicit multiplier—the measure with distribution function $F(\theta) + \theta f(\theta)$ on $[-x^*, x^*]$, uniform on the no-trade interval when F is uniform—and completing the square against it delivers the ironed schedule by pointwise maximization; no separate ironing argument is needed. On the flat the handicap is executed in full, $\bar{\Lambda}(\theta_0) = \theta_0$, so $X_t = \theta_t - \theta_0 e^{-\lambda t}$: the dealer executes none of the initial position and trades only the subsequent shocks. Off the flat the handicap decays with the impulse response of the initial report.

The first-order solution violates the date-zero condition at every interior type. On the flat the failure is immediate: by (15), a flat of D forces $\sigma_0^{\max} = 0$ there—a trader in the interior of the flat can misreport within it at no rent cost, so any exposure prices a profitable lie—while the mechanism grants first-best exposure $\sigma_0 = 1/(r + 2\lambda)$. Outside the flat, oddness of D^{FOA} makes the mirror pair $(-\theta_0, \theta_0)$ imply $\sigma_0^{\max}(\theta_0) \leq D^{\text{FOA}}(\theta_0)/\theta_0 < 1/(r + 2\lambda)$, again below the delivered exposure. For the uniform distribution, the local condition (3) is nevertheless slack outside the flat, while the integral cap evaluates in

closed form to $\sigma_0^{\max}(\theta_0) = (2 - 1/|\theta_0|)^+ / (r + 2\lambda)$ (Lemma 10 in the appendix), which lies below the delivered exposure at every interior type, with equality only at $\theta_0 = \pm 1$. Figure 2 displays the solution and its failure. As shown in Section 6.5, the finite-variation optimum in the uniform case distorts exposure at every interior type.

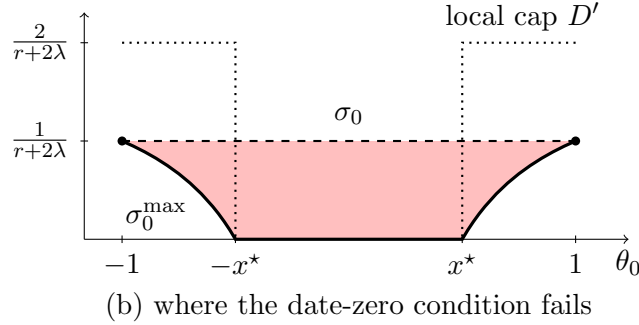
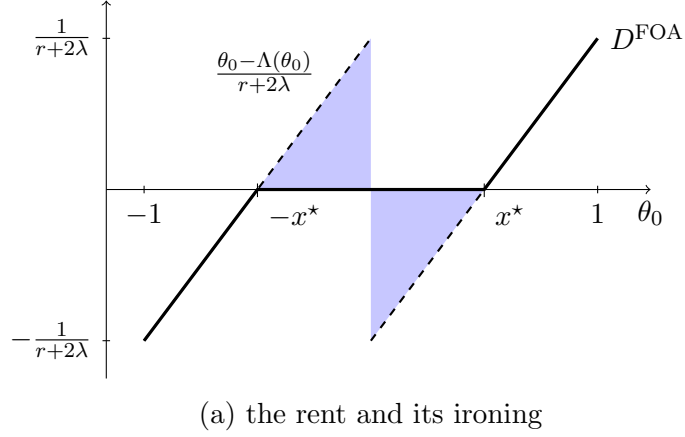


Figure 2: The first-order approach (uniform illustration). Panel (a): the rent (20) (solid) against the unironed virtual rent $(\theta_0 - \Lambda(\theta_0))/(r + 2\lambda)$ (dashed); the handicap's upward jump $1/f(0)$ at the anchor produces the downward jump of the dashed curve, and the multiplier of Proposition 4 flattens it at zero across the shaded region. Off the flat the two coincide. Panel (b): the delivered exposure $\sigma_0 = 1/(r + 2\lambda)$ (dashed) against the cap $\sigma_0^{\max}$ implied by the rent (solid, Lemma 10) and the local cap D' (dotted). On the flat the cap is zero and already the local condition fails; off the flat the local cap is twice the delivered exposure, yet the integral condition (15) still fails. The violation (shaded) covers every interior type, with equality only at $\theta_0 = \pm 1$ (dots): the failure is global, not confined to the no-trade flat.

6.4 The continuation problems in closed form

Once the date-zero rent term is fixed, the dealer maximizes continuation surplus subject to delivering the promised rent and exposure. The HJBs in Theorems 2a-2b apply after replacing the continuation flow $\hat{\theta}X$ by $\hat{\theta}X - X^2/2$. This replacement leaves the state equations and incentive constraints unchanged. Write $\rho := r + 2\lambda$. For either problem, separate first-best continuation surplus by writing

$$V(\hat{\theta}, D, \psi) = \frac{\hat{\theta}^2}{2\rho} + \frac{1}{2r\rho} + J(\tilde{D}, \psi), \quad \tilde{D} = D - \frac{\hat{\theta}}{\rho}.$$

Let $\zeta := X - \hat{\theta}$ be the tracking error and, in the local problem, let v be the diffusion coefficient of ψ . Setting $\nu := \psi - 1/\rho$, define the part common to both HJBs by

$$\begin{aligned} \mathcal{Q}J := & rJ - \sup_{\zeta \in \mathbb{R}} \left[-\frac{1}{2}\zeta^2 + J_{\tilde{D}}((r + \lambda)\tilde{D} - \zeta) \right] \\ & - \frac{1}{2}\nu^2 J_{\tilde{D}\tilde{D}} - \rho\psi J_{\psi}. \end{aligned}$$

The finite-variation HJB is

$$\min \{ \mathcal{Q}J, J_{\psi} \} = 0, \quad (23)$$

whereas the local HJB is

$$\min \left\{ \mathcal{Q}J - \sup_{v \in \mathbb{R}} \left[\frac{1}{2}v^2 J_{\psi\psi} + v\nu J_{\tilde{D}\psi} \right], J_{\psi} \right\} = 0. \quad (24)$$

Proposition 5 (Continuation value). *The finite-variation and local continuation problems have the same value. Their common adjustment to first-best surplus is*

$$J(\tilde{D}, \psi) = -\frac{\rho}{2}\tilde{D}^2 - \frac{\rho}{2}P(\psi), \quad (25)$$

where

$$P(\psi) := \int_0^\infty e^{-rs} \left[\left(\frac{1}{\rho} - e^{\rho s} \psi \right)^+ \right]^2 ds.$$

For $0 \leq \psi_0 \leq 1/\rho$, the common value is attained in both problems by

$$\zeta_t = \rho\tilde{D}_t, \quad \psi_t = \min\{e^{\rho t}\psi_0, 1/\rho\}, \quad v_t = 0.$$

For $\psi_0 > 1/\rho$, the displayed value is the common supremum but is not attained; it is approached by reducing ψ to $1/\rho$ after an arbitrarily short delay. Either date-zero program may therefore replace an initial loading above $1/\rho$ by $1/\rho$, leaving continuation value unchanged and weakly relaxing its incentive constraint.

An innovation in the trader's type changes his efficient position one for one. Under the policy in Proposition 5, $X = \hat{\theta} + \rho\tilde{D}$, so the allocation's loading on that innovation is $\rho\psi$. While $\psi < 1/\rho$, increasing ψ makes the allocation respond more fully to changes in the trader's hedging need and reduces the quadratic loss from failing to track his efficient position. The incentive condition permits a deterministic exposure path to rise no faster than $\rho\psi$. The policy uses that full rate until $\psi = 1/\rho$, where the allocation responds efficiently to new shocks.

The local problem also permits ψ to have a diffusion, but this does not raise its value. The incentive condition implies $\mathbb{E}[\psi_t] \leq e^{\rho t}\psi_0$, while the loss from imperfect exposure is quadratic in $\psi_t - 1/\rho$. Jensen's inequality therefore implies that the deterministic path in Proposition 5 minimizes expected loss. Appendix H makes this argument precise.

The penalty P is strictly decreasing and convex on $(0, 1/\rho]$, with $P = P' = 0$ at first best. As $\delta \downarrow 0$,

$$P\left(\frac{1-\delta}{\rho}\right) = \frac{\delta^3}{3\rho^3}(1 + O(\delta)). \quad (26)$$

Thus a small shortfall from first-best exposure has only a cubic continuation cost, which shapes the date-zero solutions below. At a fixed initial rent and loading, allowing rent volatility to have a diffusion after date zero does not improve continuation value. Any gap between the programs comes entirely from their different date-zero incentive constraints.

6.5 A smoothed dynamic spread

Because J is concave, substituting the common continuation value yields two convex date-zero programs over the schedules $D(\cdot)$ and $\sigma_0(\cdot)$. The finite-variation program imposes the date-zero integral condition (15), whereas the local program imposes only $\sigma_0 \leq D'$. Figure 3 shows their solutions for uniform types at $r = 1$ and $\lambda = 0$.

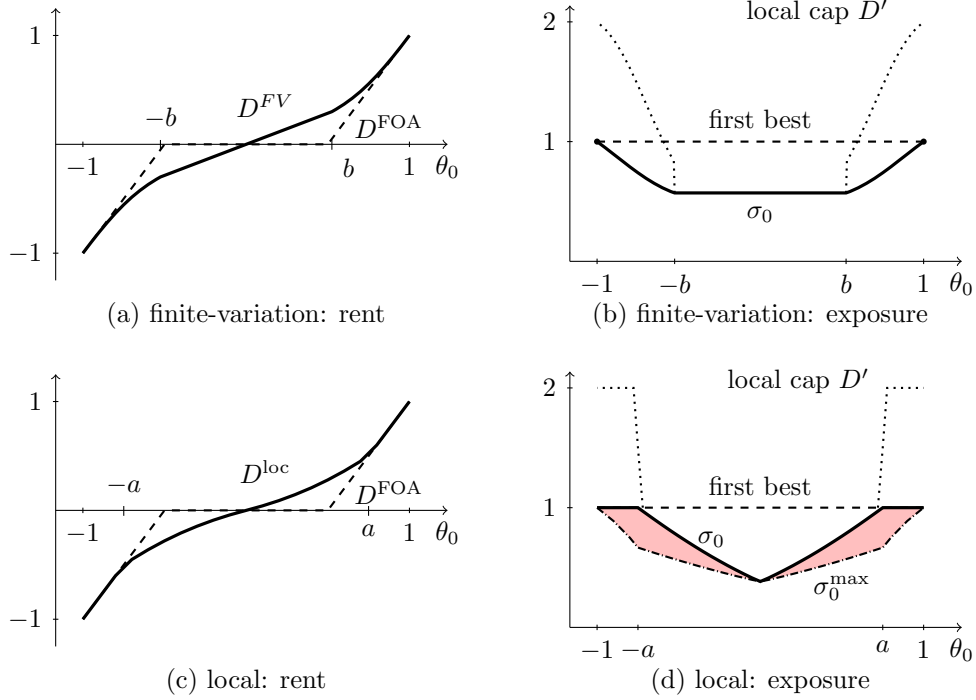


Figure 3: Optimal rents and date-zero exposure in the uniform case ($r = 1$, $\lambda = 0$). Solid curves are the program optima. In the rent panels, dashed curves show the first-order solution. In the exposure panels, horizontal dashed lines show first best, dotted curves show the local cap D' , and the dash-dotted curve in panel (d) shows the integral cap $\sigma_0^{\max}$. In the top row, the finite-variation rent is affine on $[-b, b]$, with $b = 0.5249$. In the bottom row, $a = 0.7497$ marks the point beyond which the local rent coincides with the first-order rent and exposure is first best. The shaded region is the gap between local exposure and the integral cap implied by its rent at every nonzero interior type.

A standard bid–ask spread creates a no-trade interval around zero: small trading needs are left unfilled. The first-order solution has the same date-zero geometry. Its rent is flat on $[-x^*, x^*]$, hence $X_0 = 0$, although later shocks are still traded. Global incentive compatibility requires $\sigma_0 = 0$ on the flat, but the first-order allocation grants first-best exposure $\sigma_0 = 1$; it is therefore infeasible. The finite-variation optimum instead replaces no trade with partial trade. On the bridge $[-b, b]$, where $b = 0.5249$,

$$D^{FV}(\theta_0) = s\theta_0, \quad \sigma_0^{FV}(\theta_0) = s, \quad s = 0.5712.$$

Here $X_0 = s\theta_0$: every nonzero middle type trades in the efficient direction, but at reduced scale. The middle of the market is thinned rather than shut. Later

reports still affect trade, while current-value exposure rises deterministically to first best. No type is excluded; participation binds only at the mean.

The binding pattern explains why the distortion extends beyond the bridge. In its interior, $\sigma_0^{FV} = (D^{FV})' = s$, while $D^{FV}/\theta_0 = s$ for every nonzero type in the bridge; every pairwise date-zero constraint between two distinct types in $[-b, b]$ binds. For $|\theta_0| > b$,

$$\sigma_0^{FV}(\theta_0) = \frac{D^{FV}(\theta_0)}{\theta_0} < (D^{FV})'(\theta_0).$$

The local condition is slack, but the date-zero integral condition binds; the unique binding constraint compares θ_0 with its mirror $-\theta_0$. This comparison caps exposure by the average rent slope between $-\theta_0$ and θ_0 , rather than by the local slope. Exposure is therefore distorted at every interior type and reaches first best only at the endpoints.

In the local program, $\sigma_0^{\text{loc}} = (D^{\text{loc}})'$ binds for $|\theta_0| < a$, where $a = 0.7497$, and exposure rises to first best at $|\theta_0| = a$. For $|\theta_0| > a$, the rent coincides with the first-order schedule and exposure is first best, so the local condition is slack. The rent is continuous but kinked at $\pm a$.

Yet the date-zero integral condition fails at every nonzero interior type. The mirror-type comparison determines the integral cap: for $0 < |\theta_0| < a$, strict convexity gives $(D^{\text{loc}})' > D^{\text{loc}}/\theta_0$; for $a \leq |\theta_0| < 1$, exposure is one while $D^{\text{loc}}/\theta_0 = 2 - 1/|\theta_0| < 1$. The shaded region plots the resulting gap. Thus the local test misses opposite-side comparisons throughout the interior, not only near the mean. Because the program omits these global constraints, its value is an upper bound and its displayed optimizer is not globally incentive compatible.

Table 1 compares the two program values with the first-order and optimal static benchmarks. The static benchmark fixes the allocation path from the date-zero report and ignores later reports (Appendix H.4). The uniform results are analytic (Appendix H.7); the Gaussian results are grid approximations (Appendix H.8).

	uniform $\lambda = 0$	uniform $\lambda = \frac{1}{2}$	Gaussian $\lambda = 0$	Gaussian $\lambda = \frac{1}{2}$
static	15.3%	15.1%	32.3%	31.6%
finite-variation	99.0%	99.0%	98.2%	98.4%
local program	100%	100%	100%	100%
first-order approach	107.3%	106.0%	110.8%	108.4%
Π_{loc}	0.5435	0.2752	0.6375	0.3258

Table 1: Revenue as a fraction of the local program’s value Π_{loc} , with $r = 1$. In the continuum problems, the finite-variation value is an implementable lower bound, while the local value bounds the revenue of every admissible incentive-compatible, individually rational mechanism. The first-order row reports an infeasible relaxation. Uniform values use the analytic characterizations; Gaussian values are grid approximations, not certified continuum bounds.

For the uniform cases, the analytic solutions give a rigorous bracket: the finite-variation mechanism attains 99.0% of the local upper bound, so unrestricted revenue is pinned down to within about one percent. The Gaussian grids show the same pattern, at 98.2% and 98.4%. The first-order value exceeds the local-program value by 6–11 percentage points across the reported values, so most of its apparent gain over finite variation is infeasible. Yet dynamics matter: the static mechanism earns only 15%–32% of the local value. Later reports generate most revenue; the finite-variation mechanism captures nearly all of these gains. The uniform solutions show that global constraints reshape exposure throughout the type space but cost little revenue.

6.6 From dealer to exchange

Finally, consider an exchange that reallocates the asset among N independent traders, each as in Section 6.1. A mechanism maps their reports into allocations and transfers, subject to incentive compatibility, participation, and market clearing. The setup belongs to a benchmark class of market-microstructure models in which traders have private trading needs and quadratic holding costs (e.g., Du and Zhu, 2017; Malamud and Rostek, 2017; Chen and Duffie, 2021; Antill and Duffie, 2021). Those models work with specified trading protocols. Here the exchange instead chooses a revenue-maximizing dynamic mechanism for persistent, privately observed trading needs.

The result holds separately for the unrestricted admissible IC class; the regular finite-variation class satisfying the exposure restrictions of Theorem 1a

and the date-zero inequality (15); and the regular locally IC class of Theorem 2b. For any one of these classes, let $\mathcal{V}$ consist of the allocations that can be paired with an integrable transfer to form an admissible, individually rational single-trader mechanism in that class.

Corollary 2.1 (Exchange decomposition). *Fix any one of the three classes above. Let Λ be the two-sided handicap in (17), with participation anchored at the mean. For each trader i , let*

$$h_t^i := \theta_t^i - \Lambda(\theta_0^i)e^{-\lambda t}$$

be his virtual-value target, and assume that h^i is discounted square-integrable. Suppose that each single-trader minimization problem

$$\min_{Y \in \mathcal{V}} \|Y - h^i\|^2$$

is attained. Let X^{i} denote its unique minimizer, and set*

$$\widehat{X}^i := X^{i*} - \frac{1}{N} \sum_{j=1}^N X^{j*}.$$

Then $\{\widehat{X}^i\}$, together with suitable integrable transfers, defines a feasible exchange mechanism in the selected class and maximizes total revenue among all feasible exchange mechanisms in that class.

Appendix H.6 states the design problems formally and proves the corollary. The corollary is a proof of concept: it shows how the paper’s single-agent results can be used to solve a concrete multi-agent mechanism-design problem.

7 The Master Deviation Formula

The purpose of this section is to elucidate clearly why global incentive compatibility is difficult to characterize simply in a way amenable to optimal control, why exactly the finite-variation class side-steps this difficulty, and to provide intuition for how, more broadly, to construct classes of contracts which can be easily certified in applications.

The key object is the master formula, which expresses the gain from a deviation in terms of the volatility of information rents. I present only the mean-reverting case. The formal derivation and the analogous GBM formula appear in Lemma 3 in Appendix A. The analysis here is informal.

The master formula. Fix a truthful initial report and let the agent distort the drift of his report, so the gap $\delta_t := \hat{\theta}_t - \theta_t$ obeys $d\delta_t = (-\lambda\delta_t + \gamma a_t)dt$ with $\delta_0 = 0$. Given the implementing transfers of Proposition 2, the agent's gain from an arbitrary admissible distortion a , denoted $\mathcal{G}(a) := U(\theta_0, a) - U(\theta_0, 0)$, is *exactly*

$$\mathcal{G}(a) = -\mathbb{E} \int_0^\infty e^{\lambda t} \sigma_t \delta_t a_t dt. \quad (27)$$

Thus, given the implementing transfers, rent volatility is the only feature of the mechanism that enters the gain from a deviation.

One integration by parts. Substituting $\gamma a_t dt = d\delta_t + \lambda\delta_t dt$ and integrating by parts yields

$$\mathcal{G}(a) = -\frac{1}{2\gamma} \mathbb{E} \int_0^\infty e^{\lambda t} \delta_t^2 [\lambda\sigma_t - \text{drift}(\sigma_t)] dt + \frac{1}{2\gamma} \mathbb{E} \int_0^\infty e^{\lambda t} \delta_t^2 \text{vol}(\sigma_t) a_t dt. \quad (28)$$

If σ has finite variation, then $\text{vol}(\sigma) = 0$, so the second term disappears. If also $\text{drift}(\sigma) \leq \lambda\sigma$, the first term is nonpositive.

The cascade. The second term has the same form as the master formula, with $\text{vol}(\sigma)$ in place of σ and one extra power of δ . The same two steps—the pathwise identity $\gamma \delta^{k-1} a dt = \frac{1}{k} d(\delta^k) + \lambda \delta^k dt$, followed by integration by parts—therefore apply again. Write $\mathcal{R}_k(f) := \mathbb{E} \int_0^\infty e^{\lambda t} f_t \delta_t^{k-1} a_t dt$ for any process f . Then

$$\mathcal{R}_k(f) = \frac{1}{k\gamma} \mathbb{E} \int_0^\infty e^{\lambda t} \delta_t^k [(k-1)\lambda f_t - \text{drift}(f_t)] dt - \frac{1}{k\gamma} \mathcal{R}_{k+1}(\text{vol}(f)). \quad (29)$$

By construction, $\mathcal{G}(a) = -\mathcal{R}_2(\sigma)$. Define the iterated volatilities by

$$\text{vol}^0(\sigma) := \sigma, \quad \text{vol}^{j+1}(\sigma) := \text{vol}(\text{vol}^j(\sigma)).$$

Iterating (29) gives

$$\begin{aligned} \mathcal{G}(a) = \sum_{k=2}^K \frac{(-1)^{k-1}}{k! \gamma^{k-1}} \mathbb{E} \int_0^\infty e^{\lambda t} \delta_t^k [(k-1)\lambda \text{vol}^{k-2}(\sigma_t) - \text{drift}(\text{vol}^{k-2}(\sigma_t))] dt \\ + \frac{(-1)^K}{K! \gamma^{K-1}} \mathcal{R}_{K+1}(\text{vol}^{K-1}(\sigma)). \end{aligned} \quad (30)$$

This is the cascade of higher-order terms in δ . A simple characterization of global incentive compatibility is difficult because every term must be controlled

for deviations of arbitrary size. Finite variation stops the cascade because $\text{vol}(\sigma) = 0$, leaving only the quadratic term. Otherwise, additional structure is needed to determine whether the total gain is nonpositive.

Beyond finite variation. The master formula also shows where global incentives may remain easy to check beyond finite variation: when successive volatilities follow a simple pattern, the higher-order terms can be summed. For example, consider

$$d\sigma_t = m_t dt + \alpha \sigma_t d\hat{B}_t, \quad \alpha \neq 0 \text{ constant},$$

so that $\text{vol}^j(\sigma) = \alpha^j \sigma$. The cascade then sums to

$$\mathcal{G}(a) = \frac{\gamma}{\alpha^2} \mathbb{E} \int_0^\infty e^{\lambda t} \left[m_t h(x_t) - \lambda \sigma_t g(x_t) \right] dt, \quad x_t := \frac{\alpha \delta_t}{\gamma}, \quad (31)$$

where $h(x) := e^{-x} - 1 + x$ and $g(x) := 1 - (1 + x)e^{-x}$. Both functions are nonnegative. Thus, if $\sigma_t \geq 0$ (condition (i) of Theorem 1a) and $m_t \leq 0$, the gain is nonpositive and no admissible deviation is profitable. The finite-variation condition leaves more slack: it permits $m_t \leq \lambda \sigma_t$, including some positive drift when $\sigma_t > 0$. Here the corresponding simple condition is $m_t \leq 0$.

8 Contracting without Transfers

The main implementability results, Theorems 1a and 1b, do not require the presence of transfers. Call a contract *first-order* if, under truthful continuation from every initial report, the agent's continuation value has the volatility of Proposition 2(i).

Proposition 6 (Incentives without transfers). *Suppose that $T \equiv 0$. Under the maintained admissibility and regularity conditions, the conclusions of Theorem 1a apply as stated to every first-order contract. If, in addition, the contract satisfies the date-zero envelope condition in Proposition 2(ii), then the conclusions of Theorem 1b also apply under the hypotheses stated there.*

Without transfers, however, a recursive formulation generally requires continuation utility as an additional state variable, as in Williams (2011) or Bloedel et al. (2023). Proposition 6 provides a second-order condition for local optimality in this broader setting as well.

9 Concluding Discussion

This paper has made transparent the connection between the methods of dynamic mechanism design in the Myersonian tradition and the martingale methods of continuous-time contracting, and has shown how this connection can yield new insights. The main result characterizes incentives in terms of the volatility of the agent’s information rents.

Following the initial report, and under the usual envelope conditions, a supermartingale restriction on this volatility is necessary and sufficient for local incentive compatibility. Within the class of finite-variation contracts, the same restriction also characterizes global incentive compatibility. Because the condition can be imposed pointwise through a drift constraint on the volatility process, the design problem is amenable to optimal control with volatility as a state variable: I present two programs that together bracket the designer’s payoff even where the first-order approach fails. The machinery is put to work in a benchmark setting from financial market microstructure, where that failure occurs for fundamental economic reasons.

Most of the paper carries the methods of continuous-time contracting into the Myersonian setting. The discrete-time methods of that setting, however, also yield new insights for continuous-time contracting. The integral-monotonicity condition can be used to conjecture, through elementary algebra, the correct conditions for local incentive compatibility in continuous time, and reveals the finite-variation class as the counterpart of readily interpretable “linear contracts.” Moreover, the incentive characterization applies without transfers to the first-order contracts of Section 8. When drift distortions can be positive or negative, the supermartingale condition on the volatility of rents completes the characterization of local incentive compatibility.

Several directions remain open, and the methods developed here appear more general than the paper’s setting. First, the local analysis seems readily adaptable to general utility functions. For finite-variation contracts, however, the resulting incentive conditions are likely to remain sufficient but may no longer be necessary: concavity of utility in type can introduce slack into these conditions. A second direction is to consider general Markov diffusions, which is closely linked to curvature in utility: when utility is multiplicatively separable in type and allocation, a redefinition of the type variable maps curvature into a change of diffusion, in which case the two extensions are isomorphic. Finally, the exchange application shows that the results already speak to multi-agent settings in which the optimal allocation is additively separable across agents’ reports. The case in which the volatility of each agent’s rent depends on the others’ reports is open to investigation.

A Proof of Theorem 1a

After collecting notation in Section A.1, Section A.2 derives the master deviation formula and records the model-specific algebra used below. Section A.3 then takes the candidate semimartingale representative in the theorem as given and verifies sufficiency directly. Section A.4 proves the converse: local incentive compatibility itself supplies the required supermartingale, and hence semimartingale, representative. Section A.5 assembles the two directions and completes the finite-variation statement.

A.1 Preliminaries

All objects are those of Section 4: the rent volatility σ , the rent D_t , and, in the GBM case, $Y_t = \hat{\theta}_0 \sigma_t / (\gamma \hat{\theta}_t)$ and $A_t = \gamma \hat{\theta}_t (Y_t - e^{-rt} D_t)$. Since $\mathbb{E}_t[Z]$ is a martingale with volatility σ , the discounted rent obeys

$$\begin{aligned} d(e^{-(r+\lambda)t} D_t) &= -e^{-(r+\lambda)t} X_t dt + \sigma_t d\hat{B}_t \quad (\text{OU}), \\ d(e^{-rt} \hat{\theta}_t D_t) &= -e^{-rt} \hat{\theta}_t X_t dt + \hat{\theta}_0 \sigma_t d\hat{B}_t \quad (\text{GBM}). \end{aligned} \tag{32}$$

We write $S_t := e^{-\lambda t} \sigma_t$ and, when treating the two models in parallel,

$$R_t := \begin{cases} S_t & (\text{OU}), \\ A_t & (\text{GBM}). \end{cases}$$

Also let $\delta_t := \hat{\theta}_t - \theta_t$ denote the report gap and, as in Condition 1, $\phi_t = \delta_t / \hat{\theta}_t$ and $\Psi_t = -\phi_t - \ln(1 - \phi_t)$.

Under $\mathbb{P}$, B is a Brownian motion and $\hat{B}_t = B_t + \int_0^t a_s ds$. When ξ is a semimartingale, write the absolutely continuous part of its truthful-path finite-variation component as $m_t dt$ and its continuous local-martingale part as $\ell_t d\hat{B}_t$. We write

$$\text{drift}(\xi_t) := m_t, \quad \text{drift}_B(\xi_t) := m_t + \ell_t a_t, \quad \text{vol}(\xi_t) := \ell_t$$

for its truthful-path drift, its $\mathbb{P}$ -drift, and its volatility. Expectations are under $\mathbb{P}$, where $\mathbb{E} \int H dB = 0$ for square-integrable H .

We choose the coefficient processes and their decomposition components as predictable functionals on the canonical report space, and use pathwise versions of their stochastic integrals. Consequently, identities that hold under the truthful report law continue to hold along any admissible report by (C1).

The representation theorem initially supplies σ only as a predictable $dt \otimes d\mathbb{P}$ equivalence class. No semimartingale version is assumed in the necessity argument. Whenever such a version is invoked below, it is either the candidate in the conditional verification of Section A.3 or the representative constructed in Section A.4. Independence of θ_0 and B preserves the Brownian martingale-representation property under the initial enlargement. After localization, every local martingale is therefore an $\mathcal{F}_0$ -measurable initial value plus a stochastic integral with respect to B , and hence is continuous. Local incentive compatibility is understood as in Definition 1, here with truthful initial report ($h = 0$).

A.2 The master deviation formula

Lemma 3 (Master deviation formula). *Fix an initial report $\hat{\theta}_0 \in \Theta$ and an admissible distortion $a \in \mathcal{A}$, and suppose the transfers satisfy Proposition 2(i). Write $\delta_0 = \hat{\theta}_0 - \theta_0$, and let $W_0(\hat{\theta}_0)$ be the continuation value delivered at time zero to a truthful agent of type $\hat{\theta}_0$. Then*

$$U(\hat{\theta}_0, a) = W_0(\hat{\theta}_0) - D(\hat{\theta}_0) \delta_0 - \mathbb{E} \int_0^\infty \Lambda_t \delta_t a_t dt, \quad \Lambda_t := \begin{cases} e^{\lambda t} \sigma_t & (\text{OU}), \\ A_t / \hat{\theta}_t & (\text{GBM}). \end{cases} \quad (33)$$

In particular, for a truthful initial report, $U(\theta_0, a) - U(\theta_0, 0) = -\mathbb{E} \int \Lambda_t \delta_t a_t dt$.

Proof. Let W_t be the agent's continuation value under truthful continuation from $\hat{\theta}_t$ if $\theta_t = \hat{\theta}_t$, and $g_t := \text{vol}(W_t)$, so that

$$d(e^{-rt} W_t) = -e^{-rt} (\hat{\theta}_t X_t - T_t) dt + e^{-rt} g_t d\hat{B}_t. \quad (34)$$

The OU case. If the agent stops distorting at time t , the report gap thereafter decays at rate λ :

$$\delta_s = e^{-\lambda(s-t)} \delta_t, \quad s \geq t.$$

Consequently,

$$\mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} \delta_s X_s ds \right] = D_t \delta_t.$$

The corresponding gains process is

$$G_t := \int_0^t e^{-rs} (\theta_s X_s - T_s) ds + e^{-rt} (W_t - D_t \delta_t).$$

Here $G_0 = W_0(\hat{\theta}_0) - D(\hat{\theta}_0)\delta_0$, while $\mathbb{E}[G_t] \rightarrow U(\hat{\theta}_0, a)$ by (C2) and (C4).

To compute its drift, first note that δ has finite variation. Thus

$$d(e^{-rt}D_t\delta_t) = \delta_t d(e^{-rt}D_t) + e^{-rt}D_t d\delta_t,$$

with no quadratic-covariation term. Hence, using (34), (32), and

$$d\delta_t = (-\lambda\delta_t + \gamma a_t) dt,$$

we obtain

$$\begin{aligned} dG_t &= e^{-rt}(\theta_t X_t - T_t) dt + d(e^{-rt}W_t) - d(e^{-rt}D_t\delta_t) \\ &= e^{-rt}(\theta_t X_t - T_t) dt - e^{-rt}(\hat{\theta}_t X_t - T_t) dt + e^{-rt}g_t d\hat{B}_t \\ &\quad - \delta_t \left[(-e^{-rt}X_t + \lambda e^{-rt}D_t) dt + e^{\lambda t}\sigma_t d\hat{B}_t \right] \\ &\quad - e^{-rt}D_t(-\lambda\delta_t + \gamma a_t) dt. \end{aligned}$$

Because $\theta_t - \hat{\theta}_t = -\delta_t$, the X_t -terms cancel, as do the two terms in $\lambda D_t\delta_t$. Hence

$$dG_t = -e^{-rt}\gamma D_t a_t dt + (e^{-rt}g_t - e^{\lambda t}\sigma_t\delta_t) d\hat{B}_t.$$

Substituting $d\hat{B}_t = dB_t + a_t dt$ gives

$$\begin{aligned} dG_t &= [e^{-rt}(g_t - \gamma D_t)a_t - e^{\lambda t}\sigma_t\delta_t a_t] dt \\ &\quad + (e^{-rt}g_t - e^{\lambda t}\sigma_t\delta_t) dB_t. \end{aligned}$$

Thus

$$\text{drift}_B(G_t) = e^{-rt}(g_t - \gamma D_t)a_t - e^{\lambda t}\sigma_t\delta_t a_t.$$

Proposition 2(i) gives $g_t = \gamma D_t$, so only the second term remains. The diffusion term is a $\mathbb{P}$ -martingale by (C3). Taking expectations and using (C4) therefore yields

$$U(\hat{\theta}_0, a) - G_0 = \mathbb{E} \left[\int_0^\infty \text{drift}_B(G_t) dt \right],$$

and proves the OU case of (33).

The GBM case. In the GBM case the report gap has a diffusion component, so the product formula also contains a quadratic-covariation term.

First derive the dynamics of D_t . By (32), $\hat{\theta}_t D_t$ has drift $r\hat{\theta}_t D_t - \hat{\theta}_t X_t$ and volatility

$$e^{rt}\hat{\theta}_0\sigma_t = \gamma e^{rt}\hat{\theta}_t Y_t.$$

Applying Itô's formula to

$$D_t = (\hat{\theta}_t D_t) \hat{\theta}_t^{-1},$$

and using

$$d(\hat{\theta}_t^{-1}) = \hat{\theta}_t^{-1}[(\gamma^2 - \mu) dt - \gamma d\hat{B}_t], \quad d\langle \hat{\theta} D, \hat{\theta}^{-1} \rangle_t = -\gamma^2 e^{rt} Y_t dt,$$

gives

$$dD_t = \left[(r + \gamma^2 - \mu) D_t - X_t - \gamma^2 e^{rt} Y_t \right] dt + \gamma (e^{rt} Y_t - D_t) d\hat{B}_t. \quad (35)$$

In particular,

$$\text{vol}(D_t) = \gamma (e^{rt} Y_t - D_t) = \frac{e^{rt} A_t}{\hat{\theta}_t},$$

by the definition of A_t .

Under $\mathbb{P}$, the lie evolves according to

$$d\delta_t = (\mu \delta_t + \gamma \hat{\theta}_t a_t) dt + \gamma \delta_t dB_t. \quad (36)$$

A lie is *frozen* after time t if $a_s = 0$ for all $s \geq t$. Then δ and $\hat{\theta}$ follow the same homogeneous GBM, so

$$\delta_s = \frac{\hat{\theta}_s}{\hat{\theta}_t} \delta_t, \quad s \geq t.$$

The continuation correction is therefore exact:

$$\mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} \delta_s X_s ds \right] = \frac{\delta_t}{\hat{\theta}_t} \mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} \hat{\theta}_s X_s ds \right] = D_t \delta_t.$$

As in the OU case, define

$$G_t := \int_0^t e^{-rs} (\theta_s X_s - T_s) ds + e^{-rt} (W_t - D_t \delta_t).$$

Then $G_0 = W_0(\hat{\theta}_0) - D(\hat{\theta}_0)\delta_0$, and $\mathbb{E}[G_t] \rightarrow U(\hat{\theta}_0, a)$ by (C2) and (C4).

Using $d\hat{B}_t = dB_t + a_t dt$, equation (35) implies

$$\begin{aligned} d(e^{-rt} D_t) &= e^{-rt} \left[(\gamma^2 - \mu) D_t - X_t - \gamma^2 e^{rt} Y_t + \gamma (e^{rt} Y_t - D_t) a_t \right] dt \\ &\quad + e^{-rt} \gamma (e^{rt} Y_t - D_t) dB_t. \end{aligned}$$

Moreover, because both D_t and δ_t have diffusion components,

$$d(e^{-rt} D_t \delta_t) = \delta_t d(e^{-rt} D_t) + e^{-rt} D_t d\delta_t + d\langle e^{-rt} D, \delta \rangle_t,$$

where

$$d\langle e^{-rt}D, \delta \rangle_t = e^{-rt}\gamma^2(e^{rt}Y_t - D_t)\delta_t dt.$$

Combining this product rule with (34), using $\theta_t - \hat{\theta}_t = -\delta_t$, and cancelling the transfer terms gives

$$\begin{aligned} e^{rt} \text{drift}_B(G_t) = & -\delta_t X_t + g_t a_t - D_t(\mu\delta_t + \gamma\hat{\theta}_t a_t) \\ & - \gamma^2 \delta_t (e^{rt}Y_t - D_t) \\ & - \delta_t \left[(\gamma^2 - \mu)D_t - X_t - \gamma^2 e^{rt}Y_t + \gamma(e^{rt}Y_t - D_t)a_t \right]. \end{aligned}$$

The terms independent of a_t cancel. Collecting the remaining terms gives

$$e^{rt} \text{drift}_B(G_t) = (g_t - \gamma\hat{\theta}_t D_t)a_t - \gamma(e^{rt}Y_t - D_t)\delta_t a_t.$$

Proposition 2(i) gives

$$g_t = \gamma\hat{\theta}_t D_t,$$

and the definition of A_t then yields

$$\text{drift}_B(G_t) = -e^{-rt}\gamma(e^{rt}Y_t - D_t)\delta_t a_t = -\frac{A_t}{\hat{\theta}_t}\delta_t a_t.$$

The diffusion term in G_t is a $\mathbb{P}$ -martingale by (C3). Taking expectations and using (C4) gives

$$U(\hat{\theta}_0, a) - G_0 = -\mathbb{E} \left[\int_0^\infty \frac{A_t}{\hat{\theta}_t} \delta_t a_t dt \right].$$

This proves the GBM case of (33). $\square$

We next rewrite the master term using a finite-variation function of the gap. Integration by parts then carries no Itô correction.

OU. The gap has finite variation, $d\delta_t = (-\lambda\delta_t + \gamma a_t)dt$, so δ^2 has no Itô correction and

$$\gamma \delta_t a_t dt = \frac{1}{2} d(\delta_t^2) + \lambda \delta_t^2 dt, \quad \text{equivalently} \quad e^{\lambda t} \sigma_t \gamma \delta_t a_t dt = \frac{1}{2} S_t d(e^{2\lambda t} \delta_t^2). \quad (37)$$

For each fixed initial report $\hat{\theta}_0$, $\sigma_0(\hat{\theta}_0) = S_0$ is deterministic.

GBM. Here δ carries a diffusion, so use the relative gap $\phi_t = \delta_t/\hat{\theta}_t \in (-\infty, 1)$. In the $\mathbb{P}$ -decomposition, $d(\hat{\theta}_t^{-1}) = \hat{\theta}_t^{-1}[(\gamma^2 - \mu - \gamma a_t) dt - \gamma dB_t]$ and $d\langle \delta, \hat{\theta}^{-1} \rangle = -\gamma^2 \phi_t dt$, so the product rule gives

$$\begin{aligned} d\phi_t &= [(\mu\phi_t + \gamma a_t) dt + \gamma\phi_t dB_t] + \phi_t[(\gamma^2 - \mu - \gamma a_t) dt - \gamma dB_t] - \gamma^2 \phi_t dt \\ &= \gamma a_t(1 - \phi_t) dt. \end{aligned} \tag{38}$$

Thus ϕ has finite variation and remains constant whenever $a = 0$. The potential from Condition 1,

$$\Psi_t = -\phi_t - \ln(1 - \phi_t) \geq 0,$$

is convex, is minimized at $\phi = 0$, and satisfies $\Psi_t = \frac{1}{2}\phi_t^2 + O(\phi_t^3)$. Moreover,

$$d\Psi_t = \frac{\phi_t}{1 - \phi_t} d\phi_t = \gamma \phi_t a_t dt, \tag{39}$$

so the GBM master integrand transforms as

$$\frac{A_t}{\hat{\theta}_t} \delta_t a_t dt = A_t \phi_t a_t dt = \frac{1}{\gamma} A_t d\Psi_t. \tag{40}$$

The identities above are algebraic and do not assume that S or A is a semimartingale. The next identity is used only when σ admits a semimartingale representative. In that case A and Y are semimartingales in the GBM model, and the product rule applied to $A_t = \gamma\hat{\theta}_t(Y_t - e^{-rt}D_t)$, with $\text{vol}(\hat{\theta}_t) = \gamma\hat{\theta}_t$ and $\text{vol}(e^{-rt}D_t) = A_t/\hat{\theta}_t$ from (35), gives

$$\text{vol}(A_t) = \gamma A_t + \gamma\hat{\theta}_t \text{vol}(Y_t) - \gamma A_t = \gamma\hat{\theta}_t \text{vol}(Y_t). \tag{41}$$

A.3 Verification under the supermartingale condition

We first derive the gain decompositions conditional on R being a nonnegative supermartingale. We then show that the theorem's semimartingale conditions imply this premise and read sufficiency directly from the resulting formulas.

Proposition 7 (OU gain formula). *Suppose the transfers satisfy Proposition 2(i) and $(\hat{\theta}_0, a)$ satisfies Condition 1. Suppose also that S is a non-negative supermartingale under the truthful report law, and write its Doob–Meyer decomposition as $dS_t = -dK_t + \text{vol}(S_t) d\hat{B}_t$, where K is predictable*

and increasing and $K_0 = 0$. Then $\mathbb{E} \int_0^\infty e^{2\lambda t} \delta_t^2 dK_t < \infty$, the limit $L := \lim_{T \rightarrow \infty} \mathbb{E}[e^{\lambda T} \sigma_T \delta_T^2] \geq 0$ exists, and

$$\begin{aligned} U(\hat{\theta}_0, a) - W_0(\hat{\theta}_0) + D(\hat{\theta}_0) \delta_0 &= \frac{1}{2\gamma} \sigma_0(\hat{\theta}_0) \delta_0^2 - \frac{1}{2\gamma} \mathbb{E} \int_0^\infty e^{2\lambda t} \delta_t^2 dK_t - \frac{L}{2\gamma} \\ &\quad + \frac{1}{2\gamma} \mathbb{E} \int_0^\infty e^{\lambda t} \delta_t^2 \text{vol}(\sigma_t) a_t dt. \end{aligned} \quad (42)$$

Proof. The master integral is absolutely convergent [(C4)]. On $[0, T]$, substitute (37) and integrate by parts. Since δ^2 is continuous and of finite variation and S is càdlàg, their cross-variation is zero:

$$\int_0^T S_t d(e^{2\lambda t} \delta_t^2) = e^{2\lambda T} S_T \delta_T^2 - S_0 \delta_0^2 - \int_0^T e^{2\lambda t} \delta_t^2 dS_t.$$

Substitute $dS = -dK + \text{vol}(S) d\hat{B}$ and $d\hat{B} = a dt + dB$, and take expectations. The dB -integral has zero expectation because its integrand $e^{\lambda t} \delta^2 \text{vol}(\sigma)$ is square-integrable [(C3)]. Using $e^{2\lambda T} S_T = e^{\lambda T} \sigma_T$ gives, for every $T < \infty$,

$$\begin{aligned} U(\hat{\theta}_0, a) - W_0(\hat{\theta}_0) + D(\hat{\theta}_0) \delta_0 &= \frac{1}{2\gamma} \sigma_0(\hat{\theta}_0) \delta_0^2 - \frac{1}{2\gamma} \mathbb{E}[e^{\lambda T} \sigma_T \delta_T^2] \\ &\quad - \frac{1}{2\gamma} \mathbb{E} \int_0^T e^{2\lambda t} \delta_t^2 dK_t + \frac{1}{2\gamma} \mathbb{E} \int_0^T e^{\lambda t} \delta_t^2 \text{vol}(\sigma_t) a_t dt \\ &\quad - \mathbb{E} \int_T^\infty e^{\lambda t} \sigma_t \delta_t a_t dt. \end{aligned} \quad (43)$$

As $T \rightarrow \infty$, the last term vanishes and the change-of-measure integral converges [(C3)–(C4)]. The two nonnegative quantities

$$\mathbb{E}[e^{\lambda T} \sigma_T \delta_T^2] \quad \text{and} \quad \mathbb{E} \int_0^T e^{2\lambda t} \delta_t^2 dK_t$$

must therefore have a finite sum in the limit. The second is increasing in T , so both converge separately. Denoting the first limit by $L \geq 0$ gives (42). $\square$

Proposition 8 (GBM gain formula). *Suppose the transfers satisfy Proposition 2(i) and $(\hat{\theta}_0, a)$ satisfies Condition 1. Suppose also that A is a nonnegative supermartingale under the truthful report law. Write its Doob–Meyer decomposition as*

$$dA_t = -dK_t + \text{vol}(A_t) d\hat{B}_t,$$

where K is predictable and increasing and $K_0 = 0$. Then $\mathbb{E} \int_0^\infty \Psi_t dK_t < \infty$, the limit $L^A := \lim_{T \rightarrow \infty} \mathbb{E}[A_T \Psi_T] \geq 0$ exists, and

$$U(\hat{\theta}_0, a) - W_0(\hat{\theta}_0) + D(\hat{\theta}_0) \delta_0 = \frac{1}{\gamma} A_0(\hat{\theta}_0) \Psi_0 - \frac{1}{\gamma} \mathbb{E} \int_0^\infty \Psi_t dK_t - \frac{L^A}{\gamma} + \mathbb{E} \int_0^\infty \Psi_t \hat{\theta}_t \text{vol}(Y_t) a_t dt, \quad (44)$$

where $A_0(\hat{\theta}_0)$ is deterministic for each fixed initial report.

Proof. The proof follows the preceding argument. Split the master integral [(C4)]. On $[0, T]$, use (40) and integrate by parts:

$$\int_0^T A_t d\Psi_t = A_T \Psi_T - A_0 \Psi_0 - \int_0^T \Psi_t dA_t.$$

There is no cross-variation because Ψ is continuous and of finite variation. Substitute $dA = -dK + \text{vol}(A) d\hat{B}$ and take expectations. The dB -integrand $\Psi \text{vol}(A)$ is square-integrable [(C3)], and $\text{vol}(A) = \gamma \hat{\theta} \text{vol}(Y)$ by (41). Finally let $T \rightarrow \infty$. As in the OU case, the tail vanishes, the change-of-measure integral converges, and the nonnegative boundary and dK terms converge separately. This gives (44). $\square$

The two decompositions have the same structure. With a truthful initial report, the initial-gap term vanishes; the compensator and terminal terms are nonpositive; and only the feedback term can raise the deviation payoff. This makes the sufficiency direction immediate once the candidate supermartingale representative is taken as given.

Proposition 9 (Local and global verification). *Suppose that the predictable equivalence class of σ admits a càdlàg semimartingale representative satisfying conditions (i)–(iii) of Theorem 1a. Then truthful continuation is locally incentive compatible. If σ in the OU case, respectively Y in the GBM case, is of finite variation, truthful continuation is globally incentive compatible.*

Proof. Condition (ii) makes the continuous finite-variation part of R nonincreasing. Local martingales are continuous in this filtration, so every jump of R belongs to its finite-variation part; condition (iii) makes each such jump nonpositive. Hence the entire finite-variation part is nonincreasing. Condition (i), together with the càdlàg property, makes R nonnegative up to indistinguishability. Thus R is a nonnegative local supermartingale and hence a supermartingale under the truthful report law. Write

$$dR_t = -dK_t + \text{vol}(R_t) d\hat{B}_t,$$

with K predictable and increasing and $K_0 = 0$. By (C1), the path-functional identity, the monotonicity of K , and nonnegativity continue to hold along an admissible report.

Fix $a \in \mathcal{A}$ and consider the scaled distortion εa from a truthful initial report. In the OU case, $\delta^{(\varepsilon, a)} = \varepsilon \tilde{\delta}^a$. Proposition 7 therefore gives

$$\frac{U(\theta_0, \varepsilon a) - U(\theta_0, 0)}{\varepsilon^2} \leq \frac{\varepsilon}{2\gamma} \mathbb{E} \int_0^\infty e^{\lambda t} (\tilde{\delta}_t^a)^2 |\text{vol}(\sigma_t^{(\varepsilon, a)}) a_t| dt,$$

after the two nonpositive terms are dropped. The right-hand side converges to zero by (C5).

In the GBM case, use $\text{vol}(A) = \gamma \hat{\theta} \text{vol}(Y)$ in Proposition 8. Dropping the same two nonpositive terms yields

$$\frac{U(\theta_0, \varepsilon a) - U(\theta_0, 0)}{\varepsilon^2} \leq \frac{1}{\gamma \varepsilon} \mathbb{E} \int_0^\infty \Psi_t^{(\varepsilon, a)} |\text{vol}(A_t^{(\varepsilon, a)}) a_t| dt,$$

which again converges to zero by (C5). This proves local incentive compatibility.

Now suppose that σ (OU), respectively Y (GBM), is of finite variation. A continuous local martingale of finite variation is constant, so $\text{vol}(\sigma) \equiv 0$, respectively $\text{vol}(Y) \equiv 0$, and $\text{vol}(A) \equiv 0$ in the GBM case by (41). The feedback terms in Propositions 7–8 then vanish for every $a \in \mathcal{A}$. Their remaining terms are nonpositive when the initial report is truthful, proving global incentive compatibility. $\square$

A.4 Necessity and supermartingale regularization

The preceding verification took a semimartingale representative as given. We now show that local incentive compatibility itself supplies one. This argument uses the master formula and the algebraic identities of Section A.2, but not the conditional gain decompositions above. Initially, R is only a predictable $dt \otimes d\mathbb{P}$ equivalence class; Condition 1(C3) gives its local square integrability.

Lemma 4 (Directional evaluation). *Let a be bounded with support in $[0, T_0]$, and let $\hat{\theta}^{(\varepsilon)}$ be the report induced by the distortion εa with truthful initial report. Then for every path functional G with $\mathbb{E}[G(\hat{\theta})^2] < \infty$, $\mathbb{E}[G(\hat{\theta}^{(\varepsilon)})] \rightarrow \mathbb{E}[G(\hat{\theta})]$ as $\varepsilon \downarrow 0$.*

Proof. Because a is bounded and compactly supported, Novikov's condition holds. Let $\mathbb{P}_\varepsilon$ and $\mathbb{P}_0$ denote the laws of the deviated and truthful reports on the canonical path space. Girsanov's theorem gives a density $\xi_\varepsilon = d\mathbb{P}_\varepsilon/d\mathbb{P}_0$

with $\mathbb{E}[\xi_\varepsilon] = 1$ and $\mathbb{E}[\xi_\varepsilon^2] \leq \exp(\varepsilon^2 \|a\|_\infty^2 T_0)$. If a is specified on the underlying filtered space rather than as a canonical path functional, ξ_ε is the conditional expectation of the underlying Girsanov density given the reported path, and the same bound follows from conditional Jensen. Here and below, expectations involving ξ_ε are under $\mathbb{P}_0$. Hence $\|\xi_\varepsilon - 1\|_{L^2}^2 = \mathbb{E}[\xi_\varepsilon^2] - 1 \rightarrow 0$. Therefore

$$\left| \mathbb{E}[G(\hat{\theta}^{(\varepsilon)})] - \mathbb{E}[G(\hat{\theta})] \right| = \left| \mathbb{E}[(\xi_\varepsilon - 1)G(\hat{\theta})] \right| \leq \|G\|_{L^2} \|\xi_\varepsilon - 1\|_{L^2} \rightarrow 0.$$

□

Lemma 5 (Supermartingale regularization). *Let R be a predictable process such that $\mathbb{E} \int_0^T R_t^2 dt < \infty$ for every $T < \infty$. Suppose that, on a set $\mathcal{L} \subset (0, \infty)$ of full Lebesgue measure, $R_s \geq 0$ a.s. and*

$$\mathbb{E}[\mathbf{1}_E R_t] \leq \mathbb{E}[\mathbf{1}_E R_s] \quad \text{for all } s < t \text{ in } \mathcal{L} \text{ and } E \in \mathcal{F}_s. \quad (45)$$

Then R has a nonnegative càdlàg supermartingale representative on $(0, \infty)$, equal to $R dt \otimes d\mathbb{P}$ -a.e. If $R_t \rightarrow R_0$ in L^1 as $t \downarrow 0$ for some finite $R_0 \in L^1(\mathcal{F}_0)$, the representative extends to $[0, \infty)$ with initial value R_0 .

Proof. After modification on a null set of times, joint measurability and local square integrability let us view $t \mapsto R_t$ as a strongly measurable element of $L_{\text{loc}}^2((0, \infty); L^2(\Omega))$. The vector-valued Lebesgue differentiation theorem therefore gives a full-measure set of right $L^2(\Omega)$ -Lebesgue points. Intersect $\mathcal{L}$ with this set and choose a countable dense set $\mathcal{D} \subset \mathcal{L}$. On $\mathcal{D}$, (45) defines a nonnegative supermartingale system. The upcrossing theorem, applied on successive compact intervals, gives finite right and left limits along $\mathcal{D}$. Set

$$\bar{R}_t := \lim_{\substack{q \downarrow t \\ q \in \mathcal{D}}} R_q, \quad t > 0.$$

These limits have càdlàg paths, and $\bar{R}_t$ is $\mathcal{F}_t$ -measurable because the filtration is right-continuous.

We verify the supermartingale inequality. Fix $s < t$, choose $p_n \downarrow s$ in $\mathcal{D}$, and first fix $q \in \mathcal{D}$ with $q > t$. For all large n ,

$$\mathbb{E}[R_q \mid \mathcal{F}_{p_n}] \leq R_{p_n}.$$

Reverse-martingale convergence and the definition of $\bar{R}_s$ give $\mathbb{E}[R_q \mid \mathcal{F}_s] \leq \bar{R}_s$. Now take $q_n \downarrow t$ in $\mathcal{D}$. Conditional Fatou yields

$$\mathbb{E}[\bar{R}_t \mid \mathcal{F}_s] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[R_{q_n} \mid \mathcal{F}_s] \leq \bar{R}_s.$$

Finally, at almost every t , right-hand Lebesgue differentiation gives $t_n \downarrow t$, $t_n \in \mathcal{L}$, with $R_{t_n} \rightarrow R_t$ in L^2 . Apply the same upcrossing argument to the countable set $\mathcal{D} \cup \{t_n : n \geq 1\}$. Its right limit must equal the limit already obtained along $\mathcal{D}$, so $R_{t_n} \rightarrow \bar{R}_t$ a.s. Hence $\bar{R}_t = R_t$ a.s. This proves equality $dt \otimes d\mathbb{P}$ -a.e. If $R_t \rightarrow R_0$ in L^1 as $t \downarrow 0$, then $\sup_{q \in \mathcal{D} \cap (0,1]} \mathbb{E}[R_q] < \infty$. The same upcrossing argument therefore gives an almost-sure limit as $q \downarrow 0$ through $\mathcal{D}$. Choose $q_n \downarrow 0$ among the full-measure set of times at which $\bar{R}_{q_n} = R_{q_n}$ almost surely. After passing to a subsequence, L^1 convergence gives $R_{q_n} \rightarrow R_0$ almost surely, so the right limit is R_0 . Define $\bar{R}_0 = R_0$; repeating the preceding conditional argument with $p_n \downarrow 0$ proves the supermartingale inequality from time zero. $\square$

Proposition 10 (Direct supermartingale necessity). *If reporting is locally incentive compatible for $t > 0$, then R has a nonnegative càdlàg supermartingale representative. Consequently, σ has a càdlàg semimartingale representative satisfying conditions (i)–(iii) of Theorem 1a.*

Proof. Let $\mathcal{L}$ be the full-measure set of right $L^2(\Omega)$ -Lebesgue points of R . Fix $s < t$ in $\mathcal{L}$, an event $E \in \mathcal{F}_s$, and $0 < \eta < (t - s)/2$. Let $v^{s,t,\eta}$ be the deterministic function that rises linearly from zero to one on $(s, s + \eta]$, remains one through $(s + \eta, t]$, and falls linearly to zero on $(t, t + \eta]$. The distortion below is zero through s , so the report is truthful up to that date. Thus E may be represented as an event determined by the reported history through s , and the resulting direction is progressively measurable as a functional of the report. For each fixed η , the one- and two-ramp directions used below are bounded and compactly supported. By the richness requirement in Condition 1, they satisfy (C1)–(C4) against the mechanism; their small scalings do so by (C5).

In the OU case, use the direction

$$a_u^\eta := \frac{e^{-\lambda u}}{\gamma \eta} \mathbf{1}_E (\mathbf{1}_{(s, s+\eta]}(u) - \mathbf{1}_{(t, t+\eta]}(u)). \quad (46)$$

For the scaled distortion εa^η , set $q_u = e^{\lambda u} \delta_u$. Then

$$q_u = \varepsilon \mathbf{1}_E v_u^{s,t,\eta}.$$

Since

$$e^{\lambda u} \sigma_u \delta_u \varepsilon a_u^\eta du = \frac{1}{2\gamma} S_u d(q_u^2),$$

the master formula gives the exact identity

$$\frac{U(\theta_0, \varepsilon a^\eta) - U(\theta_0, 0)}{\varepsilon^2} = -\frac{1}{2\gamma} \mathbb{E} \int_0^\infty S_u^{(\varepsilon)} d[\mathbf{1}_E (v_u^{s,t,\eta})^2]. \quad (47)$$

For GBM, omit $e^{-\lambda u}$ from (46). By (38),

$$z_u := -\log(1 - \phi_u) = \varepsilon \mathbf{1}_E v_u^{s,t,\eta}.$$

Writing $H(z) := z - 1 + e^{-z}$, we have $\Psi = H(z)$ and $H(z) = z^2/2 + O(z^3)$. Equations (40) and (39) therefore give

$$\frac{U(\theta_0, \varepsilon a^\eta) - U(\theta_0, 0)}{\varepsilon^2} = -\frac{1}{\gamma} \mathbb{E} \int_0^\infty A_u^{(\varepsilon)} d \left[\frac{H(\varepsilon \mathbf{1}_E v_u^{s,t,\eta})}{\varepsilon^2} \right]. \quad (48)$$

For fixed η , the integrals in (47)–(48) are square-integrable path functionals [(C3)]. Lemma 4 thus allows $\varepsilon \downarrow 0$. In the GBM expression, $H(\varepsilon v)/\varepsilon^2 \rightarrow v^2/2$, uniformly together with its derivative for $0 \leq v \leq 1$. The resulting finite-variation weight difference tends to zero by Cauchy–Schwarz and (C3), while Lemma 4 handles the change in report law. Both models consequently have the same second-order limit:

$$Q_\eta := \lim_{\varepsilon \downarrow 0} \frac{U(\theta_0, \varepsilon a^\eta) - U(\theta_0, 0)}{\varepsilon^2} = -\frac{1}{2\gamma} \mathbb{E} \int_0^\infty R_u d[\mathbf{1}_E (v_u^{s,t,\eta})^2].$$

The positive and negative parts of the signed measure $d(v^{s,t,\eta})^2$ each have unit mass and are supported on $(s, s + \eta]$ and $(t, t + \eta]$, respectively. They therefore form one-sided approximate identities. Local incentive compatibility gives $Q_\eta \leq 0$. Letting $\eta \downarrow 0$ and using that $s, t \in \mathcal{L}$ yields

$$\mathbb{E}[\mathbf{1}_E (R_t - R_s)] \leq 0. \quad (49)$$

It remains to establish nonnegativity. Use only the first ramp in (46), and in the GBM case again omit $e^{-\lambda u}$. The corresponding v rises from zero to one and then remains one. The same calculation, first as $\varepsilon \downarrow 0$ and then as $\eta \downarrow 0$, gives

$$-\frac{1}{2\gamma} \mathbb{E}[\mathbf{1}_E R_s] \leq 0.$$

Taking $E = \{R_s \leq -c\}$ for arbitrary $c > 0$ shows that $R_s \geq 0$ a.s. for every $s \in \mathcal{L}$.

Lemma 5, applied to (49), now gives a nonnegative càdlàg supermartingale representative $\bar{R}$ on $(0, \infty)$. The L^1 limit in (C3) supplies its value at zero. In the OU case, $\bar{\sigma}_t = e^{\lambda t} \bar{R}_t$ is the required semimartingale representative. In the GBM case,

$$A_t = \hat{\theta}_0 \sigma_t - \gamma e^{-rt} \hat{\theta}_t D_t.$$

Hence

$$\bar{\sigma}_t := \frac{\bar{R}_t + \gamma e^{-rt} \hat{\theta}_t D_t}{\hat{\theta}_0}$$

is a càdlàg semimartingale and equals the original representation coefficient $dt \otimes d\mathbb{P}$ -a.e.; the semimartingale property of the second term follows directly from (32).

Finally, write the local Doob–Meyer decomposition of $\bar{R}$ as $\bar{R} = R_0 + M - K$, where M is a local martingale and K is predictable and increasing. Every local martingale in the initially enlarged Brownian filtration is continuous. Hence the continuous finite-variation part of $\bar{R}$ is nonincreasing and

$$\Delta \bar{R} = -\Delta K \leq 0.$$

These are conditions (ii)–(iii). Its absolutely continuous part satisfies

$$\text{drift}(\sigma_t) \leq \lambda \sigma_t \quad (\text{OU}), \quad \text{drift}(A_t) \leq 0 \quad (\text{GBM})$$

$dt \otimes d\mathbb{P}$ -a.e. Nonnegativity of $\bar{R}$ is condition (i). □

A.5 Assembly and the finite-variation reduction

Propositions 9 and 10 prove sufficiency and necessity, respectively, and establish the equivalent nonnegative-supermartingale formulation.

On the finite-variation locus, Proposition 9 gives global incentive compatibility. Conversely, global incentive compatibility implies local incentive compatibility, so Proposition 10 gives conditions (i)–(iii). It remains only to verify the stated GBM reduction. Equation (41) and continuity of local martingales show that A is of finite variation if and only if Y is. On this locus, $\sigma_t = \gamma \hat{\theta}_t Y_t / \hat{\theta}_0$ and $\langle \hat{\theta}, Y \rangle = 0$. Using (32), the product rule gives

$$dA_t = \gamma \hat{\theta}_t [dY_t + \mu Y_t dt + e^{-rt} X_t dt].$$

Hence $dA_t \leq 0$ is equivalent to

$$dY_t + \mu Y_t dt \leq -e^{-rt} X_t dt,$$

which is the finite-variation reduction stated in the theorem. □

B Proof of Theorem 1b

The engine is the master deviation formula of Lemma 3, evaluated for a *joint* deviation—an initial misreport combined with a continuation distortion, i.e. a general direction of Definition 1. Under the envelope condition of Proposition 2(ii), the truthful value function satisfies $V(y) = U(c, 0) + \int_c^y D(l) dl$ for

all $y \in \Theta$, and $W_0(\hat{\theta}_0) = V(\hat{\theta}_0)$: the continuation value delivered to the report $\hat{\theta}_0$ is the truthful value of type $\hat{\theta}_0$. The master formula therefore reads

$$U(\hat{\theta}_0, a) = V(\hat{\theta}_0) + (\theta_0 - \hat{\theta}_0) D(\hat{\theta}_0) - \mathbb{E} \int_0^\infty \Lambda_t \delta_t a_t dt. \quad (50)$$

Truthful continuation is assumed locally incentive compatible after every truthful initial report, so Proposition 2(i) supplies the transfer-volatility condition and Theorem 1a makes S (OU), respectively A (GBM), a nonnegative supermartingale under the truthful law from each initial report. By (C1), the associated path-functional identities and inequalities continue to hold along admissible reports. No optimality after an initial misreport is assumed.

B.1 Local deviations at date zero

Fix $\theta_0 \in \Theta$. Given a feasible direction (h, a) , write

$$y_\varepsilon := \theta_0 + \varepsilon h$$

for its initial report. The part of the gain due only to that report has the expansion

$$\begin{aligned} & V(y_\varepsilon) + (\theta_0 - y_\varepsilon) D(y_\varepsilon) - V(\theta_0) \\ &= - \int_{y_\varepsilon}^{\theta_0} [D(l) - D(y_\varepsilon)] dl = - \frac{\varepsilon^2 h^2}{2} D'(\theta_0) + o(\varepsilon^2). \end{aligned} \quad (51)$$

The last equality uses differentiability of D at θ_0 .

The following finite-horizon consequence of (C6) will be used for the short deviations in the necessity argument.

Lemma 6 (Joint directional evaluation). *Let a be bounded with support in $[0, T]$, and let w be a deterministic element of $L^2[0, T]$. For $R = S$ in the OU model and $R = A$ in the GBM model,*

$$\mathbb{E} \int_0^T w_t R_t^{(\varepsilon, h, a)} dt \longrightarrow \mathbb{E} \int_0^T w_t R_t^{\theta_0} dt.$$

Proof. First hold the initial report fixed at y_ε . Girsanov's theorem, exactly as in Lemma 4, gives a density ξ_ε for the report distorted by εa relative to truthful continuation such that $\|\xi_\varepsilon - 1\|_{L^2} \rightarrow 0$; because a is bounded, the estimate is independent of the moving initial value y_ε . Moreover,

$$\left\| \int_0^T w_t R_t^{y_\varepsilon} dt \right\|_{L^2} \leq \|w\|_{L^2[0, T]} \left(\mathbb{E} \int_0^T |R_t^{y_\varepsilon}|^2 dt \right)^{1/2}$$

is locally bounded in ε by (C3), (C6), and the triangle inequality. The change in law therefore contributes $o(1)$. The $L^2([0, T] \times \Omega)$ continuity in (C6) then permits $y_\varepsilon \rightarrow \theta_0$, proving the claim. $\square$

Lemma 6 is needed only for the bounded short ramps in the necessity argument. Sufficiency for an arbitrary $a \in \mathcal{A}$ instead uses the joint-feedback limits in (C6) directly.

Sufficiency. Assume (3), and fix an arbitrary direction (h, a) . In the OU model,

$$\frac{\delta_t^{(\varepsilon, h, a)}}{\varepsilon} = \Delta_t^{h, a} := h e^{-\lambda t} + \tilde{\delta}_t^a.$$

Apply Proposition 7 from the initial report y_ε , and discard its nonpositive compensator and terminal terms. After division by ε^2 , this gives

$$\begin{aligned} \frac{U(y_\varepsilon, \varepsilon a) - V(\theta_0)}{\varepsilon^2} &\leq \frac{V(y_\varepsilon) + (\theta_0 - y_\varepsilon)D(y_\varepsilon) - V(\theta_0)}{\varepsilon^2} + \frac{h^2}{2\gamma} \sigma_0(y_\varepsilon) \\ &\quad + \frac{\varepsilon}{2\gamma} \mathbb{E} \int_0^\infty e^{\lambda t} (\Delta_t^{h, a})^2 \left| \text{vol}(\sigma_t^{(\varepsilon, h, a)}) a_t \right| dt. \end{aligned}$$

By (51) and (C6), the last term converges to zero and $\sigma_0(y_\varepsilon) \rightarrow \sigma_0(\theta_0)$. Hence

$$\limsup_{\varepsilon \downarrow 0} \frac{U(y_\varepsilon, \varepsilon a) - V(\theta_0)}{\varepsilon^2} \leq \frac{h^2}{2} \left(\frac{\sigma_0(\theta_0)}{\gamma} - D'(\theta_0) \right) \leq 0.$$

For GBM, let

$$\phi_0^\varepsilon = \frac{\varepsilon h}{y_\varepsilon}, \quad \Psi_0^\varepsilon = -\phi_0^\varepsilon - \log(1 - \phi_0^\varepsilon) = \log\left(\frac{y_\varepsilon}{\theta_0}\right) + \frac{\theta_0}{y_\varepsilon} - 1.$$

Then $\Psi_0^\varepsilon/\varepsilon^2 \rightarrow h^2/(2\theta_0^2)$. Proposition 8, followed by the same deletion of nonpositive terms, yields

$$\begin{aligned} \frac{U(y_\varepsilon, \varepsilon a) - V(\theta_0)}{\varepsilon^2} &\leq \frac{V(y_\varepsilon) + (\theta_0 - y_\varepsilon)D(y_\varepsilon) - V(\theta_0)}{\varepsilon^2} + \frac{A_0(y_\varepsilon) \Psi_0^\varepsilon}{\gamma \varepsilon^2} \\ &\quad + \frac{1}{\gamma \varepsilon} \mathbb{E} \int_0^\infty \Psi_t^{(\varepsilon, h, a)} \left| \text{vol}(A_t^{(\varepsilon, h, a)}) a_t \right| dt. \end{aligned}$$

Condition (C6) again makes the last term $o(1)$ and gives continuity of A_0 . Since $A_0(\theta_0) = \gamma \theta_0 [Y_0(\theta_0) - D(\theta_0)]$,

$$\limsup_{\varepsilon \downarrow 0} \frac{U(y_\varepsilon, \varepsilon a) - V(\theta_0)}{\varepsilon^2} \leq \frac{h^2}{2} \left(\frac{A_0(\theta_0)}{\gamma \theta_0^2} - D'(\theta_0) \right) \leq 0$$

by the GBM line of (3). This proves local sufficiency in both models.

Necessity. Assume local incentive compatibility, and fix a nonzero feasible initial direction h . For each $\tau > 0$, consider first the OU direction

$$a_t^\tau := -\frac{he^{-\lambda t}}{\gamma\tau} \mathbf{1}_{(0,\tau]}(t).$$

This direction is bounded and compactly supported. On $[0, \tau]$, its transformed gap satisfies the exact identity

$$e^{\lambda t} \frac{\delta_t^{(\varepsilon, h, a^\tau)}}{\varepsilon} = h \left(1 - \frac{t}{\tau}\right);$$

after τ , the gap is zero. Substitution in the master formula (50) gives

$$\begin{aligned} \frac{U(y_\varepsilon, \varepsilon a^\tau) - V(\theta_0)}{\varepsilon^2} &= \frac{V(y_\varepsilon) + (\theta_0 - y_\varepsilon)D(y_\varepsilon) - V(\theta_0)}{\varepsilon^2} \\ &\quad + \frac{h^2}{\gamma\tau} \mathbb{E} \int_0^\tau S_t^{(\varepsilon, h, a^\tau)} \left(1 - \frac{t}{\tau}\right) dt. \end{aligned} \quad (52)$$

Lemma 6 and (51) show that the limit as $\varepsilon \downarrow 0$ exists and equals

$$-\frac{h^2}{2} D'(\theta_0) + \frac{h^2}{\gamma\tau} \mathbb{E} \int_0^\tau S_t^{\theta_0} \left(1 - \frac{t}{\tau}\right) dt.$$

By (C3), $S_t^{\theta_0} \rightarrow S_0^{\theta_0}$ in L^1 as $t \downarrow 0$. Local incentive compatibility, followed by $\tau \downarrow 0$, therefore implies

$$0 \geq \frac{h^2}{2} \left(\frac{\sigma_0(\theta_0)}{\gamma} - D'(\theta_0) \right).$$

This is the OU condition in (3).

For GBM, take

$$a_t^\tau := -\frac{h}{\gamma\theta_0\tau} \mathbf{1}_{(0,\tau]}(t).$$

Set $z_t^\varepsilon := -\log(1 - \phi_t^\varepsilon)$. Equation (38) gives, uniformly on $[0, \tau]$,

$$z_t^\varepsilon = \log\left(\frac{y_\varepsilon}{\theta_0}\right) - \frac{\varepsilon h t}{\theta_0 \tau}, \quad \frac{\phi_t^\varepsilon}{\varepsilon} \longrightarrow \frac{h}{\theta_0} \left(1 - \frac{t}{\tau}\right).$$

At $t = \tau$, the residual log-gap is $O(\varepsilon^2)$; it contributes nothing further to the master integral because a^τ is then zero. By (C6), these scaled joint reports

satisfy (C1)–(C4) against the mechanism, so the master formula applies despite the residual gap. The master formula now gives

$$\frac{U(y_\varepsilon, \varepsilon a^\tau) - V(\theta_0)}{\varepsilon^2} = \frac{V(y_\varepsilon) + (\theta_0 - y_\varepsilon)D(y_\varepsilon) - V(\theta_0)}{\varepsilon^2} + \frac{h}{\gamma\theta_0\tau} \mathbb{E} \int_0^\tau A_t^{(\varepsilon, h, a^\tau)} \frac{\phi_t^\varepsilon}{\varepsilon} dt. \quad (53)$$

To make the varying weight explicit, set

$$w_\varepsilon(t) := \frac{\phi_t^\varepsilon}{\varepsilon}, \quad w(t) := \frac{h}{\theta_0} \left(1 - \frac{t}{\tau}\right).$$

The preceding convergence is uniform on $[0, \tau]$. The density estimate in Lemma 6 and the locally uniform L^2 bound from (C3), (C6), and the triangle inequality give the explicit bound below, where ξ_ε is the density used in that lemma:

$$\begin{aligned} & \left| \mathbb{E} \int_0^\tau A_t^{(\varepsilon, h, a^\tau)} [w_\varepsilon(t) - w(t)] dt \right| \\ & \leq \|w_\varepsilon - w\|_\infty \|\xi_\varepsilon\|_{L^2} \sqrt{\tau} \left(\mathbb{E} \int_0^\tau |A_t^{y_\varepsilon}|^2 dt \right)^{1/2} \longrightarrow 0. \end{aligned}$$

Applying Lemma 6 to the fixed weight w , the limit as $\varepsilon \downarrow 0$ in (53) is thus

$$-\frac{h^2}{2} D'(\theta_0) + \frac{h^2}{\gamma\theta_0^2\tau} \mathbb{E} \int_0^\tau A_t^{\theta_0} \left(1 - \frac{t}{\tau}\right) dt.$$

Letting $\tau \downarrow 0$, using (C3), and invoking local incentive compatibility yields

$$0 \geq \frac{h^2}{2} \left(\frac{A_0(\theta_0)}{\gamma\theta_0^2} - D'(\theta_0) \right).$$

Because $A_0(\theta_0) = \gamma\theta_0[Y_0(\theta_0) - D(\theta_0)]$, this is precisely the GBM condition in (3). Since θ_0 was arbitrary, necessity follows for every initial type.

B.2 Global verification on the finite-variation locus

For this subsection, retain the envelope and truthful-continuation conditions of Theorem 1b, but do not assume that D is differentiable. Suppose that σ in the OU model, respectively Y in the GBM model, is of finite variation and that (4), respectively (5), holds. Then $\text{vol}(\sigma) = 0$, respectively $\text{vol}(A) = 0$, so the

feedback terms in Propositions 7–8 vanish for every admissible deviation. As above, local incentive compatibility after each truthful initial report implies, by Theorem 1a, the nonnegative supermartingale condition required by those propositions. Condition (C1) permits the resulting path-functional formulas to be evaluated along deviated reports.

OU. Apply Proposition 7 with $W_0(\hat{\theta}_0) = V(\hat{\theta}_0)$ and an arbitrary initial gap. Dropping the nonpositive compensator and terminal terms gives

$$U(\hat{\theta}_0, a) \leq V(\hat{\theta}_0) + (\theta_0 - \hat{\theta}_0) D(\hat{\theta}_0) + \frac{1}{2\gamma} \sigma_0(\hat{\theta}_0) (\theta_0 - \hat{\theta}_0)^2 \quad \text{for every } a \in \mathcal{A}. \quad (54)$$

This bound holds uniformly over continuation strategies, including adaptive ones. Using $V(\theta_0) - V(\hat{\theta}_0) = \int_{\hat{\theta}_0}^{\theta_0} D(l) dl$ and $\frac{1}{2}(\theta_0 - \hat{\theta}_0)^2 = \int_{\hat{\theta}_0}^{\theta_0} (l - \hat{\theta}_0) dl$,

$$V(\theta_0) - U(\hat{\theta}_0, a) \geq \int_{\hat{\theta}_0}^{\theta_0} \left[D(l) - D(\hat{\theta}_0) - \frac{\sigma_0(\hat{\theta}_0)}{\gamma} (l - \hat{\theta}_0) \right] dl \geq 0$$

by (4).

GBM. Here

$$\phi_0 = \frac{\hat{\theta}_0 - \theta_0}{\hat{\theta}_0}, \quad \Psi_0 = -\phi_0 - \log(1 - \phi_0).$$

Dropping the nonpositive terms in Proposition 8 yields

$$U(\hat{\theta}_0, a) \leq V(\hat{\theta}_0) + (\theta_0 - \hat{\theta}_0) D(\hat{\theta}_0) + \frac{1}{\gamma} A_0(\hat{\theta}_0) \Psi_0.$$

The initial-gap correction has the exact integral representation

$$\Psi_0 = \int_{\hat{\theta}_0}^{\theta_0} \frac{l - \hat{\theta}_0}{\hat{\theta}_0 l} dl \quad (55)$$

because both sides vanish at $\theta_0 = \hat{\theta}_0$ and have θ_0 -derivative $(\theta_0 - \hat{\theta}_0)/(\hat{\theta}_0 \theta_0)$. Since $A_0(\hat{\theta}_0) = \gamma \hat{\theta}_0 (Y_0(\hat{\theta}_0) - D(\hat{\theta}_0))$,

$$V(\theta_0) - U(\hat{\theta}_0, a) \geq \int_{\hat{\theta}_0}^{\theta_0} \left[D(l) - D(\hat{\theta}_0) - (Y_0(\hat{\theta}_0) - D(\hat{\theta}_0)) \frac{l - \hat{\theta}_0}{l} \right] dl \geq 0$$

by (5). This proves global incentive compatibility at $t = 0$ in both models. $\square$

Remark 1 (Parallel shift: monotonicity of the rent). *Taking $a \equiv 0$ in (50) gives*

$$U(\hat{\theta}_0, 0) = V(\hat{\theta}_0) + (\theta_0 - \hat{\theta}_0)D(\hat{\theta}_0)$$

exactly. Incentive compatibility at $t = 0$ therefore requires

$$\int_{\hat{\theta}_0}^{\theta_0} [D(l) - D(\hat{\theta}_0)] dl \geq 0 \quad \text{for every } \hat{\theta}_0, \theta_0 \in \Theta.$$

Equivalently, D is nondecreasing. This conclusion is independent of the volatility restrictions.

C Strict Local Incentives and Approximation

Definition 1 permits equality in the quadratic incentive conditions. At such a boundary, higher-order terms may determine the sign of a deviation. This section records the precise sense in which that qualification does not change the principal's local design problem. The argument perturbs the mechanism, rather than treating its rent volatility as a freely chosen process, so the resulting strict inequality is implementable.

For a mechanism M , write U_M for the agent's payoff. Say that M has *strict local incentives* if, at every initial type and for every nonzero admissible direction (h, a) ,

$$\limsup_{\varepsilon \downarrow 0} \frac{U_M(\theta_0 + \varepsilon h, \varepsilon a) - U_M(\theta_0, 0)}{\varepsilon^2} < 0. \quad (56)$$

This condition makes truth-telling optimal for every sufficiently small scaling of each fixed direction. It does not require a common neighborhood that is uniform across directions.

Proposition 11 (Strict approximation). *Fix either the OU or the GBM model, and consider a class of admissible, individually rational mechanisms with differentiable rents that satisfies the transfer conditions of Proposition 2, the regularity requirements used in Theorems 1a–1b, and is closed under deterministic convex mixtures. Suppose the class contains a finite-revenue benchmark mechanism $\bar{M}$ satisfying the relevant date-zero inequality in (3). Suppose also that there is a deterministic, integrable function $\bar{k}_t > 0$ such that, after every initial report,*

$$\bar{S}_t = \int_t^\infty \bar{k}_s ds \quad (\text{OU}), \quad \bar{A}_t = \int_t^\infty \bar{k}_s ds \quad (\text{GBM}). \quad (57)$$

Let $M = (X, T)$ be any mechanism in the class satisfying conditions (i)–(iii) of Theorem 1a and the relevant date-zero inequality in (3). For $\eta \in (0, 1)$, form the pointwise mixture

$$M^\eta = (X^\eta, T^\eta) := (1 - \eta)(X, T) + \eta(\bar{X}, \bar{T}).$$

Then M^η has strict local incentives. Moreover,

$$\Pi(M^\eta) = (1 - \eta)\Pi(M) + \eta\Pi(\bar{M}) \longrightarrow \Pi(M) \quad \text{as } \eta \downarrow 0.$$

Consequently, within any such class, imposing strict local incentives does not change the supremum of the principal's local design problem.

Proof. Rents, martingale-representation coefficients, and continuation values are linear in the mechanism. Thus $D^\eta = (1 - \eta)D + \eta\bar{D}$, and, writing $R = S$ in the OU model and $R = A$ in the GBM model,

$$R^\eta = (1 - \eta)R + \eta\bar{R}.$$

Theorem 1a makes R a nonnegative supermartingale. If $dR = -dK + \text{vol}(R)dB$ is its Doob–Meyer decomposition, then (57) gives

$$dK_t^\eta = (1 - \eta)dK_t + \eta\bar{k}_t dt. \quad (58)$$

The transfer conditions and the date-zero inequalities are affine and hence are preserved by the mixture. Individual rationality is preserved as well, because truthful utility is linear in the mechanism.

Fix an initial type x and a nonzero direction (h, a) admissible against M^η . We retain the strictly positive component of (58) in the gain bounds used in the sufficiency proof of Theorem 1b, rather than discarding the entire compensator term.

In the OU model, recall

$$\Delta_t^{h,a} = he^{-\lambda t} + \gamma \int_0^t e^{-\lambda(t-s)} a_s ds.$$

For every finite T , Proposition 7, (51), and Condition 1(C6) give

$$\begin{aligned} & \limsup_{\varepsilon \downarrow 0} \frac{U_{M^\eta}(x + \varepsilon h, \varepsilon a) - U_{M^\eta}(x, 0)}{\varepsilon^2} \\ & \leq \frac{h^2}{2} \left(\frac{\sigma_0^\eta(x)}{\gamma} - (D^\eta)'(x) \right) - \frac{\eta}{2\gamma} \mathbb{E} \int_0^T e^{2\lambda t} (\Delta_t^{h,a})^2 \bar{k}_t dt. \end{aligned} \quad (59)$$

The first term on the right is nonpositive by (3). The integral is strictly positive for some finite T : indeed,

$$d\Delta_t^{h,a} = (-\lambda\Delta_t^{h,a} + \gamma a_t)dt, \quad \Delta_0^{h,a} = h,$$

so $\Delta^{h,a} = 0$ almost everywhere would imply $h = 0$ and $a = 0$ almost everywhere. Hence (59) is strictly negative.

For GBM, set

$$z_t^\varepsilon := -\log(1 - \phi_t^{(\varepsilon, h, a)}).$$

The type and report have the same diffusion coefficient, so their log ratio satisfies, pathwise,

$$z_t^\varepsilon = \log\left(\frac{x + \varepsilon h}{x}\right) + \gamma\varepsilon \int_0^t a_s ds.$$

It follows that

$$\frac{\Psi_t^{(\varepsilon, h, a)}}{\varepsilon^2} \longrightarrow \frac{1}{2} \left(\frac{h}{x} + \gamma \int_0^t a_s ds \right)^2 \quad \text{pointwise.}$$

Retaining the benchmark component of K^η in Proposition 8, using the date-zero expansion from the proof of Theorem 1b, and applying Fatou's lemma on $[0, T]$ yields

$$\begin{aligned} & \limsup_{\varepsilon \downarrow 0} \frac{U_{M^\eta}(x + \varepsilon h, \varepsilon a) - U_{M^\eta}(x, 0)}{\varepsilon^2} \\ & \leq \frac{h^2}{2} \left(\frac{A_0^\eta(x)}{\gamma x^2} - (D^\eta)'(x) \right) - \frac{\eta}{2\gamma} \mathbb{E} \int_0^T \left(\frac{h}{x} + \gamma \int_0^t a_s ds \right)^2 \bar{k}_t dt. \end{aligned} \quad (60)$$

Again the first term is nonpositive by (3), while the integral is strictly positive for some finite T unless $h = 0$ and $a = 0$ almost everywhere. Thus (60) is strictly negative. In both models, the terms suppressed in (59)–(60) are either nonpositive or $o(1)$; the latter are precisely the feedback terms controlled by Condition 1(C6).

This proves (56). Finally, the pointwise mixture of transfers gives the displayed revenue identity. Since every weakly locally incentive-compatible mechanism has strict approximants converging to it in revenue, and the strict class is contained in the weak class, the two suprema coincide. $\square$

The benchmark condition. Condition (57) requires the strict drift to be induced by an actual admissible mechanism. It is straightforward when allocations are unrestricted and the relevant moments are finite. Let $q_t, m_t > 0$ be deterministic and suppose the following allocations and their implementing transfers are admissible; the coefficients may be made to decay as rapidly as needed. In the OU model, set

$$\bar{X}_t = q_t \hat{\theta}_t + m_t \hat{\theta}_0,$$

and define

$$Q_O := \int_0^\infty q_t e^{-(r+2\lambda)t} dt, \quad M_O := \int_0^\infty m_t e^{-(r+\lambda)t} dt.$$

Direct calculation gives

$$\bar{D}'(x) = Q_O + M_O, \quad \bar{S}_t = \gamma \int_t^\infty q_s e^{-(r+2\lambda)s} ds.$$

Thus $\bar{k}_t = \gamma q_t e^{-(r+2\lambda)t} > 0$, and the OU date-zero inequality is strict: $\bar{\sigma}_0/\gamma = Q_O < \bar{D}'$.

In the GBM model, set

$$\bar{X}_t = -\frac{q_t}{\hat{\theta}_t} + m_t \hat{\theta}_0,$$

and define

$$Q_G := \int_0^\infty e^{-rt} q_t dt, \quad M_G := \int_0^\infty e^{-(r-\mu)t} m_t dt.$$

Then

$$\bar{D}(x) = -\frac{Q_G}{x} + M_G x, \quad \bar{A}_t = \gamma \int_t^\infty e^{-rs} q_s ds.$$

Hence $\bar{k}_t = \gamma e^{-rt} q_t > 0$, and

$$\frac{\bar{A}_0(x)}{\gamma x^2} = \frac{Q_G}{x^2} < \frac{Q_G}{x^2} + M_G = \bar{D}'(x).$$

Proposition 2 supplies implementing transfers. In both models, the envelope utility is bounded below, so its free constant can be chosen to ensure individual rationality. With a constrained allocation range, the proposition applies whenever the admissible class contains a benchmark satisfying (57). The benchmark hypothesis isolates this feasibility and admissibility requirement; it is not needed for the weak characterization in Theorems 1a–1b.

D Proof of Proposition 2

Fix the allocation X , and write

$$g_t := \text{vol}(\hat{W}_t), \quad g_t^\star := \begin{cases} \gamma D_t & (\text{OU}), \\ \gamma \hat{\theta}_t D_t & (\text{GBM}). \end{cases}$$

Necessity of (i). We use the gains-process calculation from Lemma 3 before the restriction $g = g^\star$ is imposed. Let a be bounded and supported in $[0, T]$, and use a superscript (ε, a) for evaluation along the report generated by the distortion εa . In the OU model, that calculation gives

$$\begin{aligned} U(\theta_0, \varepsilon a) - V(\theta_0) &= \varepsilon \mathbb{E} \int_0^T e^{-rt} [g_t^{(\varepsilon, a)} - \gamma D_t^{(\varepsilon, a)}] a_t dt \\ &\quad - \varepsilon^2 \mathbb{E} \int_0^T e^{\lambda t} \sigma_t^{(\varepsilon, a)} \tilde{\delta}_t^a a_t dt. \end{aligned} \quad (61)$$

For GBM, the corresponding identity is

$$\begin{aligned} U(\theta_0, \varepsilon a) - V(\theta_0) &= \varepsilon \mathbb{E} \int_0^T e^{-rt} [g_t^{(\varepsilon, a)} - \gamma \hat{\theta}_t^{(\varepsilon, a)} D_t^{(\varepsilon, a)}] a_t dt \\ &\quad - \varepsilon \mathbb{E} \int_0^T A_t^{(\varepsilon, a)} \phi_t^{(\varepsilon, a)} a_t dt. \end{aligned} \quad (62)$$

Here

$$\phi_t^{(\varepsilon, a)} = 1 - \exp\left(-\gamma \varepsilon \int_0^t a_s ds\right),$$

so $\phi^{(\varepsilon, a)} = O(\varepsilon)$ uniformly on $[0, T]$. After localization, (C3) and Lemma 4 imply that the second line of each display is $O(\varepsilon^2)$.

Set $c_t := e^{-rt}(g_t - g_t^\star)$ on the truthful path. If $c \neq 0$ on a set of positive $dt \otimes d\mathbb{P}$ -measure, there are $n \in \mathbb{N}$ and $T < \infty$ such that the predictable set

$$E := \{t \leq T : 1/n \leq |c_t| \leq n\}$$

has positive product measure. Regard $a_t := \text{sign}(c_t) \mathbf{1}_E$ as a predictable functional of the reported history. It is an admissible direction by (C1). Define

$$c_t^{(\varepsilon, a)} := e^{-rt} [g_t^{(\varepsilon, a)} - (g_t^\star)^{(\varepsilon, a)}],$$

so Lemma 4 gives

$$\mathbb{E} \int_0^T c_t^{(\varepsilon, a)} a_t dt \longrightarrow \mathbb{E} \int_E |c_t| dt =: \kappa > 0.$$

Equations (61)–(62) therefore imply

$$U(\theta_0, \varepsilon a) - V(\theta_0) = \varepsilon[\kappa + o(1)] + O(\varepsilon^2).$$

Dividing by ε^2 gives $[\kappa + o(1)]/\varepsilon + O(1) \rightarrow +\infty$, contradicting local incentive compatibility. Hence $g = g^*$, $dt \otimes d\mathbb{P}$ -a.e., as claimed.

Necessity of (ii). Fix a true initial type x and an initial report y , and set $\delta_0 := y - x$. If the agent makes no continuation distortion, the report path is the truthful path issued from y , while the report–type gap is

$$\delta_t = \begin{cases} \delta_0 e^{-\lambda t} & (\text{OU}), \\ \delta_0 \hat{\theta}_t / y & (\text{GBM}). \end{cases}$$

Thus linearity of the allocation payoff in type, together with the definition of $D(y)$, gives the exact identity

$$U(y, 0; x) = V(y) + (x - y)D(y). \quad (63)$$

Here $U(y, 0; x)$ makes the dependence on the true type explicit; it is the quantity denoted by $U(y, 0)$ elsewhere. Indeed, relative to a truthful type y , the allocation loss is

$$\delta_0 \begin{cases} \mathbb{E}_y \int_0^\infty e^{-(r+\lambda)t} X_t dt & (\text{OU}), \\ \frac{1}{y} \mathbb{E}_y \int_0^\infty e^{-rt} \hat{\theta}_t X_t dt & (\text{GBM}), \end{cases} = \delta_0 D(y),$$

where $\mathbb{E}_y$ denotes expectation under the report law starting from y .

Incentive compatibility at $t = 0$ and (63) imply, for every $x, y \in \Theta$,

$$V(x) \geq V(y) + (x - y)D(y),$$

with equality at $x = y$. Equivalently,

$$V(x) = \sup_{y \in \Theta} \{V(y) + (x - y)D(y)\}.$$

Hence V is convex and $D(y) \in \partial V(y)$. On every compact subinterval of Θ , V is absolutely continuous and $V' = D$ almost everywhere. Therefore

$$V(\theta_0) - V(c) = \int_c^{\theta_0} D(l) dl.$$

Finally, rearranging

$$V(\theta_0) = \hat{W}_0^X(\theta_0) - \mathbb{E} \left[\int_0^\infty e^{-rt} T_t dt \middle| \theta_0 \right]$$

gives the expected-payment formula in (ii).

Construction. Let

$$H_t := g_t^* - \text{vol}(\hat{W}_t^X), \quad \kappa_0 := \hat{W}_0^X(\hat{\theta}_0) - \int_c^{\hat{\theta}_0} D(l)dl - U(c, 0),$$

and define

$$M_t := - \int_0^t H_s d\hat{B}_s, \quad T_t := r(\kappa_0 + M_t).$$

By the admissibility qualifications in the proposition, M is a martingale on every finite horizon under truthful reporting and all conditional expectations below are well defined.

Because $\mathbb{E}[M_t \mid \theta_0] = 0$, conditional Fubini gives

$$\mathbb{E} \left[\int_0^\infty e^{-rt} T_t dt \mid \theta_0 \right] = \kappa_0.$$

Along the truthful path, $\hat{\theta}_0 = \theta_0$, and hence

$$U(\theta_0, 0) = \hat{W}_0^X(\theta_0) - \kappa_0 = U(c, 0) + \int_c^{\theta_0} D(l)dl.$$

This verifies (ii).

For $s \geq t$, the martingale property gives $\mathbb{E}_t[T_s] = r(\kappa_0 + M_t)$. Consequently,

$$\mathbb{E}_t \int_t^\infty e^{-r(s-t)} T_s ds = \kappa_0 + M_t,$$

and therefore

$$\hat{W}_t = \hat{W}_t^X - \kappa_0 - M_t = \hat{W}_t^X - \kappa_0 + \int_0^t H_s d\hat{B}_s.$$

It follows that

$$\text{vol}(\hat{W}_t) = \text{vol}(\hat{W}_t^X) + H_t = g_t^*,$$

which is (i). Since c and $U(c, 0)$ were arbitrary, the construction works for every admissible choice specified in the proposition. $\square$

E Proofs of Theorems 2a and 2b

We prove the OU statements. Write $\kappa := r + 2\lambda$ and $\Xi_t := (\hat{\theta}_t, D_t, \psi_t)$. Under truthful reporting,

$$d\hat{\theta}_t = \lambda(\bar{\theta} - \hat{\theta}_t)dt + \gamma d\hat{B}_t, \quad dD_t = [(r + \lambda)D_t - X_t]dt + \psi_t d\hat{B}_t.$$

The finite-variation constraint and the semimartingale conditions of Theorem 1a can be written in a common form:

$$d\psi_t = \kappa\psi_t dt + v_t d\hat{B}_t - dK_t, \quad \Delta\psi_t = -\Delta K_t \leq 0, \quad (64)$$

where K is nondecreasing, $K_0 = 0$, and $v = 0$ in the finite-variation class. Thus Ξ_0 is the actual initial state delivered by the mechanism; the continuation does not make a separate adjustment at date zero.

The delivery requirement in the theorem rules out solutions of the promise equation that contain a bubble. A convenient sufficient check is that

$$e^{-(r+\lambda)T} D_T \longrightarrow 0 \quad \text{in } L^1$$

and that $\int_0^T e^{-(r+\lambda)s} X_s ds$ converges in L^1 . Also suppose that

$$\int_0^t e^{-(r+\lambda)s} \psi_s d\hat{B}_s$$

is a true martingale on each finite horizon, as follows, for example, from finite-horizon square integrability of its integrand. The finite-horizon martingale identity for

$$e^{-(r+\lambda)t} D_t + \int_0^t e^{-(r+\lambda)s} X_s ds$$

then gives the displayed rent formula in the theorem. These are sufficient conditions for checking delivery, not additional restrictions when delivery is verified directly. To see the decomposition explicitly, write

$$Q_t := e^{-\kappa t} \psi_t = e^{-\lambda t} \sigma_t, \quad dQ_t = \eta_t d\hat{B}_t - dC_t$$

for the Doob–Meyer decomposition of the nonnegative supermartingale in Theorem 1a, normalized by $C_0 = 0$. Then $v_t = e^{\kappa t} \eta_t$ and $dK_t = e^{\kappa t} dC_t$, which gives (64). If the nonnegative supermartingale Q reaches zero at a stopping time τ , optional sampling gives, for every $t \geq 0$,

$$\mathbf{1}_{\{\tau \leq t\}} \mathbb{E}[Q_t \mid \mathcal{F}_\tau] \leq \mathbf{1}_{\{\tau \leq t\}} Q_\tau = 0.$$

Thus $Q_t = 0$ on $\{\tau \leq t\}$, first for rational t and then, by right-continuity, at all later times. Hence ψ remains zero after τ , and uniqueness of the semimartingale decomposition forces $v = 0$ and K to remain constant after τ . This is the state-constraint interpretation used at the boundary $\psi = 0$.

E.1 A common verification identity

For a smooth function $\Phi(\vartheta, d, p)$, define the payoff-inclusive controlled operator

$$\begin{aligned} \mathcal{H}^{x,v}\Phi &:= x(\vartheta - \Phi_D) + \lambda(\bar{\theta} - \vartheta)\Phi_\vartheta + (r + \lambda)d\Phi_D + \kappa p\Phi_\psi \\ &\quad + \frac{\gamma^2}{2}\Phi_{\vartheta\vartheta} + \gamma p\Phi_{\vartheta D} + \frac{p^2}{2}\Phi_{DD} + \frac{v^2}{2}\Phi_{\psi\psi} + v(\gamma\Phi_{\vartheta\psi} + p\Phi_{D\psi}). \end{aligned} \quad (65)$$

When $v = 0$, the last line's final two terms are absent and $\sup_x \mathcal{H}^{x,0}\Phi = \mathcal{L}\Phi$. In the local program,

$$\sup_{x,v} \mathcal{H}^{x,v}\Phi = \mathcal{L}\Phi + \sup_v \left\{ \frac{v^2}{2}\Phi_{\psi\psi} + v(\gamma\Phi_{\vartheta\psi} + p\Phi_{D\psi}) \right\}.$$

Let

$$R_t^\Phi := r\Phi(\Xi_{t-}) - \mathcal{H}^{X_t, v_t}\Phi(\Xi_{t-})$$

be the Bellman residual along an admissible policy, and let

$$J_t^\Phi := \Phi(\hat{\theta}_t, D_t, \psi_{t-}) - \Phi(\hat{\theta}_t, D_t, \psi_t)$$

denote the value loss at a jump of ψ . Since $\hat{\theta}$ and D are continuous, these are the only jump terms. Generalized Itô's formula, applied after the usual localization, gives

$$\begin{aligned} \Phi(\Xi_0) &= \int_0^T e^{-rt} \hat{\theta}_t X_t dt + e^{-rT} \Phi(\Xi_T) + \int_0^T e^{-rt} R_t^\Phi dt \\ &\quad + \int_0^T e^{-rt} \Phi_\psi(\Xi_{t-}) dK_t^c + \sum_{0 < t \leq T} e^{-rt} J_t^\Phi - M_T^\Phi, \end{aligned} \quad (66)$$

where K^c is the continuous part of K and

$$M_T^\Phi := \int_0^T e^{-rt} (\gamma\Phi_\vartheta + \psi_{t-}\Phi_D + v_t\Phi_\psi)(\Xi_{t-}) d\hat{B}_t. \quad (67)$$

For Theorem 2a, $v = 0$, so neither the ψ -second derivatives in (65) nor the last term in the integrand of (67) is needed. Condition (T1) makes the stopped stochastic integrals true martingales on every finite horizon. Admissibility and the same condition justify removing the localization. Notice that the jump contribution in (66) is the exact value difference, not a first-order approximation.

If ψ reaches zero, apply Itô's formula up to its hitting time and then to the C^2 boundary trace $\Phi^0(\vartheta, d) := \Phi(\vartheta, d, 0)$. The boundary is absorbing, and the two identities paste together by continuity of Φ ; a jump to zero is already included through J^Φ . Thus the boundary residual is precisely $r\Phi^0 - \mathcal{L}_0\Phi^0$, as specified by the state-constraint condition.

E.2 Proof of Theorem 2a

The variational inequality (11) implies

$$rV - \mathcal{L}V \geq 0, \quad V_\psi \geq 0.$$

Since $\mathcal{H}^{X_t, 0}V \leq \mathcal{L}V$,

$$R_t^V = (rV - \mathcal{L}V)(\Xi_{t-}) + (\mathcal{L}V - \mathcal{H}^{X_t, 0}V)(\Xi_{t-}) \geq 0.$$

Moreover, K^c is nondecreasing, and every jump lowers ψ . Monotonicity of V in ψ therefore gives

$$V_\psi(\Xi_{t-})dK_t^c \geq 0, \quad J_t^V \geq 0.$$

Taking expectations in (66) yields

$$\mathbb{E} \int_0^T e^{-rt} \hat{\theta}_t X_t dt + e^{-rT} \mathbb{E}[V(\Xi_T)] \leq V(\Xi_0).$$

Because the zero distortion is admissible against the mechanism, (C2) gives absolute convergence of the payoff integral. Letting $T \rightarrow \infty$ and using the $\liminf$ in (T1) proves

$$\mathbb{E} \int_0^\infty e^{-rt} \hat{\theta}_t X_t dt \leq V(\Xi_0).$$

Thus V dominates every admissible finite-variation continuation.

Now consider the controls in the attainment clause of the theorem. Condition (b) says that the allocation attains the supremum in $\mathcal{L}$ and that $rV - \mathcal{L}V = 0$ almost everywhere along the resulting path. Hence $R_t^V = 0$ almost everywhere. Condition (c) is exactly

$$\int_0^\infty e^{-rt} V_\psi(\Xi_{t-}) dK_t^c = 0, \quad V(\hat{\theta}_t, D_t, \psi_t) = V(\hat{\theta}_t, D_t, \psi_{t-}) \quad \text{at every jump.} \quad (68)$$

All nonnegative defects in (66) consequently vanish. Condition (T2) then gives

$$V(\Xi_0) = \mathbb{E} \int_0^\infty e^{-rt} \hat{\theta}_t X_t dt.$$

Together with the preceding domination inequality, this proves both attainment and optimality. One familiar way to obtain (68) is to let ψ grow at its capped rate below a free boundary and use K to keep it from crossing that

boundary. Then K is constant below the boundary, $V_\psi = 0$ where continuous adjustment occurs, and any downward jump stays within an interval on which V is constant. The theorem states these equality conditions directly because the variational inequality alone does not impose a particular free-boundary geometry.

It remains to verify the date-zero conclusion. For any feasible choice, use σ_0 itself when attainment is direct and otherwise take the measurable $\tilde{\sigma}_0$ specified in the theorem. In the latter case, replacing σ_0 by $\tilde{\sigma}_0$ leaves V , and hence the objective, unchanged; because $\tilde{\sigma}_0 \leq \sigma_0$, it weakly relaxes (4). Starting from this actual initial exposure, use the attaining continuations, chosen measurably across initial types as assumed in the theorem, and the transfers from Proposition 2. σ —equivalently, the exposure ψ —is of finite variation, so Theorem 1a gives global continuation incentive compatibility. The constraint (4) and Theorem 1b give incentive compatibility of the initial report, while (7) gives participation. Its revenue is precisely the objective in (12), by (6). Hence every such choice is implementable. Taking the supremum shows that the program's value is a lower bound on the unrestricted principal's value; if a maximizer exists, the resulting mechanism attains its value. $\square$

E.3 Proof of Theorem 2b

For $z = (\vartheta, d, p)$, set

$$Q_F(z, v) := \frac{v^2}{2} F_{\psi\psi}(z) + v(\gamma F_{\vartheta\psi}(z) + p F_{D\psi}(z)).$$

On the interior $\{p > 0\}$, the local variational inequality implies

$$\rho(z) := r\bar{V}(z) - \mathcal{L}\bar{V}(z) - \sup_v Q_{\bar{V}}(z, v) \geq 0, \quad \bar{V}_\psi(z) \geq 0.$$

Writing

$$\begin{aligned} \Delta_t^X &:= \sup_x x(\hat{\theta}_t - \bar{V}_D(\Xi_{t-})) - X_t(\hat{\theta}_t - \bar{V}_D(\Xi_{t-})), \\ \Delta_t^v &:= \sup_w Q_{\bar{V}}(\Xi_{t-}, w) - Q_{\bar{V}}(\Xi_{t-}, v_t), \end{aligned}$$

we have

$$R_t^{\bar{V}} = \rho(\Xi_{t-}) + \Delta_t^X + \Delta_t^v \geq 0 \quad \text{on } \{\psi_{t-} > 0\}.$$

On $\{\psi_{t-} = 0\}$, admissibility requires $v_t = 0$, and the state-constraint condition gives

$$R_t^{\bar{V}} = (r\bar{V} - \mathcal{L}_0\bar{V})(\Xi_{t-}) + (\mathcal{L}_0\bar{V} - \mathcal{H}^{X_t, 0}\bar{V})(\Xi_{t-}) \geq 0.$$

Thus the residual is nonnegative everywhere. The continuous-adjustment and jump terms in (66) are also nonnegative, exactly as in the finite-variation proof. Condition (T1) therefore gives

$$\mathbb{E} \int_0^\infty e^{-rt} \hat{\theta}_t X_t dt \leq \bar{V}(\Xi_0)$$

for every admissible locally incentive-compatible continuation.

We pause to address the potentially vanishing curvature in the supremum over v in (13). Write

$$q := \bar{V}_{\psi\psi}, \quad c := \gamma \bar{V}_{\hat{\theta}\psi} + \psi \bar{V}_{D\psi}.$$

Pointwise,

$$\sup_{v \in \mathbb{R}} \left\{ \frac{1}{2} q v^2 + c v \right\} = \begin{cases} -c^2/(2q) & q < 0, \\ 0 & q = 0, \ c = 0, \\ +\infty & q > 0 \text{ or } (q = 0, \ c \neq 0). \end{cases} \quad (69)$$

Because a classical solution of (13) is finite, the last case cannot occur. Thus $v^* = -c/q$ only on $\{q < 0\}$. On the degenerate set $\{q = c = 0\}$, every v attains the pointwise supremum; one must choose a measurable value that produces an admissible state process rather than divide by q . At every point of $\{\bar{V}_\psi = 0\}$ with $\psi > 0$, this degeneracy is automatic: the nonnegative C^1 function $\bar{V}_\psi$ attains its minimum there, so

$$\bar{V}_{\psi\psi} = \bar{V}_{\hat{\theta}\psi} = \bar{V}_{D\psi} = 0.$$

The possible disappearance of curvature therefore creates no missing term in the verification inequality. It does make admissibility of the chosen v a genuine condition: the ratio $-c/q$ need not remain bounded as $q \uparrow 0$. At the absorbing boundary $\psi = 0$, the only admissible choice is $v = 0$. No relaxed control is needed.

Under the attainment hypotheses, both pointwise controls attain their respective suprema. Moreover, $\rho(\Xi_t) = 0$ at interior states almost everywhere, the selected boundary residual vanishes, and the analogues of (68) hold for $\bar{V}$. Every defect in (66) is then zero. Condition (T2) gives

$$\bar{V}(\Xi_0) = \mathbb{E} \int_0^\infty e^{-rt} \hat{\theta}_t X_t dt,$$

so the selected policy attains the local continuation bound. The allocation X must likewise attain the supremum in (10). When its range is noncompact,

that supremum need not be attained; when $\bar{V}_D = \hat{\theta}$, the HJB may not identify X . This is why the theorem directly assumes that the proposed controls produce an admissible continuation, rather than inferring one from a formal first-order formula.

Finally, fix an incentive-compatible, individually rational mechanism. Its continuation surplus is bounded by $\bar{V}$. Moreover, incentive compatibility implies that D is nondecreasing by Remark 1, so D is differentiable Lebesgue-almost everywhere and hence F -almost everywhere. At each differentiability point, the necessity argument in the proof of Theorem 1b gives $\sigma_0 \leq \gamma D'$, while Theorem 1a gives $\sigma_0 \geq 0$. Thus the mechanism's date-zero data are feasible for the relaxed program (14). Substitution in (6) shows that its revenue cannot exceed that program's value. This proves the upper bound on the unrestricted principal's value.

Conversely, take a maximizer satisfying the all-type conditions in the theorem, combine it with the attaining continuations chosen measurably from its selected initial states, and use the transfers of Proposition 2. Theorems 1a–1b give local incentive compatibility after and at date zero, respectively, and (7) gives participation. The resulting mechanism attains the local-program value. Any admissible, locally incentive-compatible, individually rational comparator whose date-zero rent is nondecreasing and differentiable on Θ , whose initial exposure satisfies the two inequalities in the theorem at every type, and whose utilities satisfy the envelope condition of Proposition 2(ii), is feasible for the relaxation and is subject to the continuation bound. No such comparator can yield more. $\square$

F Proof of Proposition 6

When $T \equiv 0$, the agent's continuation value is $\hat{W} = \hat{W}^X$. In the gains-process calculation of Lemma 3, the transfer terms cancel between current utility and the continuation-value dynamics. Their only remaining role in the proof of Theorem 1a is through the volatility of $\hat{W}$. For a first-order contract, that volatility is precisely the one specified in Proposition 2(i). The master deviation formula and all subsequent necessity and verification arguments in the proof of Theorem 1a therefore apply unchanged.

At date zero, the additional condition in the proposition is exactly the envelope representation of Proposition 2(ii), now with $T \equiv 0$. It supplies the identity used in (50); every subsequent step in the proof of Theorem 1b is independent of transfers. Subject to the remaining hypotheses of that theorem, its proof therefore applies unchanged as well. $\square$

G The Geometric Case of Proposition 3

We complete the proof of Proposition 3 for the geometric Brownian motion, in parallel with the Ornstein–Uhlenbeck case of Section 5. Two features distinguish it: the innovation loading is state-dependent, and the impulse response is stochastic.

Under the multiplicative Euler scheme $\theta_{t+1} = \theta_t + \mu\theta_t\Delta t + \gamma\theta_t\varepsilon_{t+1}$, the per-step innovation is

$$\theta_{t+1} - \theta_t - \mu\theta_t\Delta t = \gamma\theta_t\varepsilon_{t+1} = \eta_t\varepsilon_{t+1}, \quad \eta_t := \gamma\theta_t,$$

so the volatility coefficient $\eta_t = \gamma\theta_t$ is state-dependent, in contrast to the constant $\eta_t = \gamma$ of the mean-reverting case. The solution $\theta_t = \theta_0 \exp((\mu - \frac{1}{2}\gamma^2)t + \gamma B_t)$ is proportional to its initial condition, so the impulse response $I_{t,0} = \partial\theta_t/\partial\theta_0 = \theta_t/\theta_0$ is stochastic and remains inside the rent integral. With bilinear utility $u_\theta(\theta, X) = X$, the discounted marginal rent is

$$Z = \frac{1}{\theta_0} \sum_{t=0}^{\infty} \beta^t \theta_t X_t \Delta t, \quad \beta = e^{-r\Delta t},$$

and, for a linear contract (Definition 2), the representation $Z = D(\theta_0) + \sum_t \sigma_t \varepsilon_{t+1}$ reads

$$Z = D(\theta_0) + \sum_{t=0}^{\infty} \frac{\sigma_t}{\eta_t} (\theta_{t+1} - \theta_t - \mu\theta_t\Delta t), \quad \frac{1}{\eta_t} (\theta_{t+1} - \theta_t - \mu\theta_t\Delta t) = \varepsilon_{t+1}.$$

Let $D_t(\hat{\theta}_t, \theta_t)$ be the current-value continuation rent under a one-shot misreport $\hat{\theta}_t$, normalized by the current type as in Section 4. Consider the one-shot misreport $\hat{\theta}_t$ and write $\hat{\theta}_t = \theta_t + (\hat{\theta}_t - \theta_t)$ in the two innovation terms that flank period t . The term carrying σ_{t-1} contributes

$$\hat{\theta}_t - \theta_{t-1} - \mu\theta_{t-1}\Delta t = [\theta_t - \theta_{t-1} - \mu\theta_{t-1}\Delta t] + (\hat{\theta}_t - \theta_t),$$

while the term carrying σ_t , whose loading $\eta_t = \gamma\hat{\theta}_t$ is frozen at the reported value, contributes

$$\theta_{t+1} - \hat{\theta}_t - \mu\hat{\theta}_t\Delta t = [\theta_{t+1} - \theta_t - \mu\theta_t\Delta t] - (\hat{\theta}_t - \theta_t)(1 + \mu\Delta t).$$

The perturbation thus enters with coefficient $(1 + \mu\Delta t)$ —the geometric analogue of the mean-reverting factor $(1 - \lambda\Delta t)$. Because the continuation is normalized by the current type, the stochastic impulse response leaves a residual flow term $u_\theta(\hat{\theta}_t, X_t) \Delta t = X_t \Delta t$ that the Ornstein–Uhlenbeck case does not

produce; carried to present value it appears as $e^{-rt}X_t \Delta t$. Proceeding as in Section 5, cancelling the common martingale and history terms, and dividing through by the positive impulse-response weight θ_t/θ_0 ,

$$\beta^t [D_t(l, l) - D_t(\hat{\theta}_t, l)] = \frac{1}{l} \left(Y_{t-1} - Y_t(1 + \mu \Delta t) - e^{-rt} X_t \Delta t \right) (l - \hat{\theta}_t), \quad Y_t := \frac{\theta_0 \sigma_t}{\gamma \theta_t},$$

the rescaled loading of Section 4 (on the truthful path, $\hat{\theta} = \theta$), with l in the role of the true current type. Write $C_t := Y_{t-1} - Y_t(1 + \mu \Delta t) - e^{-rt} X_t \Delta t$ for the coefficient, which does not depend on l .

Integral monotonicity requires $\int_{\hat{\theta}_t}^{\theta_t} [D_t(l, l) - D_t(\hat{\theta}_t, l)] dl \geq 0$ for all $\theta_t, \hat{\theta}_t > 0$. Since the type space is positive under GBM, $1/l > 0$ throughout, and

$$\int_{\hat{\theta}_t}^{\theta_t} \frac{C_t}{l} (l - \hat{\theta}_t) dl = C_t \left[(\theta_t - \hat{\theta}_t) - \hat{\theta}_t \ln \frac{\theta_t}{\hat{\theta}_t} \right],$$

whose bracket is nonnegative for all $\theta_t, \hat{\theta}_t > 0$. Hence the integral is nonnegative for every pair if and only if $C_t \geq 0$, that is,

$$Y_{t-1} \geq Y_t(1 + \mu \Delta t) + e^{-rt} X_t \Delta t \quad \text{for all } t \geq 1.$$

Equivalently $-(Y_t - Y_{t-1}) - \mu Y_t \Delta t - e^{-rt} X_t \Delta t \geq 0$; letting $\Delta t \rightarrow 0$,

$$dY_t + \mu Y_t dt \leq -e^{-rt} X_t dt, \quad Y_t = \frac{\theta_0 \sigma_t}{\gamma \theta_t},$$

the geometric bound of Theorem 1a. As in the Ornstein–Uhlenbeck case, the pointwise inequality $Y_{t-1} \geq Y_t(1 + \mu \Delta t) + e^{-rt} X_t \Delta t$ forbids upward jumps in Y —a positive increment not shrinking with the step violates it—so the limit delivers both the drift bound (ii) and the no-upward-jumps condition (iii) of Theorem 1a. This completes the proof of Proposition 3.

H Proofs for the Application

Throughout this appendix the normalization of the application (Section 6) is in force: OU dynamics with $\bar{\theta} = 0$ and $\gamma = 1$, flow utility $\theta X - \frac{1}{2} X^2$, F symmetric with mean zero, f continuous and strictly positive on the interior of Θ , $\mathbb{E}[\theta_0^2] < \infty$, and F regular in the sense of Section 6.3. Abbreviate $\rho := r + 2\lambda$ and let

$$v_+(\theta) := \theta - \frac{1 - F(\theta)}{f(\theta)}, \quad x^* := \sup\{\theta : v_+(\theta) \leq 0\},$$

so that v_+ is nondecreasing, continuous, $v_+(x^*) = 0$, and $v_+ > 0$ strictly above x^* ; by symmetry $v_-(\theta) := \theta + F(\theta)/f(\theta) = -v_+(-\theta)$ is nondecreasing with zero at $-x^*$, and $x^* > 0$ since $v_+(0) = -1/(2f(0)) < 0$. Write $v(\theta) := \theta - \bar{\Lambda}(\theta) = \rho D^{\text{FOA}}(\theta)$ for the ironed virtual value of (20): $v = v_-$ below $-x^*$, $v = 0$ on $[-x^*, x^*]$, and $v = v_+$ above x^* . Since $F(\theta)/f(\theta) \geq 0$ and $(1 - F(\theta))/f(\theta) \geq 0$, the sign structure gives, for every interior θ ,

$$|v(\theta)| \leq |\theta|, \quad |\bar{\Lambda}(\theta)| \leq |\theta|, \quad (70)$$

so the candidate quantities $v(\theta_0)$ and $\bar{\Lambda}(\theta_0)$ have finite second moments. We first reduce the first-order relaxation to a date-zero program in the rent schedule. We then solve that program using a participation multiplier and construct the allocation that attains the resulting bound.

H.1 Reduction of the first-order relaxation

The relaxed (first-order) program of Section 6.3 is

$$\Pi^{\text{FOA}} = \sup_{X, u} \mathbb{E} \left[\int_0^\infty e^{-rt} (\theta_t X_t - \tfrac{1}{2} X_t^2) dt \right] - \mathbb{E}[U(\theta_0)] \quad (71)$$

over admissible allocations and $u \in \mathbb{R}$, subject to: $D(\theta_0) = \mathbb{E}[Z \mid \theta_0]$ admits a nondecreasing version, and, with $U(\theta_0) := u + \int_{x^*}^{\theta_0} D(l) dl$, individual rationality $U \geq 0$ holds on Θ . (The base type x^* is a normalization of u .) By Lemma 1 and Proposition 2(ii), applied to the effective transfer $\tilde{T} = T + X^2/2$, every admissible incentive-compatible, individually rational mechanism induces a feasible point of (71) with the same revenue, so Π^{FOA} bounds the principal's value.

For $\lambda > 0$, $\mathbb{E}[\theta_s^2 \mid \theta_0] = \theta_0^2 e^{-2\lambda s} + (1 - e^{-2\lambda s})/(2\lambda)$; when $\lambda = 0$, the last term is interpreted as s . Hence the first-best surplus is

$$\tfrac{1}{2} \mathbb{E} \left[\int_0^\infty e^{-rs} \theta_s^2 ds \mid \theta_0 \right] = \frac{\theta_0^2}{2\rho} + \frac{1}{2r\rho} \quad (72)$$

where the formula at $\lambda = 0$ follows by continuity. The tracking identity eliminates the remaining dynamic choice.

Lemma 7 (Cost of a promise). *For every initial type and admissible mechanism for which the conditional tracking cost is finite,*

$$\tfrac{1}{2} \mathbb{E} \left[\int_0^\infty e^{-rs} \zeta_s^2 ds \mid \theta_0 \right] \geq \frac{\rho}{2} \tilde{D}_0^2,$$

with equality if and only if $\zeta_s = \rho \tilde{D}_0 e^{-\lambda s}$ for $dt \otimes d\mathbb{P}$ -almost every (s, ω) .

Proof. Drop the two nonnegative terms in (19). Equality requires both to vanish: $\nu \equiv 0$ and $\zeta_s = \rho \tilde{D}_s$ a.e. The promise equation (18) then reads $d\tilde{D}_t = -\lambda \tilde{D}_t dt$, so $\tilde{D}_s = \tilde{D}_0 e^{-\lambda s}$ and $\zeta_s = \rho \tilde{D}_0 e^{-\lambda s}$. Conversely, if the tracking error has this path, the definition of the promise gives $\tilde{D}_s = \tilde{D}_0 e^{-\lambda s}$; its volatility is therefore zero, and both nonnegative terms in (19) vanish. $\square$

Using $\theta X - \frac{1}{2}X^2 = \frac{1}{2}\theta^2 - \frac{1}{2}\zeta^2$, (72), and Lemma 7, the objective of (71) is at most

$$\frac{\mathbb{E}[\theta_0^2]}{2\rho} + \frac{1}{2r\rho} + \mathcal{S}(D, u), \quad \mathcal{S}(D, u) := -\frac{\rho}{2} \mathbb{E}\left[\left(D(\theta_0) - \frac{\theta_0}{\rho}\right)^2\right] - \mathbb{E}[U(\theta_0)], \quad (73)$$

with equality at any feasible (D, u) delivered by an admissible allocation attaining Lemma 7. Maximizing $\mathcal{S}$ over nondecreasing D and u with $U \geq 0$ is the *static program*; without loss $\mathbb{E}[U] < \infty$, since otherwise the objective is $-\infty$.

H.2 Proof of Proposition 4

Step 1: the multiplier. Define $M : \Theta \rightarrow [0, 1]$ by

$$M(\theta) := \begin{cases} 0, & \theta < -x^*, \\ F(\theta) + \theta f(\theta), & -x^* \leq \theta < x^*, \\ 1, & \theta \geq x^*. \end{cases}$$

Lemma 8. *M is the distribution function of a Borel probability measure on Θ , continuous and supported on $[-x^*, x^*]$, and on the interior of Θ*

$$\frac{M(\theta) - F(\theta)}{f(\theta)} = \bar{\Lambda}(\theta), \quad \text{equivalently} \quad \theta - \frac{M(\theta) - F(\theta)}{f(\theta)} = v(\theta). \quad (74)$$

Proof. Write $G(\theta) := F(\theta) + \theta f(\theta)$ on $[-x^*, x^*]$ and note the two identities $G = f v_-$ and $G = 1 + f v_+$.

Endpoint values. Symmetry gives $G(-x^*) = F(-x^*) - x^* f(-x^*) = (1 - F(x^*)) - x^* f(x^*) = -f(x^*) v_+(x^*) = 0$ and, directly from the definition of x^* , $G(x^*) = F(x^*) + x^* f(x^*) = 1 + f(x^*) v_+(x^*) = 1$; continuity of f makes M continuous.

Monotonicity. On $[-x^*, x^*]$, $v_- \geq 0$ and $v_+ \leq 0$. Take $\theta < \theta'$ in this interval. If $f(\theta') \geq f(\theta)$, then $G(\theta') = f(\theta') v_-(\theta') \geq f(\theta) v_-(\theta) = G(\theta)$, because both factors at θ' are weakly larger. If $f(\theta') < f(\theta)$, then, since $v_+(\theta') \leq 0$,

$$G(\theta') = 1 + f(\theta') v_+(\theta') \geq 1 + f(\theta) v_+(\theta') \geq 1 + f(\theta) v_+(\theta) = G(\theta).$$

Identity. Below $-x^*$, $(M - F)/f = -F/f = \Lambda = \bar{\Lambda}$; on the flat, $(M - F)/f = \theta = \bar{\Lambda}$; above x^* , $(M - F)/f = (1 - F)/f = \Lambda = \bar{\Lambda}$. $\square$

Step 2: pricing participation. For any (D, u) feasible in the static program with $\mathbb{E}[U] < \infty$:

Lemma 9. $\int_{\Theta} |D(l)| |F(l) - M(l)| dl < \infty$ and

$$-\mathbb{E}[U(\theta_0)] + \int_{\Theta} U dM = \int_{\Theta} D(l)(F(l) - M(l)) dl = -\mathbb{E}[\bar{\Lambda}(\theta_0) D(\theta_0)]. \quad (75)$$

Proof. Integrability. Let $H(l) := \mathbf{1}\{l \geq x^*\}$. Since D is nondecreasing, D^+ is nondecreasing and D^- is nonincreasing. For $\theta_0 > x^*$,

$$\int_{x^*}^{\theta_0} |D| = \int_{x^*}^{\theta_0} D + 2 \int_{x^*}^{\theta_0} D^- \leq (U(\theta_0) - u) + 2D^-(x^*)(\theta_0 - x^*),$$

and the analogous bound for $\theta_0 < x^*$ uses D^+ . Taking expectations, using $\mathbb{E}|\theta_0| < \infty$ and $\mathbb{E}[U] < \infty$, and applying Tonelli gives

$$\int_{\Theta} |D(l)| |H(l) - F(l)| dl = \mathbb{E} \int_{\theta_0 \wedge x^*}^{\theta_0 \vee x^*} |D(l)| dl < \infty.$$

Moreover, $H - M$ vanishes outside the compact interval $[-x^*, x^*]$, on which the monotone function D is bounded. Thus $\int |D| |H - M| < \infty$, and hence also $\int |D| |F - M| < \infty$.

Identity. The preceding bounds justify Fubini under both distributions:

$$\begin{aligned} \mathbb{E}[U(\theta_0)] &= u + \int_{\Theta} D(l)(H(l) - F(l)) dl, \\ \int_{\Theta} U dM &= u + \int_{\Theta} D(l)(H(l) - M(l)) dl. \end{aligned}$$

Subtracting cancels u and yields the first equality in (75). The second follows from $M - F = f\bar{\Lambda}$, by (74). $\square$

Step 3: weak duality. For any feasible (D, u) ,

$$\begin{aligned} \mathcal{S}(D, u) &\leq \mathcal{S}(D, u) + \int_{\Theta} U dM && [U \geq 0, M \geq 0] \\ &= -\frac{\rho}{2} \mathbb{E} \left[\left(D - \frac{\theta_0}{\rho} \right)^2 \right] - \mathbb{E}[\bar{\Lambda}(\theta_0) D] && [\text{Lemma 9}] \\ &= -\frac{\rho}{2} \mathbb{E} \left[\left(D - \frac{v(\theta_0)}{\rho} \right)^2 \right] + \frac{\mathbb{E}[v(\theta_0)^2 - \theta_0^2]}{2\rho} \\ &\leq \frac{\mathbb{E}[v(\theta_0)^2 - \theta_0^2]}{2\rho}, \end{aligned} \quad (76)$$

where the third line completes the square at $v = \theta_0 - \bar{\Lambda}(\theta_0)$, using (74). All terms are well defined: $\mathbb{E}[v^2] \leq \mathbb{E}[\theta_0^2] < \infty$ by (70), and Lemma 9 makes the cross term integrable. The last line is the unconstrained pointwise bound. Its maximizer $D = v/\rho = D^{\text{FOA}}$ is nondecreasing—continuous, equal to $v_-/\rho \leq 0$ below $-x^*$, zero on the flat, $v_+/\rho \geq 0$ above x^* —so monotonicity requires no further ironing.

Step 4: the candidate attains the chain. Take $D = D^{\text{FOA}}$ and $u = 0$, so $U = U^{\text{FOA}}(\theta_0) = \int_{x^*}^{|\theta_0|} D^{\text{FOA}}(l) dl$ by symmetry of D^{FOA} (odd) about zero. Then $U^{\text{FOA}} \geq 0$, with equality precisely on $[-x^*, x^*]$ (the integrand is strictly positive above x^* by the definition of x^*), so individual rationality holds. Moreover, (70) gives

$$0 \leq U^{\text{FOA}}(\theta_0) \leq \frac{\theta_0^2}{2\rho},$$

so $\mathbb{E}[U^{\text{FOA}}] < \infty$. Since $\text{supp } M \subseteq [-x^*, x^*]$, $\int U^{\text{FOA}} dM = 0$: both inequalities in (76) bind. Hence

$$\sup \mathcal{S} = \mathcal{S}(D^{\text{FOA}}, 0) = \frac{\mathbb{E}[v(\theta_0)^2] - \mathbb{E}[\theta_0^2]}{2\rho}.$$

Step 5: the allocation delivers the bound. For $X_t = \theta_t - \bar{\Lambda}(\theta_0)e^{-\lambda t}$ as in (21), stochastic Fubini on $\theta_t = \theta_0 e^{-\lambda t} + \int_0^t e^{-\lambda(t-s)} dB_s$ gives

$$Z = \frac{\theta_0 - \bar{\Lambda}(\theta_0)}{\rho} + \int_0^\infty \frac{e^{-(r+\lambda)s}}{\rho} dB_s, \quad (77)$$

so $Z \in L^2$ by (70) and $\mathbb{E}[Z \mid \theta_0] = D^{\text{FOA}}(\theta_0)$. Its martingale loading is the deterministic smooth process $\sigma_s = e^{-(r+\lambda)s}/\rho$, so the required square-integrability and regularity are immediate. In current-value units, $\psi \equiv 1/\rho$: exposure is first best, as claimed. The tracking error is $\zeta_s = -\bar{\Lambda}(\theta_0)e^{-\lambda s} = \rho \tilde{D}_0 e^{-\lambda s}$, since $\tilde{D}_0 = D^{\text{FOA}} - \theta_0/\rho = -\bar{\Lambda}/\rho$, so Lemma 7 holds with equality and (73) is an equality at the candidate. Combining with Steps 3–4,

$$\Pi^{\text{FOA}} = \frac{\mathbb{E}[\theta_0^2]}{2\rho} + \frac{1}{2r\rho} + \frac{\mathbb{E}[v^2] - \mathbb{E}[\theta_0^2]}{2\rho} = \frac{1}{2\rho} \left(\frac{1}{r} + \mathbb{E}[v(\theta_0)^2] \right),$$

which is (22).

Step 6: uniqueness. $\mathcal{S}$ is strictly concave in $D \in L^2(F)$ and the feasible set is convex, so the maximizing D is unique F -a.e. Once $D = D^{\text{FOA}}$, the value U equals the constant u on the support of M . Hence $U \geq 0$ and $\int U dM = 0$ force $u = 0$. Finally, equality in Lemma 7 pins $\zeta_s = \rho \tilde{D}_0 e^{-\lambda s}$, hence X , up to $dt \otimes d\mathbb{P}$ -null sets. $\square$

H.3 The cap implied by the first-order rent: uniform case

Lemma 10. *Let F be uniform on $[-1, 1]$, so $x^* = \frac{1}{2}$ and $v(\theta) = \text{sgn}(\theta)(2|\theta| - 1)^+$. The cap (15) implied by $D^{\text{FOA}} = v/\rho$ is*

$$\sigma_0^{\max}(\hat{\theta}_0) = \begin{cases} 0, & |\hat{\theta}_0| \leq \frac{1}{2}, \\ \frac{2 - 1/|\hat{\theta}_0|}{\rho}, & \frac{1}{2} < |\hat{\theta}_0| \leq 1. \end{cases}$$

For $|\hat{\theta}_0| > \frac{1}{2}$ the infimum defining the cap is attained at the mirror type $m = -\hat{\theta}_0$, and only there; on the flat it is attained by every distinct m in the flat.

Proof. The antiderivative of v is $A(\theta) := [(|\theta| - \frac{1}{2})^+]^2$, so in units of $1/\rho$ the quotient in the definition of $\sigma_0^{\max}$ is

$$Q(m) := \frac{2[A(m) - A(\hat{\theta}_0) - v(\hat{\theta}_0)(m - \hat{\theta}_0)]}{(m - \hat{\theta}_0)^2},$$

and by symmetry we may take $\hat{\theta}_0 \geq 0$.

Reports on the flat. If $\hat{\theta}_0 \leq \frac{1}{2}$, then $A(\hat{\theta}_0) = v(\hat{\theta}_0) = 0$, and $Q(m) = 2A(m)/(m - \hat{\theta}_0)^2 \geq 0$. It vanishes for every $m \in [-\frac{1}{2}, \frac{1}{2}] \setminus \{\hat{\theta}_0\}$, so the cap is zero.

Reports above the flat. Write $\hat{\theta}_0 = \frac{1}{2} + s$, where $s \in (0, \frac{1}{2}]$, and consider three regions for m . For a same-side lie $m \in [\frac{1}{2}, 1] \setminus \{\hat{\theta}_0\}$, A is quadratic with $A'' = 2$, so $Q(m) = 2$. For $m \in [-\frac{1}{2}, \frac{1}{2})$, let $d := \hat{\theta}_0 - m \in (s, 1 + s]$. Then

$$Q = \frac{2s(2d - s)}{d^2}, \quad \frac{\partial Q}{\partial d} = \frac{4s(s - d)}{d^3} \leq 0,$$

so Q decreases as the lie moves toward the opposite outer region. Finally, for $m = -\frac{1}{2} - w$, $w \in [0, \frac{1}{2}]$,

$$Q = \frac{2[(w + s)^2 + 2s]}{(1 + s + w)^2},$$

whose derivative in w has the sign of $w - s$: the minimum is at $w = s$, i.e. at $m = -\hat{\theta}_0$, where $Q = 4s/(1 + 2s)$. Since $4s/(1 + 2s) \leq 2s(2 + s)/(1 + s)^2$ (cross-multiplying, this reduces to $0 \leq s$) and $4s/(1 + 2s) < 2$, the overall minimum is $4s/(1 + 2s) = 2 - 1/\hat{\theta}_0$, attained at $m = -\hat{\theta}_0$ only. $\square$

H.4 The static benchmark

To isolate the value of responding to reports after date zero, call a mechanism *static* if the date-zero report fixes its entire allocation path and later reports are ignored: $X_t = x_t(\hat{\theta}_0)$. Incentive compatibility then reduces to static screening in the discounted allocation. The following result supplies the static row of Table 1.

Proposition 12 (The static benchmark). *Under the assumptions of Proposition 4, the optimal static mechanism is the conditional expectation of the first-order allocation,*

$$X_t = (\theta_0 - \bar{\Lambda}(\theta_0))e^{-\lambda t} = \mathbb{E}[X_t^{\text{FOA}} \mid \theta_0],$$

with the same rent schedule (20), and its revenue falls short of the first-order value by exactly the shock surplus:

$$\Pi_{\text{stat}} = \frac{\mathbb{E}[(\theta_0 - \bar{\Lambda}(\theta_0))^2]}{2(r + 2\lambda)} = \Pi^{\text{FOA}} - \frac{1}{2r(r + 2\lambda)}.$$

Proof. Step 1: static incentive compatibility. Suppose an initial report m fixes the allocation path $x(m)$, and define

$$Z(m) := \int_0^\infty e^{-(r+\lambda)t} x_t(m) dt, \quad C(m) := \frac{1}{2} \int_0^\infty e^{-rt} x_t(m)^2 dt.$$

Because $\mathbb{E}[\theta_t \mid \theta_0] = \theta_0 e^{-\lambda t}$, the trader's expected gross utility from this path is $\theta_0 Z(m) - C(m)$. Lemma 1 therefore implies that every incentive-compatible static mechanism has Z nondecreasing and $U' = Z$ a.e. Its revenue is

$$\mathbb{E}[\theta_0 Z(\theta_0) - C(\theta_0) - U(\theta_0)].$$

Conversely, let Z be nondecreasing, choose its representative so that $Z(m) \in \partial U(m)$, and let U be the associated convex value with $U' = Z$ a.e. Define the present value of the payment by

$$p(m) := mZ(m) - C(m) - U(m)$$

and charge the constant transfer flow $rp(m)$; ignore all later reports. A type θ_0 reporting m then obtains

$$U(m) + (\theta_0 - m)Z(m) \leq U(\theta_0),$$

where the inequality is the subgradient inequality for U . Thus the static screening conditions are also sufficient.

Step 2: the least-cost path. Fix Z . Cauchy–Schwarz gives

$$Z^2 \leq \left(\int_0^\infty e^{-rt} x_t^2 dt \right) \left(\int_0^\infty e^{-(r+2\lambda)t} dt \right) = \frac{1}{\rho} \int_0^\infty e^{-rt} x_t^2 dt.$$

Equality holds exactly when $x_t = \rho Z e^{-\lambda t}$ a.e. Hence this exponential path minimizes C for a given promise Z . Writing $D := Z$, its surplus is

$$\theta_0 D - \frac{\rho}{2} D^2 = \frac{\theta_0^2}{2\rho} - \frac{\rho}{2} \left(D - \frac{\theta_0}{\rho} \right)^2,$$

so the static problem is $\mathbb{E}[\theta_0^2]/(2\rho) + \sup \mathcal{S}(D, u)$: the reduction (73) with the first-best surplus of the initial position, $\mathbb{E}[\theta_0^2]/(2\rho)$, in place of the full first-best term (72).

Step 3: solution and value. Steps 1–4 and 6 in the proof of Proposition 4 now apply without change. The unique optimal rent is D^{FOA} , and its least-cost path is $x_t = \rho D^{\text{FOA}}(\theta_0) e^{-\lambda t} = (\theta_0 - \bar{\Lambda}(\theta_0)) e^{-\lambda t} = \mathbb{E}[X_t^{\text{FOA}} \mid \theta_0]$, and

$$\Pi_{\text{stat}} = \frac{\mathbb{E}[\theta_0^2]}{2\rho} + \frac{\mathbb{E}[v^2] - \mathbb{E}[\theta_0^2]}{2\rho} = \frac{\mathbb{E}[(\theta_0 - \bar{\Lambda}(\theta_0))^2]}{2\rho} = \Pi^{\text{FOA}} - \frac{1}{2r\rho}.$$

□

The last equality decomposes the first-order value into the static revenue and the full surplus from future shocks, $1/(2r\rho)$. Together with Table 1, it shows that the tight finite-variation/local bracket remains tight even though responding to later reports creates substantial value.

H.5 Proof of Proposition 5

Recall that $\rho = r + 2\lambda$.

Lemma 11 (Deterministic exposure is optimal). *Fix $0 \leq \psi_0 \leq 1/\rho$. Among the exposure processes allowed by Theorem 1a with initial value ψ_0 ,*

$$\mathbb{E} \int_0^\infty e^{-rs} \left(\psi_s - \frac{1}{\rho} \right)^2 ds \geq P(\psi_0).$$

Equality is attained by $\psi_s = \min\{e^{\rho s} \psi_0, 1/\rho\}$, whose corresponding rent volatility is deterministic and of finite variation.

Proof. Theorem 1a implies that $e^{-\rho s}\psi_s$ is a nonnegative supermartingale. It follows that

$$0 \leq \mathbb{E}[\psi_s] \leq e^{\rho s}\psi_0.$$

Jensen's inequality therefore gives

$$\mathbb{E} \left[\left(\psi_s - \frac{1}{\rho} \right)^2 \right] \geq \left(\mathbb{E}[\psi_s] - \frac{1}{\rho} \right)^2 \geq \left[\left(\frac{1}{\rho} - e^{\rho s}\psi_0 \right)^+ \right]^2.$$

Integrating yields the bound. The displayed deterministic path attains both inequalities at every s . $\square$

Step 1: verification of both HJBs. Write $\alpha := r/\rho \in (0, 1]$. For $0 < \psi < 1/\rho$, the substitution $u = e^{\rho s}\psi$ in the definition of P gives

$$P(\psi) = \frac{\psi^\alpha}{\rho} \int_\psi^{1/\rho} \left(\frac{1}{\rho} - u \right)^2 u^{-\alpha-1} du. \quad (78)$$

Differentiating (78), $P'(\psi) = (\alpha/\psi)P(\psi) - (1/\rho - \psi)^2/(\rho\psi)$. Thus, with $\nu = \psi - 1/\rho$,

$$rP(\psi) = \nu^2 + \rho\psi P'(\psi), \quad 0 < \psi < \frac{1}{\rho}; \quad (79)$$

while differentiation of the original integral gives $P'(\psi) = -2 \int_0^{s^*(\psi)} e^{(\rho-r)s} \left(\frac{1}{\rho} - e^{\rho s}\psi \right) ds < 0$ on $(0, 1/\rho)$, where $s^*(\psi) = \rho^{-1} \ln(1/(\rho\psi))$, and $P'(\psi) \rightarrow 0$ as $\psi \uparrow 1/\rho$. Differentiating once more, with no boundary term because the integrand vanishes at $s^*(\psi)$, gives

$$P''(\psi) = 2 \int_0^{s^*(\psi)} e^{(2\rho-r)s} ds > 0,$$

and $P''(\psi) \rightarrow 0$ as $\psi \uparrow 1/\rho$. Extending P by zero on $[1/\rho, \infty)$ therefore makes it C^2 on $(0, \infty)$, nonincreasing, and convex. Hence $J = -\frac{\rho}{2}\tilde{D}^2 - \frac{\rho}{2}P(\psi)$ is nondecreasing in ψ , as required.

It remains to check the first branch of (23); its term $-\zeta^2/2$ is the application's quadratic holding cost after writing $x = \hat{\theta} + \zeta$. For the candidate, $J_{\tilde{D}} = -\rho\tilde{D}$, $J_{\tilde{D}\tilde{D}} = -\rho$, $J_\psi = -\frac{\rho}{2}P' \geq 0$. The maximizer in (23) is $\zeta^* = -J_{\tilde{D}} = \rho\tilde{D}$. Substitution cancels the terms in $\tilde{D}^2$ and leaves the residual

$$\frac{\rho}{2} \left[\nu^2 - rP(\psi) + \rho\psi P'(\psi) \right],$$

which is zero on $\{0 < \psi < 1/\rho\}$ by (79) and equals $\frac{\rho}{2}\nu^2 \geq 0$ on $\{\psi \geq 1/\rho\}$, where $J_\psi = 0$. Thus (23) holds, with free boundary $\{\psi = 1/\rho\}$. For the local HJB, $J_{\tilde{D}\psi} = 0$ and $J_{\psi\psi} = -(\rho/2)P'' \leq 0$, so

$$\sup_{v \in \mathbb{R}} \left[\frac{1}{2}v^2 J_{\psi\psi} + v\nu J_{\tilde{D}\psi} \right] = 0,$$

attained at $v = 0$. Hence (24) holds as well. At the absorbing boundary $\psi = 0$, $P(0^+) = 1/(r\rho^2)$ and $\nu(0)^2 = 1/\rho^2$, so the state-constraint inequality holds with equality when $v = 0$.

Step 2: a common upper bound. A continuation with payoff $-\infty$ satisfies the desired bound trivially. Fix, therefore, a continuation state $(\hat{\theta}, D, \psi_0)$ and any finite-payoff admissible continuation in either class. Its rent volatility satisfies the conditions of Theorem 1a. Since first-best surplus is finite, square completion implies that its tracking cost is finite. By a time shift, apply the tracking identity (19) and Lemma 11 from that state. Continuation surplus equals first best less half the tracking cost, so, dropping the nonnegative amortization term,

$$\mathbb{E} \left[\int_0^\infty e^{-rs} \hat{\theta}_s X_s ds - \frac{1}{2} \int_0^\infty e^{-rs} X_s^2 ds \right] \leq \frac{\hat{\theta}^2}{2\rho} + \frac{1}{2r\rho} - \frac{\rho}{2} \tilde{D}^2 - \frac{\rho}{2} \mathbb{E} \int_0^\infty e^{-rs} \nu_s^2 ds.$$

If $\psi_0 \leq 1/\rho$, Lemma 11 bounds the last expectation below by $P(\psi_0)$, giving the bound $V(\hat{\theta}, D, \psi_0)$; if $\psi_0 > 1/\rho$, the last term is simply nonnegative and $P(\psi_0) = 0$, giving the bound again.

Step 3: attainment in both classes for $\psi_0 \leq 1/\rho$. Define the deterministic exposure $\psi_t = \min\{e^{\rho t}\psi_0, 1/\rho\}$ with $K_t = \int_0^t \rho\psi_s \mathbf{1}\{\psi_s = 1/\rho\} ds$, so that (9) holds with $v = 0$, K continuous and nondecreasing, increasing only where $J_\psi = 0$; let $\tilde{D}$ solve $d\tilde{D}_t = -\lambda\tilde{D}_t dt + \nu_t d\tilde{B}_t$ from $\tilde{D}_0 = D - \hat{\theta}/\rho$, and set $X_t = \hat{\theta}_t + \rho\tilde{D}_t$, $D_t = \tilde{D}_t + \hat{\theta}_t/\rho$. Then $dD_t = [(r + \lambda)D_t - X_t]dt + \psi_t d\tilde{B}_t$ by direct substitution, so $M_t := e^{-(r+\lambda)t}D_t + \int_0^t e^{-(r+\lambda)s}X_s ds$ is a true martingale (ψ bounded), and since the explicit Gaussian solution gives $e^{-(r+\lambda)T}D_T \rightarrow 0$ in L^1 , D_t is indeed the continuation rent of X and $\sigma_t = e^{-(r+\lambda)t}\psi_t$ its volatility: the continuation delivers (D, ψ_0) , lies in the finite-variation class, and satisfies the conditions of Theorem 1a ($\sigma \geq 0$ deterministic with $\text{drift}(\sigma) \leq \lambda\sigma$). To verify admissibility, note that ψ and ν are bounded and deterministic, while $\hat{\theta}$ and $\tilde{D}$ solve linear Gaussian equations whose second moments grow at most linearly. Exponential discounting therefore gives $Z \in L^2$ and the required terminal limits. Moreover, σ is deterministic and of finite variation, so its diffusion coefficient—and hence the feedback terms in Condition 1—vanish. The construction in Proposition 2 then supplies implementing transfers. Finally,

both inequalities in Step 2 bind: $\zeta_t = \rho \tilde{D}_t$ eliminates the amortization square, and the displayed deterministic exposure path attains Lemma 11. Thus continuation surplus equals $V(\hat{\theta}, D, \psi_0)$. This continuation is feasible in both classes, so the bound in Step 2 is the common value.

Step 4: states above first-best exposure. For $\psi_0 > 1/\rho$, the common bound in Step 2 is $V(\hat{\theta}, D, \psi_0) = V(\hat{\theta}, D, 1/\rho)$ but cannot be attained. Indeed, set $\delta := (\psi_0 - 1/\rho)/2 > 0$ and let

$$\tau := \inf\{t > 0 : |\psi_t - \psi_0| \geq \delta\} \wedge 1.$$

Right continuity gives $\tau > 0$ a.s., and $|\nu_t| \geq \delta$ on $[0, \tau)$. Hence $\int_0^\infty e^{-rt} \nu_t^2 dt > 0$ a.s., so the exposure-cost term dropped in Step 2 is strictly positive.

The bound is nevertheless approachable. For $\varepsilon > 0$, set $\psi_t = e^{\rho t} \psi_0$ on $[0, \varepsilon)$, jump down to $1/\rho$ at ε , and keep ψ there thereafter; use $\zeta_t = \rho \tilde{D}_t$ throughout. This finite-variation path is admissible in both classes. Its extra cost relative to the boundary policy is

$$\frac{\rho}{2} \int_0^\varepsilon e^{-rt} \left(e^{\rho t} \psi_0 - \frac{1}{\rho} \right)^2 dt \longrightarrow 0.$$

Thus the displayed value is the immediate-intervention supremum. Finally, replacing a date-zero selection $\sigma_0 > 1/\rho$ by $1/\rho$ leaves $V(\theta_0, D, \cdot)$ unchanged and weakly relaxes both (15) and the local constraint $\sigma_0 \leq D'$. $\square$

H.6 Proof of Corollary 2.1

Setup and design problems. Trader i 's initial type and Brownian shock are (θ_0^i, B^i) , independent across i , with $\theta_0^i \sim F$ symmetric about zero. Each trader observes only his own type path. The handicap Λ is the function in (17) anchored at zero, with $\Lambda(0) = 0$. Fix a mechanism-independent set $\mathfrak{R}$ of reporting strategies, common to all traders and mechanisms. A strategy consists of an initial report and an own-filtration-adapted drift distortion satisfying Condition 1(C1). The set contains truth, includes every initial report paired with a bounded compactly supported drift distortion, and is invariant under simultaneous sign reflection of the type, Brownian shock, and report. It is also closed under the small scaled and joint perturbations considered in Condition 1(C5)–(C6). A mechanism assigns adapted allocations X^i and transfer flows T^i to every profile in $\mathfrak{R}^N$. Under every such profile, allocations are discounted square-integrable and transfers have finite discounted expected absolute value. Trader i 's interim mechanism is obtained by averaging over truthfully reporting opponents, as in Step 1 below. A mechanism is admitted

only if every rule in $\mathfrak{R}$ satisfies Condition 1(C1)–(C4) against each interim mechanism, which must also satisfy (C5)–(C6). The finite-variation and local classes impose the additional restrictions below.

When trader i uses $r^i \in \mathfrak{R}$ and all other traders report truthfully, his interim payoff at initial type x is

$$J^i(r^i \mid x) := \mathbb{E} \left[\int_0^\infty e^{-rt} \left(\theta_t^i X_t^i(r^i, \theta^{-i}) - \frac{1}{2} X_t^i(r^i, \theta^{-i})^2 - T_t^i(r^i, \theta^{-i}) \right) dt \mid \theta_0^i = x \right].$$

All three exchange problems require

$$\begin{aligned} J^i(\text{truth} \mid x) &\geq 0 \quad \text{for } F\text{-almost every } x, \\ \sum_{i=1}^N X_t^i(r) &= 0 \quad \text{for every } r \in \mathfrak{R}^N \text{ and } t \geq 0, \quad \text{a.s.} \end{aligned}$$

The three classes are:

- (G) in the unrestricted admissible class, $J^i(r^i \mid x) \leq J^i(\text{truth} \mid x)$ for every $r^i \in \mathfrak{R}$ and almost every x ;
- (FV) in the regular finite-variation class, each trader's interim rent volatility has finite variation, satisfies the exposure restrictions of Theorem 1a, and satisfies the date-zero inequality (15), together with the transfer and admissibility conditions of Theorem 2a;
- (L) in the regular local class, each trader's interim mechanism belongs to the locally incentive-compatible comparison class of Theorem 2b, including its envelope, monotonicity, date-zero, and admissibility requirements.

Theorems 1a–1b make every mechanism in (FV) globally incentive compatible. For each of the three classes, the exchange maximizes

$$\sup_{(X,T)} \sum_{i=1}^N \mathbb{E} \int_0^\infty e^{-rt} T_t^i dt \tag{80}$$

subject to the corresponding restrictions above. Thus (G) is the unrestricted admissible problem, (FV) is the finite-variation problem with the date-zero integral inequality, and (L) is the regular local problem covered by Theorem 2b.

On the truthful path, square integrability also follows directly from individual rationality and transfer integrability. Indeed,

$$\theta^i X^i - \frac{1}{2}(X^i)^2 = \frac{1}{2}(\theta^i)^2 - \frac{1}{2}(X^i - \theta^i)^2,$$

so, after integrating individual rationality over initial types,

$$\frac{1}{2}\|X^i - \theta^i\|^2 \leq \frac{1}{2}\|\theta^i\|^2 + \mathbb{E} \int_0^\infty e^{-rt} |T_t^i| dt < \infty.$$

Thus X^i is discounted square-integrable. Here $\mathcal{H}$ is the Hilbert space of progressively measurable processes generated by all traders' private histories, with

$$\|Y\|^2 := \mathbb{E} \int_0^\infty e^{-rt} Y_t^2 dt < \infty.$$

Let $\mathcal{H}_i$ be the closed subspace of processes adapted to trader i 's filtration, and let

$$h_t^i := \theta_t^i - \Lambda(\theta_0^i) e^{-\lambda t}$$

be trader i 's virtual target; $\|h^i\| < \infty$ by the hypothesis of the corollary. Fix one of (G), (FV), or (L), and let $\mathcal{V} \subset \mathcal{H}_i$ consist of the single-trader allocations that can be paired with a transfer to satisfy the corresponding restrictions, including individual rationality and the reportwise integrability requirements. We suppress the class label below.

Step 1: averaging a feasible exchange. Fix a feasible exchange mechanism and trader i . For each report path $\hat{\theta}^i$, average over the other traders' independent, truthfully reported private histories and define

$$Y_t^i(\hat{\theta}^i) := \mathbb{E}_{-i} \left[X_t^i(\hat{\theta}^i, \theta^{-i}) \right], \quad \tilde{T}_t^{i,\text{red}}(\hat{\theta}^i) := \mathbb{E}_{-i} \left[T_t^i(\hat{\theta}^i, \theta^{-i}) + \frac{1}{2} X_t^i(\hat{\theta}^i, \theta^{-i})^2 \right].$$

Pair Y^i with the raw transfer

$$T^{i,\text{red}} := \tilde{T}^{i,\text{red}} - \frac{1}{2}(Y^i)^2.$$

For almost every fixed initial type θ_0^i and every reporting strategy of trader i , with all expectations below conditional on θ_0^i ,

$$\begin{aligned} & \mathbb{E} \int_0^\infty e^{-rt} \left[\theta_t^i X_t^i - \frac{1}{2}(X_t^i)^2 - T_t^i \right] dt \\ &= \mathbb{E} \int_0^\infty e^{-rt} \left[\theta_t^i Y_t^i - \tilde{T}_t^{i,\text{red}} \right] dt = \mathbb{E} \int_0^\infty e^{-rt} \left[\theta_t^i Y_t^i - \frac{1}{2}(Y_t^i)^2 - T_t^{i,\text{red}} \right] dt. \end{aligned}$$

Thus averaging preserves the payoff from every report. The reduced single-trader mechanism therefore preserves global incentive compatibility in (G) and (FV), local incentive compatibility in (L), and individual rationality in all three classes. Its truthful value, rent, and rent volatility are exactly the corresponding interim objects of the exchange mechanism. In particular, the finite-variation and date-zero restrictions in (FV), and the envelope, monotonicity, and date-zero restrictions in (L), are preserved. Under truth,

$$Y_t^i = \mathbb{E}[X_t^i \mid \mathcal{F}_t^i] =: (\mathcal{P}_i X^i)_t.$$

Conditional Jensen's inequality gives $Y^i \in \mathcal{H}_i$, and the reduced transfer is integrable because $X^i \in \mathcal{H}$ and T^i is integrable. The same argument applies under every report. Hence

$$\mathcal{P}_i X^i \in \mathcal{V}.$$

This is a consequence of the original mechanism's class-specific incentive and participation restrictions, not a separate restriction on the exchange.

Because $Y^i = \mathcal{P}_i X^i$, the difference $X^i - Y^i$ is orthogonal to every process in $\mathcal{H}_i$. In particular,

$$\begin{aligned} \|X^i - h^i\|^2 &= \|X^i - Y^i\|^2 + \|Y^i - h^i\|^2, \\ \|X^i - X^{i*}\|^2 &= \|X^i - Y^i\|^2 + \|Y^i - X^{i*}\|^2. \end{aligned} \tag{81}$$

Step 2: the single-trader geometry. Write $\tilde{T} = T + X^2/2$. If $X, X' \in \mathcal{V}$, adding their effective transfers implements $X + X'$; multiplying an effective transfer by $c \geq 0$ implements cX . Because the same reports are available in each mechanism, every global or local incentive comparison is correspondingly added or multiplied by c . Individual rationality is preserved as well. The truthful values, rents, and rent volatilities add or scale in the same way. Hence finite variation and the dynamic exposure restrictions are preserved; (15) is additive and homogeneous; and the envelope, monotonicity, and date-zero restrictions of the local class are preserved. In raw-transfer coordinates, the transfers are

$$T^\oplus = T + T' - XX', \quad T^c = cT + \frac{1}{2}(c - c^2)X^2.$$

Under every report, the relevant allocations are square-integrable, so these transfers are integrable by Cauchy–Schwarz. When $c = 0$, the zero mechanism is immediate. Linearity of the interim value, rent, volatility, and their semi-martingale coefficients, together with the triangle inequality, also preserves Condition 1(C3)–(C6); (C1) is unchanged and (C2) follows from the report-wise integrability just established. Thus $\mathcal{V}$ is a convex cone.

By assumption, X^{i*} attains the distance minimum. Convexity makes it unique. Differentiating $\|X^{i*} + \varepsilon(Y - X^{i*}) - h^i\|^2$ at $\varepsilon = 0+$, and then using $Y = 0$ and $Y = 2X^{i*}$, gives

$$\langle h^i - X^{i*}, Y - X^{i*} \rangle \leq 0 \quad \forall Y \in \mathcal{V}, \quad \langle h^i - X^{i*}, X^{i*} \rangle = 0. \quad (82)$$

Step 3: symmetry and single-trader optimality. Let R reflect trader i 's initial type and Brownian shock, $(\theta_0^i, B^i) \mapsto (-\theta_0^i, -B^i)$, a measure-preserving map under which the type path flips sign, and for an allocation Y define $(\mathcal{R}Y)_t(\omega) := -Y_t(R\omega)$. Then $\mathcal{R}$ is a linear isometry. The same sign change maps every permitted reporting strategy to another permitted strategy, and the reflected mechanism gives the same payoffs as the original. The finite-variation, date-zero, and local restrictions are also invariant under this reflection. Thus $\mathcal{R}$ maps $\mathcal{V}$ to itself. It also fixes the target: $-h_t^i(R\omega) = \theta_t^i(\omega) - \Lambda(\theta_0^i(\omega))e^{-\lambda t} = h_t^i(\omega)$, using that Λ is odd. Hence $\mathcal{R}X^{i*}$ is also a distance minimizer, and uniqueness gives $\mathcal{R}X^{i*} = X^{i*}$. Consequently $\mathbb{E}[X_t^{i*}] = 0$ for dt-a.e. t , the rent D^{i*} is odd with $D^{i*}(0) = 0$, the truthful value U^{i*} is even and convex, minimized at zero, and the transfers may set $U^{i*}(0) = 0$, so that $\mathbb{E}[U^{i*}] = \mathbb{E}[\Lambda(\theta_0)D^{i*}(\theta_0)]$ exactly. This also identifies the distance minimizer with the optimal single-trader mechanism. In classes (G) and (FV), Lemma 1 gives $U' = D_Y$ almost everywhere and makes D_Y nondecreasing. In class (L), these are part of the comparison class in Theorem 2b. Thus, for every $Y \in \mathcal{V}$, U is convex and is pinned down up to a constant. The best transfers set its minimum equal to zero and give

$$\mathbb{E}[U] = \mathbb{E}[\Lambda D_Y] - \inf_{\theta \in \Theta} \int_0^\theta D_Y(l) dl.$$

The exchange of expectation and integration in $\mathbb{E} \int_0^{\theta_0} D_Y = \mathbb{E}[\Lambda D_Y]$ is justified in Step 4 below. Single-trader revenue is

$$\Pi(Y) = \frac{1}{2}\|h^i\|^2 - \frac{1}{2}\|Y - h^i\|^2 + \inf_{\theta \in \Theta} \int_0^\theta D_Y(l) dl.$$

The last term is nonpositive. It is zero for X^{i*} , whose rent is odd. Hence $\Pi(X^{i*}) \geq \Pi(Y)$ for every $Y \in \mathcal{V}$. Equality requires Y to be another distance minimizer, so X^{i*} is the unique optimum.

Step 4: an upper bound for every feasible exchange. For any feasible exchange mechanism, Step 1 reduces each trader's problem to a single-trader mechanism in the selected class. In (G) and (FV), Lemma 1 implies that U^i is convex and $(U^i)' = D^i$ almost everywhere; in (L), the same conclusions follow

from the imposed envelope condition and monotonicity of D^i . Convexity and individual rationality for F -almost every type imply $U^i(0) \geq 0$. Hence

$$\mathbb{E}[U^i] \geq \mathbb{E}\left[\int_0^{\theta_0^i} D^i(l) dl\right] = \mathbb{E}[\Lambda(\theta_0^i) D^i(\theta_0^i)],$$

where exchanging expectation and integration is valid because D^i is nondecreasing and $\int_0^{\theta_0^i} |D^i| \leq |\theta_0|(|D^i(0)| + |D^i(\theta_0)|)$, which is integrable by Cauchy–Schwarz ($D^i = \mathbb{E}[Z^i | \theta_0^i] \in L^2$). Using $\mathbb{E}[\Lambda D^i] = \mathbb{E} \int_0^\infty e^{-rt} \Lambda(\theta_0^i) e^{-\lambda t} X_t^i dt$ and $\theta X - \frac{1}{2}X^2$, total revenue satisfies

$$\Pi \leq \sum_{i=1}^N \left[\frac{1}{2} \|h^i\|^2 - \frac{1}{2} \|X^i - h^i\|^2 \right], \quad (83)$$

with equality if and only if $\mathbb{E}[U^i] = \mathbb{E}[\Lambda D^i]$ for every i .

Step 5: the quadratic bound. Apply (82) to the allocation $Y^i = \mathcal{P}_i X^i \in \mathcal{V}$ obtained in Step 1. Together with (81), this gives

$$\begin{aligned} \|X^i - h^i\|^2 &\geq \|X^i - Y^i\|^2 + \|X^{i*} - h^i\|^2 + \|Y^i - X^{i*}\|^2 \\ &= \|X^{i*} - h^i\|^2 + \|X^i - X^{i*}\|^2. \end{aligned}$$

Because $\sum_i X^i = 0$, mean subtraction minimizes $\sum_i \|X^i - X^{i*}\|^2$ over all clearing profiles. Hence, with $\bar{X}^* := \frac{1}{N} \sum_j X^{j*}$,

$$\sum_i \|X^i - h^i\|^2 \geq \sum_i \|X^{i*} - h^i\|^2 + N \|\bar{X}^*\|^2.$$

Step 6: attainment and implementation. For $\hat{X}^i = X^{i*} - \bar{X}^*$,

$$\sum_i \|\hat{X}^i - h^i\|^2 = \sum_i \|X^{i*} - h^i\|^2 + N \|\bar{X}^*\|^2 - 2 \sum_i \langle \bar{X}^*, X^{i*} - h^i \rangle,$$

and the final sum is zero. To see this, substitute $\bar{X}^* = N^{-1} \sum_j X^{j*}$. The $j = i$ term is zero by the second identity in (82). For $j \neq i$, independence and $\mathbb{E}[X_t^{j*}] = 0$ (Step 3) give

$$\langle X^{j*}, h^i - X^{i*} \rangle = \int_0^\infty e^{-rt} \mathbb{E}[X_t^{j*}] \mathbb{E}[h_t^i - X_t^{i*}] dt = 0.$$

It remains to verify incentives. Choose an effective transfer $\tilde{T}^{i*}$ implementing X^{i*} , normalized so that $U^{i*}(0) = 0$, and set

$$c := 1 - \frac{1}{N}, \quad q^i := \frac{1}{N} \sum_{j \neq i} X^{j*}, \quad \hat{X}^i = cX^{i*} - q^i.$$

Give trader i the raw transfer

$$T^{i,\text{ex}} := c\tilde{T}^{i*} - \frac{1}{2}(\hat{X}^i)^2.$$

The effective transfer $\tilde{T}^{i*}$ is integrable because its raw transfer is integrable and $X^{i*} \in \mathcal{H}_i$; moreover, $\hat{X}^i \in \mathcal{H}$. Hence $T^{i,\text{ex}}$ is integrable, and the same argument applies under every report. Its effective transfer is $c\tilde{T}^{i*}$. If $X^{i*,a}$ and $\tilde{T}^{i*,a}$ are the allocation and effective transfer induced by a reporting strategy a , then, conditional on almost every fixed initial type θ_0^i , the trader's expected payoff in the exchange is

$$\mathbb{E} \int_0^\infty e^{-rt} \left[c(\theta_t^i X_t^{i*,a} - \tilde{T}_t^{i*,a}) - \theta_t^i q_t^i \right] dt = cU^{i*}(a).$$

The equality uses the independence of q^i from trader i 's private history and $\mathbb{E}[q_t^i] = 0$, and it holds for every reporting strategy. It preserves global incentive compatibility in (G) and (FV), and local incentive compatibility in (L). Individual rationality holds, and the interim objects satisfy

$$U^i = cU^{i*} \geq 0, \quad D^i = cD^{i*}, \quad \sigma^i = c\sigma^{i*}.$$

The same scaling applies to the semimartingale coefficients in Condition 1, so (C3)–(C6) are inherited from the single-trader mechanism; (C1) is unchanged and (C2) follows from the reportwise integrability above. Thus the candidate remains in the selected class: finite variation, the exposure restrictions, and (15) are preserved in (FV), while the envelope, monotonicity, and date-zero restrictions are preserved in (L). It follows that $\mathbb{E}[U^i] = \mathbb{E}[\Lambda D^i]$, so (83) holds with equality at the candidate. Market clearing is immediate. Combining the upper bound in Step 4 with Step 5 and the equality just established shows that no feasible exchange mechanism earns more. Thus $\{\hat{X}^i\}$, paired with the displayed transfers, is optimal in the selected class. Since the class was arbitrary, this proves the unrestricted, finite-variation, and regular local versions of the corollary. $\square$

H.7 The optima of the two programs: uniform case

This subsection proves the structure described in Section 6.5 and plotted in Figure 3, for the uniform distribution on $[-1, 1]$ at every (r, λ) . Abbreviate $\rho := r + 2\lambda$: the first-best exposure is $1/\rho$, the handicap is $\Lambda(\theta_0) = \text{sgn}(\theta_0)(1 - |\theta_0|)$, the ironing point is $x^* = \frac{1}{2}$, and P is the exposure penalty of Proposition 5, whose closed form is recorded in Appendix H.8 and whose

value at $\sigma = 0$ is defined by continuity. Four properties of P , immediate from its defining integral, are used below: on $(0, 1/\rho)$, P is strictly decreasing with $P'(0^+) = -\infty$, and strictly convex with

$$P''(\sigma) = 2 \int_0^{T(\sigma)} e^{(2\rho-r)s} ds > 0, \quad e^{\rho T(\sigma)} := \frac{1}{\rho\sigma};$$

and at first best $P(1/\rho) = P'(1/\rho) = P''(1/\rho) = 0$ with $P'''(1/\rho^-) = -2$ (differentiate P'' once more, using $e^{\rho T} = 1/(\rho\sigma)$), the cubic vanishing (26). At $r = 1$, $\lambda = 0$ these specialize to $P(\sigma) = 1 - \sigma^2 + 2\sigma \ln \sigma$ and $P'(\sigma) = 2(1 - \sigma + \ln \sigma)$. By the reduction of Section H the date-zero stage of either program maximizes

$$\mathcal{J}(D, \sigma_0) = \mathbb{E} \left[-\frac{\rho}{2} \left(D(\theta_0) - \frac{\theta_0}{\rho} \right)^2 - \frac{\rho}{2} P(\sigma_0(\theta_0)) - \Lambda(\theta_0) D(\theta_0) \right] \quad (84)$$

over nondecreasing D with $D(0) = 0$ and $\sigma_0 \in [0, 1/\rho]$, subject to the family (15) (finite-variation program) or to $\sigma_0 \leq D'$ alone (local program). The common first-best term $\mathbb{E}[\theta_0^2]/(2\rho) + 1/(2r\rho)$ is omitted from $\mathcal{J}$ and restored when reporting total revenue.

The first step extends Lemma 10 from the first-order rent to every rent schedule that is *star-shaped*: odd, nondecreasing, with the chord $\theta \mapsto D(\theta)/\theta$ nondecreasing on $(0, 1]$.

Lemma 12 (Chord certification). *Let D be odd, nondecreasing, locally absolutely continuous, with $D(0) = 0$, and star-shaped on $[-1, 1]$. Define*

$$c(\theta) := \begin{cases} D(\theta)/\theta, & \theta \neq 0, \\ \lim_{x \downarrow 0} D(x)/x, & \theta = 0, \end{cases}$$

where the limit exists by star-shapedness. Then, for every report $\hat{\theta}_0 \in [-1, 1]$ and every $m \neq \hat{\theta}_0$,

$$\frac{2\mathcal{B}(m, \hat{\theta}_0)}{(m - \hat{\theta}_0)^2} \geq c(\hat{\theta}_0) = c(|\hat{\theta}_0|).$$

Thus $\sigma_0(\theta) = c(\theta)$ satisfies the entire family (15), and $\sigma_0^{\max}(\hat{\theta}_0) = c(\hat{\theta}_0)$. If $\hat{\theta}_0 \neq 0$, equality holds precisely when c is constant between $|m|$ and $|\hat{\theta}_0|$; in particular, the mirror lie $m = -\hat{\theta}_0$ always binds. At $\hat{\theta}_0 = 0$, the cap is obtained as $m \rightarrow 0$, and a nonzero m attains it precisely when c is constant on $[0, |m|]$.

Proof. Take $\hat{\theta}_0 > 0$ (the case $\hat{\theta}_0 < 0$ is symmetric), write $c^* := c(\hat{\theta}_0)$, and set

$$\Phi(m) := 2\mathcal{B}(m, \hat{\theta}_0) - c^*(m - \hat{\theta}_0)^2.$$

Since $\mathcal{B}(m, \hat{\theta}_0) = \int_{\hat{\theta}_0}^m [D(l) - D(\hat{\theta}_0)] dl$,

$$\Phi'(m) = 2[D(m) - D(\hat{\theta}_0)] - 2c^*(m - \hat{\theta}_0) = 2[D(m) - c^*m] = 2m[c(|m|) - c^*],$$

using $D(\hat{\theta}_0) = c^*\hat{\theta}_0$ and $D(m)/m = c(|m|)$. Star-shapedness gives the sign pattern: $\Phi' \leq 0$ on $(0, \hat{\theta}_0)$ and on $(-1, -\hat{\theta}_0)$, and $\Phi' \geq 0$ on $(\hat{\theta}_0, 1)$ and on $(-\hat{\theta}_0, 0)$. Hence on $(0, 1]$, Φ is minimized at $m = \hat{\theta}_0$, where $\Phi(\hat{\theta}_0) = 0$; and on $[-1, 0)$, Φ is minimized at $m = -\hat{\theta}_0$, where $\Phi(-\hat{\theta}_0) = 2[2\hat{\theta}_0 D(\hat{\theta}_0)] - 4c^*\hat{\theta}_0^2 = 0$, using $\mathcal{B}(-\hat{\theta}_0, \hat{\theta}_0) = 2\hat{\theta}_0 D(\hat{\theta}_0)$ for odd D . So $\Phi \geq 0$ everywhere, which is the displayed inequality; combined with the choice $\sigma_0 = \text{chord}$, the cap equals the chord. For the equality clause, $\Phi(m)$ is the integral of Φ' from the nearest minimizer ($\hat{\theta}_0$ or $-\hat{\theta}_0$), and Φ' vanishes on an interval exactly where c is constant on the corresponding range of $|\cdot|$.

It remains to consider $\hat{\theta}_0 = 0$. For $m \neq 0$, oddness gives

$$\frac{2\mathcal{B}(m, 0)}{m^2} = \frac{2}{m^2} \int_0^{|m|} l c(l) dl \geq c(0).$$

The quotient converges to $c(0)$ as $m \rightarrow 0$, proving the cap claim. Because c is nondecreasing, equality for a fixed nonzero m holds exactly when c is constant on $[0, |m|]$. $\square$

Two consequences frame the construction. First, on any interval where the local condition $\sigma_0 \leq D'$ and the chord condition $\sigma_0 \leq D/\theta_0$ bind together, $\theta D'(\theta) = D(\theta)$, whose solutions are exactly the linear schedules through the anchor: affinity is forced, not assumed. Second, the equality clause delivers the binding structure: every constraint between two types in an interval with constant chord binds; where the chord is strictly increasing, only the mirror lie binds.

Proposition 13 (Solution of the finite-variation program, uniform case). *Let $s \in (0, 1/\rho)$ be the unique root of*

$$P'(s) = -\frac{3}{8\rho(2 - \rho s)}, \tag{85}$$

and set $b = 3/(4(2 - \rho s)) \in (\frac{3}{8}, \frac{3}{4})$. At $r = 1$, $\lambda = 0$, (85) reads $1 - s + \ln s = -3/(16(2 - s))$ with $(s, b) = (0.5712, 0.5249)$; at $r = 1$, $\lambda = \frac{1}{2}$, $(s, b) =$

(0.2393, 0.4929). Define the odd schedule D^{FV} by $D^{FV}(\theta_0) = s\theta_0$ on $[0, b]$ and $D^{FV}(\theta_0) = \theta_0\varsigma(\theta_0)$ on $(b, 1]$, where $\varsigma(\theta_0) \in (0, 1/\rho]$ is the unique solution of

$$2\rho\theta_0^2\varsigma + \rho P'(\varsigma) = 2\theta_0(2\theta_0 - 1), \quad (86)$$

and set

$$\sigma_0^{FV}(\theta_0) = \begin{cases} D^{FV}(\theta_0)/\theta_0, & \theta_0 \neq 0, \\ s, & \theta_0 = 0. \end{cases}$$

Then (D^{FV}, σ_0^{FV}) uniquely solves the finite-variation program up to F -null sets; the displayed schedules give its continuous version. Moreover, ς is strictly increasing on $[b, 1]$ with $\varsigma(b) = s$ and $\varsigma(1) = 1/\rho$, so exposure is distorted below first best at every interior type and reaches first best exactly at $\theta_0 = \pm 1$; every pairwise constraint between two distinct types in $[-b, b]$ binds; and at every report $|\hat{\theta}_0| > b$ the unique binding constraint is the mirror pair $(-\hat{\theta}_0, \hat{\theta}_0)$.

Proof. Step 1: reduction and symmetrization. Let $\mathcal{P}$ denote the finite-variation date-zero program and $\mathcal{R}$ the reduced program in which the family (15) is replaced by the two subfamilies it implies: the local condition $\sigma_0 \leq D'$ a.e. (the $m \rightarrow \hat{\theta}_0$ limit at Lebesgue points) and the mirror-pair condition, which for arbitrary D reads $2\hat{\theta}_0^2\sigma_0(\hat{\theta}_0) \leq 2\hat{\theta}_0 D(\hat{\theta}_0) - \int_{-\hat{\theta}_0}^{\hat{\theta}_0} D$. Every $\mathcal{P}$ -feasible point is $\mathcal{R}$ -feasible, so $\text{val}(\mathcal{P}) \leq \text{val}(\mathcal{R})$. Both programs are invariant under the reflection carrying (D, σ_0) to $(-D(-\cdot), \sigma_0(-\cdot))$, and their feasible sets are convex. Because the objective is concave, averaging any feasible point with its reflection weakly raises its value. Thus it is enough to solve $\mathcal{R}$ among odd D and even σ_0 ; strict concavity makes any maximizer unique up to F -null sets. If the resulting schedule is star-shaped, Lemma 12 makes it $\mathcal{P}$ -feasible, and the value ranking then makes it optimal for $\mathcal{P}$ as well. For odd D , the mirror-pair condition is the chord condition. We therefore work on $[0, 1]$, where $2f \equiv 1$ and $\Lambda(\theta) = 1 - \theta$.

Step 2: the pair (s, b) and the shoulder. The map $s \mapsto P'(s) + 3/(8\rho(2 - \rho s))$ is strictly increasing on $(0, 1/\rho)$ (both terms are, since $P'' > 0$), tends to $-\infty$ as $s \downarrow 0$, and equals $3/(8\rho) > 0$ at $s = 1/\rho$: (85) has a unique root $s \in (0, 1/\rho)$, and $b = 3/(4(2 - \rho s)) \in (\frac{3}{8}, \frac{3}{4})$. Note for later use that (85) is $-\frac{\rho}{2}P'(s) = b/4$, equivalently $|P'(s)| = b/(2\rho)$. The left side of (86) is strictly increasing in ς (derivative $\rho(2\theta_0^2 + P'') > 0$), tends to $-\infty$ as $\varsigma \downarrow 0$, and at $\varsigma = 1/\rho$ exceeds the right side by $2\theta_0(1 - \theta_0) \geq 0$: a unique solution $\varsigma(\theta_0) \in (0, 1/\rho]$ exists for $\theta_0 \in (0, 1]$, with $\varsigma < 1/\rho$ on $(0, 1)$ and $\varsigma(1) = 1/\rho$. At $\theta_0 = b$, substituting $\varsigma = s$ and $\rho P'(s) = -b/2$ reduces (86) to $2\rho b^2 s - b/2 = 2b(2b - 1)$, equivalently $b(4 - 2\rho s) = \frac{3}{2}$, an identity by the definition of b : hence $\varsigma(b) = s$, and D^{FV} is

continuous. Implicit differentiation of (86) gives

$$\varsigma'(\theta_0) = \frac{8\theta_0 - 2 - 4\rho\theta_0\varsigma}{\rho(2\theta_0^2 + P''(\varsigma))} = \frac{2[1 - (\rho/\theta_0)|P'(\varsigma)|]}{\rho(2\theta_0^2 + P''(\varsigma))},$$

the second form after eliminating ς with (86), which gives $2\rho\theta_0\varsigma = 2(2\theta_0 - 1) + (\rho/\theta_0)|P'(\varsigma)|$. The denominator is positive, and the numerator is positive along the entire shoulder: at $\theta_0 = b$ it equals $2[1 - (\rho/b)|P'(s)|] = 2[1 - \frac{1}{2}] = 1$; and if it vanished first at some $T \in (b, 1]$, then ς would be nondecreasing on $[b, T]$, so $|P'(\varsigma(T))| \leq |P'(s)|$ (as $|P'|$ decreases in ς) and $\rho/T < \rho/b$, making the numerator strictly greater than 1 at T —a contradiction. Hence ς is strictly increasing. In particular the chord of D^{FV} is s on $[0, b]$ and ς on $(b, 1]$: D^{FV} is star-shaped, and on the shoulder $(D^{FV})' = \varsigma + \theta_0\varsigma' > \varsigma$, so the local condition is slack there.

Step 3: multipliers. Assign densities $\eta \geq 0$ to $\sigma_0 \leq D'$ and $\nu \geq 0$ to $\theta_0\sigma_0 \leq D$. The stationarity conditions for (84) on $[0, 1]$ are, pointwise,

$$-\frac{\rho}{2}P'(\sigma_0) = \eta + \theta_0\nu, \quad 2\theta_0 - 1 - \rho D = \eta' - \nu, \quad (87)$$

with complementary slackness, $\eta(1) = 0$, and $D(0) = 0$. On the shoulder set $\eta \equiv 0$; then the second condition reads $\nu = \rho D - (2\theta_0 - 1)$, and the first becomes $\rho D = 2\theta_0 - 1 - \rho P'(D/\theta_0)/(2\theta_0)$, which is (86) divided through by $2\theta_0$: stationarity holds by construction, with $\nu = -\rho P'(\varsigma)/(2\theta_0) \geq 0$ and $\nu(1) = 0$. On the bridge, solving (87) with $D = s\theta_0$ gives the linear ODE $(\theta_0^2\nu)' = \theta_0[1 - (2 - \rho s)\theta_0]$, whose solution bounded at the origin, and the associated η , are

$$\nu(\theta_0) = \frac{1}{2} - \frac{(2 - \rho s)\theta_0}{3}, \quad \eta(\theta_0) = \frac{b}{4} - \frac{\theta_0}{2} + \frac{(2 - \rho s)\theta_0^2}{3},$$

using $-\frac{\rho}{2}P'(s) = b/4$ from (85). Both are admissible: ν is decreasing with $\nu(b) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} > 0$ (and continuous at b , since $-\rho P'(s)/(2b) = \frac{1}{4}$); η is a convex parabola whose vertex is at $\theta_0 = 3/(4(2 - \rho s)) = b$ with $\eta(b) = b/4 - b/2 + b/4 = 0$, so $\eta \geq 0$ on $[0, b]$, continuous at b , and vanishing on the shoulder as required.

Step 4: sufficiency and conclusion. The integrand of (84) is concave in (D, σ_0) and the constraints of $\mathcal{R}$ are linear, so for any feasible (D, σ_0) ,

$$\mathcal{J}(D, \sigma_0) - \mathcal{J}(D^{FV}, \sigma_0^{FV}) \leq \int_0^1 \left[(2\theta_0 - 1 - \rho D^{FV}) \delta D - \frac{\rho}{2} P'(\sigma_0^{FV}) \delta \sigma \right] d\theta_0,$$

with $\delta D := D - D^{FV}$, $\delta \sigma := \sigma_0 - \sigma_0^{FV}$. Substituting (87), integrating the $\eta' \delta D$ term by parts (boundary terms vanish: $\delta D(0) = 0$, $\eta(1) = 0$), the right side

equals $\int_0^1 [\eta(\sigma_0 - D') + \nu(\theta_0 \sigma_0 - D)] d\theta_0 \leq 0$ by complementary slackness at the candidate and feasibility of (D, σ_0) . Thus the candidate solves the symmetric reduction of $\mathcal{R}$, and the symmetrization argument in Step 1 makes it optimal for $\mathcal{R}$. Because it is star-shaped, Lemma 12 makes it $\mathcal{P}$ -feasible; hence it also solves $\mathcal{P}$, uniquely up to F -null sets. The binding structure follows from the equality clause of Lemma 12: for $|\hat{\theta}_0| \leq b$ the chord is constant on $[-b, b]$, so every distinct lie with $|m| \leq b$ attains the cap; for $|\hat{\theta}_0| > b$ the chord is strictly increasing on the relevant range, so the mirror is the unique binding lie. $\square$

Remark 2 (Bridge versus flat). *Whether the bridge covers the first-order flat is parameter-dependent. At $r = 1$, $\lambda = 0$, the root of (85) satisfies $s > \frac{1}{2}$ (the left side of its logarithmic form is negative at $s = \frac{1}{2}$), so $b > x^*$: the smoothed spread extends strictly beyond the no-trade interval it replaces. At $r = 1$, $\lambda = \frac{1}{2}$, $b = 0.4929 < x^*$: the bridge ends strictly inside the flat, and the mirror-binding shoulders take over where the first-order approach would still pool. Numerically, b decreases in λ , crossing x^* between $\lambda = 0.1$ and $\lambda = \frac{1}{2}$.*

Proposition 14 (Solution of the local program, uniform case). *Define the function $g(w) := \rho^2(wP'(w) - P(w)) - 2\rho P'(w)$ on $(0, 1/\rho]$: since $g'(w) = \rho P''(w)(\rho w - 2) < 0$, g is strictly decreasing, with $g(0^+) = +\infty$ and a double zero at $w = 1/\rho$. Let $w_0 \in (0, 1/\rho)$ be the unique root of $g(w_0) = 1$. At $r = 1$, $\lambda = 0$, $g(w) = 6w - w^2 - 4\ln w - 5$ and $w_0 = 0.3811$; at $r = 1$, $\lambda = \frac{1}{2}$, $w_0 = 0.1474$. Set*

$$\vartheta(w) := \int_{w_0}^w \frac{\rho P''(u)}{2\sqrt{g(u)}} du, \quad a := \vartheta(1/\rho) \in (\tfrac{1}{2}, 1);$$

at $r = 1$, $a = 0.7497$ for $\lambda = 0$ and $a = 0.7121$ for $\lambda = \frac{1}{2}$. Up to F -null sets, the local program is uniquely solved by the following continuous schedule. On $[0, a]$, it is given parametrically by

$$(\theta_0, D^{\text{loc}}, \sigma_0^{\text{loc}}) = \left(\vartheta(w), \frac{2\vartheta(w) - 1 + \sqrt{g(w)}}{\rho}, w \right), \quad w \in [w_0, 1/\rho],$$

on which $\sigma_0^{\text{loc}} = (D^{\text{loc}})'$ (the local condition binds), together with $D^{\text{loc}} = (2\theta_0 - 1)/\rho$ and $\sigma_0^{\text{loc}} = 1/\rho$ on $(a, 1]$: the first-order schedule at first-best exposure. Equivalently, on $(0, a)$ the solution is characterized by the first integral $(2\theta_0 - 1 - \rho D)^2 = g(D')$. In particular $\sigma_0^{\text{loc}}(0) = w_0$, exposure is continuous, and D' jumps from $1/\rho$ to $2/\rho$ at a .

Proof. Step 1: stationarity and the two regions. Symmetrization as in Step 1 above applies verbatim (the local constraint set is reflection-invariant), so work

with odd D on $[0, 1]$. We construct a candidate with one binding region and then verify all first-order and complementary-slackness conditions; concavity makes that verification sufficient. With a multiplier density $\eta \geq 0$ on $\sigma_0 \leq D'$, the stationarity conditions are $-\frac{\rho}{2}P'(\sigma_0) = \eta$ (the multiplier on $\sigma_0 \leq 1/\rho$ is zero since $P'(1/\rho) = 0$) and $\eta' = 2\theta_0 - 1 - \rho D$, with $\eta(1) = 0$, $D(0) = 0$, and complementary slackness. Where $\eta = 0$: $P'(\sigma_0) = 0$ forces $\sigma_0 = 1/\rho$, and $\eta' = 0$ forces $D = (2\theta_0 - 1)/\rho$, whose slope $2/\rho$ keeps the constraint slack: consistent. Where $\eta > 0$: $\sigma_0 = D' =: w < 1/\rho$ and the two conditions combine into $\frac{d}{d\theta_0}[-\frac{\rho}{2}P'(w)] = 2\theta_0 - 1 - \rho D$.

Step 2: a first integral. Let $m := -\frac{\rho}{2}P'(w)$, so $m' = 2\theta_0 - 1 - \rho D$ and $m'' = 2 - \rho w$: the system is autonomous in (m, m') . Multiplying by m' and integrating, using $dm = -\frac{\rho}{2}P''(w)dw$ and $\int wP''(w)dw = wP'(w) - P(w)$,

$$(m')^2 = 4m - 2\rho \int w dm + \text{const} = -2\rho P'(w) + \rho^2(wP'(w) - P(w)) + \text{const}.$$

The pasting values pin the constant: at the junction $\theta_0 = a$, matching the outer region requires $D(a) = (2a - 1)/\rho$ and, by continuity of η with $\eta \equiv 0$ outside, $\eta(a) = 0$, i.e. $w(a) = 1/\rho$; both sides then vanish with $\text{const} = 0$ (P, P' vanish at first best), giving the first integral with g as defined. At $\theta_0 = 0$, $D = 0$ forces $g(w_0) = 1$; the root is unique because g is strictly decreasing with $g(0^+) = +\infty$ and $g(1/\rho) = 0$.

Step 3: construction and verification. On $(0, a)$ the solution has $\rho D > 2\theta_0 - 1$ (the branch $\rho D - (2\theta_0 - 1) = +\sqrt{g(w)}$, forced by $\rho D(0) - (2 \cdot 0 - 1) = 1 = \sqrt{g(w_0)}$), and differentiating the first integral along the solution, using $g'(w) = \rho P''(w)(\rho w - 2)$, yields the separable equation $dw/d\theta_0 = 2\sqrt{g(w)}/(\rho P''(w)) > 0$, whose inverse is ϑ . Direct differentiation of $\rho D = 2\vartheta(w) - 1 + \sqrt{g(w)}$ gives $dD/d\vartheta = w$, as required. The integral defining a is finite because g has a quadratic zero at $1/\rho$: by the cubic vanishing (26), $P''(u) \rightarrow 2(1/\rho - u)$ and $g(u) \rightarrow \rho(1/\rho - u)^2$ to leading order as $u \uparrow 1/\rho$, so the integrand tends to $\sqrt{\rho}$.

The bounds on a are immediate from the first integral: integrating $(2\theta_0 - 1 - \rho D)' = 2 - \rho w$ over $[0, a]$ gives $1 = \int_0^a (2 - \rho w) d\theta_0$, and since $0 < w < 1/\rho$ on $[0, a)$ the integrand lies strictly between 1 and 2 there, so $\frac{1}{2} < a < 1$.

Along the parametric solution $D' = w$ increases from w_0 to $1/\rho$, so D^{loc} is convex on $[0, a]$ with $D'(a^-) = 1/\rho$; and $a > \frac{1}{2} = x^*$, so the outer schedule $(2\theta_0 - 1)/\rho$ is positive, increasing, and coincides with the first-order rent (20). The multiplier $\eta = -\frac{\rho}{2}P'(w) \geq 0$ vanishes only at $w = 1/\rho$, i.e. at $\theta_0 = a$: signs and complementary slackness hold throughout, and the concavity argument of Step 4 above (with $\nu \equiv 0$) gives optimality; strict concavity gives uniqueness up to F -null sets. $\square$

Evaluating (84) and adding the common first-best term gives the uniform values reported in Table 1: at $r = 1$, $\lambda = 0$, $\Pi_{\text{loc}} = 0.5435$ and $\Pi_{FV} = 0.5379$; at $r = 1$, $\lambda = \frac{1}{2}$, $\Pi_{\text{loc}} = 0.2753$ and $\Pi_{FV} = 0.2725$. The preceding propositions determine both program values analytically at every (r, λ) , and the grid solutions below agree with them at the reported precision.

H.8 Computational details for Section 6.5

The date-zero programs are approximated on a grid $\theta_1 < \dots < \theta_n$. The uniform grid covers $[-1, 1]$; for the Gaussian, the integrals are truncated at $[-4, 4]$, whose omitted probability is approximately 6.3×10^{-5} . The variables are $\{D_i, \sigma_i\}$, and the nonconstant part of the objective is

$$\sum_i w_i \left[-\frac{\rho}{2} \left(D_i - \frac{\theta_i}{\rho} \right)^2 - \frac{\rho}{2} P(\sigma_i) - \Lambda(\theta_i) D_i \right],$$

where the F -weights w_i use the trapezoid rule. Total revenue adds the common first-best term

$$\frac{\mathbb{E}[\theta_0^2]}{2\rho} + \frac{1}{2r\rho}.$$

The continuation term comes from Proposition 5, and rents are priced by the symmetric-anchor handicap (17). Direct integration gives the penalty in elementary form. With $x := \rho\sigma \in (0, 1]$ and $\alpha := r/\rho$,

$$P(\sigma) = \begin{cases} \frac{1 - x^2 + 2x \ln x}{r^3}, & \lambda = 0, \\ \frac{1}{\rho^2} \left[\frac{1 - x^\alpha}{r} - \frac{x^\alpha - x}{\lambda} + \frac{x^\alpha - x^2}{r + 4\lambda} \right], & \lambda > 0, \end{cases}$$

with the value at $x = 0$ defined by continuity.

The finite-variation discretization imposes monotonicity of D , $D(0) = 0$, $0 \leq \sigma_i \leq 1/\rho$, and all $n(n-1)$ nodal counterparts of (15), computing $\mathcal{B}$ by the trapezoid rule. The local discretization replaces the pairwise family by central-difference inequalities $\sigma_i \leq D'_i$ at the interior nodes. These are finite-dimensional concave-maximization problems with linear constraints. A trust-region constrained solver was used on grids with $n = 81$ – 161 , and also $n = 321$ for the local program, with the gradient tolerance set to 10^{-10} .

The refinements display approximately second-order convergence. For example, the total local value for the uniform case is 0.54364, 0.54356, and 0.54354 at $n = 81, 161, 321$, respectively, extrapolating to 0.54353. The ratios in Table 1 vary by about 0.1 percentage points across these grids. The static

and first-order rows use the closed forms in Appendix H.4 and Proposition 4, respectively, evaluated by quadrature.

These refinements assess the discretized programs. Except in the uniform cases certified analytically above, nodal feasibility and grid convergence do not by themselves certify a feasible interpolation or an upper bound for the corresponding continuum program.

I On Numerics for the Control Programs

When closed forms are unavailable, the continuation HJBs can be solved by a monotone semi-Lagrangian scheme. We use the current-value exposure $\psi = \text{vol}(D)$ from Section 4; with $\rho := r + 2\lambda$, its maximal drift is $\rho\psi$. These coordinates keep the infinite-horizon problem stationary.

The finite-variation scheme. Let $z = (\vartheta, d, \psi)$, choose a time step $h > 0$, and place a rectangular grid on a bounded state domain. For a grid function W , let $\mathcal{I}_\Delta W$ be multilinear interpolation, using nonnegative weights. At z , choose an allocation x and a post-adjustment exposure $\bar{\psi} \in [0, \psi]$; choosing $\bar{\psi} < \psi$ implements the free downward adjustment. Define

$$z_\pm(x, \bar{\psi}) := \begin{pmatrix} \vartheta + \lambda(\bar{\theta} - \vartheta)h \pm \gamma\sqrt{h} \\ d + [(r + \lambda)d - x]h \pm \bar{\psi}\sqrt{h} \\ e^{\rho h \bar{\psi}} \end{pmatrix}, \quad \Delta_r(h) := \frac{1 - e^{-rh}}{r},$$

and the one-step operator

$$(\mathcal{T}_{h,\Delta}W)(z) := \sup_{\substack{x \\ 0 \leq \bar{\psi} \leq \psi}} \left\{ \Delta_r(h) \vartheta x + \frac{e^{-rh}}{2} \sum_{s \in \{-, +\}} \mathcal{I}_\Delta W(z_s(x, \bar{\psi})) \right\}. \quad (88)$$

If the model includes an allocation cost $c(x)$, replace ϑx by $\vartheta x - c(x)$. Unbounded control sets must be truncated using an a priori bound, or expanded until the truncation is inactive.

The same sign is used for the report and promise innovations. The two-point lottery therefore reproduces the rank-one covariance in (10), including the mixed term $\gamma\psi V_{\hat{\theta}D}$. A naive Cartesian stencil for this mixed derivative can generate negative weights; upwinding the drift does not repair that defect. In (88), interpolation, averaging, and maximization all preserve order, so $\mathcal{T}_{h,\Delta}$ is monotone.

At $\psi = 0$, the scheme forces $\bar{\psi} = 0$, as required by the absorbing boundary. At the artificial edges of the other state variables, one must impose boundary

data consistent with verified growth bounds and avoid extrapolation with negative weights. The grid and time step should be refined so that interpolation error is $o(h)$. Monotonicity, consistency, stability, a comparison principle, and controlled domain truncation then give convergence to the viscosity solution of (11).

Iteration and discrete bounds. On a finite grid, the operator in (88) is an e^{-rh} -contraction in the sup norm. A convenient initial upper vector U^0 is the value, on the same grid, of the relaxation that allows future exposure at elapsed time s to be reselected below $\psi e^{\rho s}$, ignoring the irreversibility of past reductions. This relaxation contains every admissible grid policy, and dynamic programming for the relaxation gives $\mathcal{T}_{h,\Delta}U^0 \leq U^0$. More generally, any initial vector satisfying $\mathcal{T}_{h,\Delta}U^0 \leq U^0$ may be used. Hence

$$U^{n+1} = \mathcal{T}_{h,\Delta}U^n$$

decreases to the unique value $V_{h,\Delta}$ of the discretized problem.

The maximizers at an iteration define a grid policy. Fixing that policy and solving its linear system gives a lower bound $L^n \leq V_{h,\Delta}$; retaining $\max_{k \leq n} L^k$ makes the reported lower sequence nondecreasing. Thus

$$\max_{k \leq n} L^k \leq V_{h,\Delta} \leq U^n \quad \text{on the grid}$$

and the width of this pointwise bracket measures solution error for the discretized problem. It does not, by itself, bound the discretization error relative to the continuous-time model.

Recovering a mechanism. To obtain a continuous-time candidate, apply the interpolated grid policy at the dates $0, h, 2h, \dots$. At each positive decision date, reduce ψ to the selected $\bar{\psi}$, hold the selected allocation until the next date, and let ψ grow at rate ρ between dates. This produces only finitely many downward jumps on each finite horizon and hence a finite-variation exposure process. Evolve the other states under the actual diffusion and construct transfers from Proposition 2. The resulting policy gives a genuine continuous-time lower bound only after verifying well-posedness, admissibility, transversality, and the date-zero conditions of Theorem 1b, and evaluating its actual payoff. A grid-policy linear system certifies only its value in the discretized model. Because $K_0 = 0$, an initial reduction must instead be incorporated in the date-zero choice of exposure, subject to the no-loss condition in Theorem 2a.

The local upper-bound program. The companion program bounds the larger class of locally incentive-compatible mechanisms. Its monotone step is obtained by also choosing v : replace the last coordinate of $z_{\pm}$ by

$$e^{\rho h} \bar{\psi} \pm v \sqrt{h}$$

and require both next-period exposure values to be nonnegative. The same common sign across all three state coordinates reproduces the covariance terms in (13); the admissible range for v expands as $h \downarrow 0$. A numerical fixed point is not itself a certified continuous-time upper bound. The verification argument for Theorem 2b instead requires a smooth supersolution satisfying the Bellman inequality, the state-constraint boundary condition, and (T1) throughout the domain. Merely fitting a smooth function above the grid values is insufficient unless these inequalities are verified. The difference between a certified upper bound and the payoff of an admissible finite-variation policy bounds the possible cost of that restriction.

J The GBM Control Programs

This section states and proves the geometric Brownian counterparts to Theorems 2a and 2b. The verification structure is the same as in the OU case. Under GBM, however, homogeneity makes the continuation value immediate: the rent promise already determines continuation surplus. We first derive the state equations and HJBs, then state the resulting date-zero programs.

J.1 States and equations

Define the current-value GBM exposure

$$p_t := e^{rt} A_t = \hat{\theta}_t \text{vol}(D_t), \quad p_t \geq 0. \quad (89)$$

This choice gives a fixed boundary at $p = 0$. Using (35) and $\text{vol}(D_t) = e^{rt} A_t / \hat{\theta}_t$, the report and rent satisfy

$$\begin{aligned} d\hat{\theta}_t &= \mu \hat{\theta}_t dt + \gamma \hat{\theta}_t d\hat{B}_t, \\ dD_t &= \left[(r - \mu) D_t - X_t - \gamma \frac{p_{t-}}{\hat{\theta}_t} \right] dt + \frac{p_{t-}}{\hat{\theta}_t} d\hat{B}_t. \end{aligned} \quad (90)$$

To express the incentive restriction, write the Doob–Meyer decomposition of the nonnegative supermartingale A as

$$dA_t = \eta_t d\hat{B}_t - dC_t,$$

where C is nondecreasing and $C_0 = 0$. Setting

$$v_t := e^{rt}\eta_t, \quad dK_t := e^{rt}dC_t$$

gives

$$dp_t = rp_t dt + v_t d\hat{B}_t - dK_t, \quad p_t \geq 0, \quad (91)$$

where K is adapted, càdlàg, and nondecreasing, and $K_0 = 0$. Conversely, (91) implies

$$dA_t = e^{-rt}v_t d\hat{B}_t - e^{-rt}dK_t,$$

so $A = e^{-rt}p$ is a nonnegative local supermartingale and hence a supermartingale. Thus (91) is exactly the GBM condition in Theorem 1a. Moreover, $\text{vol}(A) = \gamma\hat{\theta} \text{vol}(Y)$ by (41); hence Y is of finite variation exactly when $v = 0$. At $p = 0$, the nonnegative supermartingale A is absorbed, so $v = 0$ and K cannot increase.

Throughout, a continuation delivers rent d if $D_0 = d$ and D is the promised-rent process:

$$\hat{\theta}_t D_t = \mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} \hat{\theta}_s X_s ds \right]. \quad (92)$$

This requirement is substantive: the state equations alone also admit solutions that do not represent the conditional payoff.

Write $\Xi_t^G := (\hat{\theta}_t, D_t, p_t)$. For a smooth function $W(\vartheta, d, p)$, define

$$\begin{aligned} \mathcal{L}^G W := & \sup_x x(\vartheta - W_D) + \mu\vartheta W_\vartheta + \left[(r - \mu)d - \gamma \frac{p}{\vartheta} \right] W_D + rpW_p \\ & + \frac{\gamma^2 \vartheta^2}{2} W_{\vartheta\vartheta} + \gamma p W_{\vartheta D} + \frac{p^2}{2\vartheta^2} W_{DD}, \end{aligned} \quad (93)$$

where the supremum is over the allocation's range. The finite-variation HJB is

$$\min \{ rV^G - \mathcal{L}^G V^G, V_p^G \} = 0, \quad (94)$$

whereas the local HJB is

$$\min \left\{ r\bar{V}^G - \mathcal{L}^G \bar{V}^G - \sup_{v \in \mathbb{R}} \left[\frac{v^2}{2} \bar{V}_{pp}^G + v \left(\gamma \vartheta \bar{V}_{\vartheta p}^G + \frac{p}{\vartheta} \bar{V}_{Dp}^G \right) \right], \bar{V}_p^G \right\} = 0. \quad (95)$$

They are posed on $(0, \infty) \times \mathbb{R} \times (0, \infty)$. The type boundary at zero is not reached in finite time. At $p = 0$, let $\mathcal{L}_0^G$ denote $\mathcal{L}^G$ with $p = 0$. The state-constraint condition is

$$rW(\vartheta, d, 0) - \mathcal{L}_0^G W(\vartheta, d, 0) \geq 0,$$

with equality for attainment; the only admissible boundary choice in the local program is $v = 0$.

At date zero,

$$p_0(\theta) = A_0(\theta) = \gamma\theta[Y_0(\theta) - D(\theta)], \quad (96)$$

and therefore

$$Y_0(\theta) = D(\theta) + \frac{p_0(\theta)}{\gamma\theta}, \quad \sigma_0(\theta) = \gamma D(\theta) + \frac{p_0(\theta)}{\theta}.$$

Thus the global condition (5) becomes

$$\int_{\hat{\theta}}^{\theta} \left[D(l) - D(\hat{\theta}) - \frac{p_0(\hat{\theta})}{\gamma\hat{\theta}} \frac{l - \hat{\theta}}{l} \right] dl \geq 0 \quad \text{for all } \hat{\theta}, \theta \in \Theta. \quad (97)$$

Theorem 2c (The finite-variation program: GBM). *Define $V^G(\vartheta, d, p) := \vartheta d$. This function solves (94), including the boundary equation at $p = 0$. Every admissible finite-variation continuation that starts from $\Xi_0^G = \xi = (\vartheta, d, p)$ and delivers rent d satisfies*

$$V^G(\xi) = \mathbb{E} \int_0^\infty e^{-rt} \hat{\theta}_t X_t dt.$$

Thus every such continuation attains the same continuation surplus; the HJB does not select among them.

For the date-zero stage, consider

$$\sup_{D(\cdot), p_0(\cdot), U_c} \int_{\Theta} [\theta - h_c(\theta)] D(\theta) dF(\theta) - U_c \quad (98)$$

subject to (7), $p_0 \geq 0$, and (97). Assume that, for each feasible choice, there is a measurable $0 \leq \tilde{p}_0 \leq p_0$ such that admissible finite-variation continuations from

$$(\theta, D(\theta), \tilde{p}_0(\theta))$$

that deliver the selected rent $D(\theta)$ can be chosen measurably with respect to θ and, together with transfers satisfying Proposition 2(i)–(ii), form an admissible mechanism. Then every feasible objective value in (98) is attained by a globally incentive-compatible, individually rational mechanism. Hence the program is an implementable lower bound on the principal's value; if its supremum is attained under the stated continuation requirement, a mechanism attaining the program's value exists.

Theorem 2d (The local program: GBM). *The function $\bar{V}^G(\vartheta, d, p) := \vartheta d$ solves (95), including the boundary equation. It equals the continuation surplus of every admissible continuation that starts from $\Xi_0^G = (\vartheta, d, p)$ and delivers rent d . It is therefore an upper bound for every admissible locally incentive-compatible continuation from that state.*

Consider the relaxed date-zero program

$$\sup_{D(\cdot), p_0(\cdot), U_c} \int_{\Theta} [\theta - h_c(\theta)] D(\theta) dF(\theta) - U_c \quad (99)$$

subject to (7), D nondecreasing, and

$$0 \leq p_0(\theta) \leq \gamma \theta^2 D'(\theta) \quad F\text{-almost everywhere.} \quad (100)$$

Every incentive-compatible, individually rational mechanism is feasible for this relaxation, so its value is an upper bound on the unrestricted principal's value.

Finally, suppose (99) has a maximizer for which D is differentiable on Θ , (100) holds at every type, and admissible continuations delivering the chosen rent and satisfying (90)–(91) can be chosen measurably from the selected initial states. Suppose these continuations, together with transfers satisfying Proposition 2(i)–(ii), form an admissible mechanism. Then the mechanism is locally incentive compatible and individually rational and attains the local-program value. It is optimal among admissible, locally incentive-compatible, individually rational mechanisms whose date-zero rent is nondecreasing and differentiable on Θ , whose initial exposure satisfies (100) at every type, and whose utilities satisfy the envelope condition of Proposition 2(ii).

J.2 Proofs of Theorems 2c and 2d

Continuation value and the HJBs. For $V^G(\vartheta, d, p) = \vartheta d$,

$$V_D^G = \vartheta, \quad V_{\vartheta}^G = d, \quad V_{\vartheta D}^G = 1, \quad V_p^G = 0,$$

and every other second derivative appearing in (94)–(95) is zero. Hence

$$\mathcal{L}^G V^G = \mu \vartheta d + \left[(r - \mu) d - \gamma \frac{p}{\vartheta} \right] \vartheta + \gamma p = r \vartheta d = r V^G.$$

Also $V_p^G = 0$, and all of the terms optimized over v in (95) vanish. Thus both HJBs and the boundary equation hold identically. Neither HJB selects X or v ; admissibility and deliverability remain substantive. Because V^G is independent of p , continuous and jump adjustments through K also leave its value unchanged.

Now fix $\xi = (\vartheta, d, p)$ and an admissible continuation that starts from $\Xi_0^G = \xi$ and delivers rent d . By the definition of the GBM rent, (92) holds. The rent dynamics in (32) provide the corresponding martingale verification:

$$d(e^{-rt}\hat{\theta}_t D_t) = -e^{-rt}\hat{\theta}_t X_t dt + \vartheta \sigma_t d\hat{B}_t.$$

The stochastic integral is a square-integrable martingale by admissibility and the representation of discounted gross surplus. Moreover, Condition 1(C2) and (92) imply

$$\left| e^{-rT} \mathbb{E}[\hat{\theta}_T D_T] \right| \leq \mathbb{E} \int_T^\infty e^{-rs} |\hat{\theta}_s X_s| ds \longrightarrow 0.$$

Integrating the rent dynamics, taking expectations, and then letting $T \rightarrow \infty$ therefore gives

$$\vartheta d = \mathbb{E} \int_0^\infty e^{-rt} \hat{\theta}_t X_t dt.$$

This proves the continuation claims in both theorems.

Date zero: finite variation. For $\hat{\theta}, \theta > 0$, set

$$J(\hat{\theta}, \theta) := \int_{\hat{\theta}}^\theta \frac{l - \hat{\theta}}{l} dl = \theta - \hat{\theta} - \hat{\theta} \log \frac{\theta}{\hat{\theta}} \geq 0.$$

Thus replacing p_0 by $\tilde{p}_0 \leq p_0$ weakly relaxes (97). It leaves the objective unchanged because V^G is independent of p .

Use the measurable finite-variation continuations assumed in Theorem 2c. With $v = 0$, (91) makes $A = e^{-rt}p$ nonnegative and nonincreasing, and Y is of finite variation. Theorem 1a therefore gives global incentive compatibility after the initial report. Equation (97) is exactly (5) after substituting (96); Theorem 1b gives global incentive compatibility at date zero. The transfers in the theorem supply the two envelope conditions, while (7) gives participation.

Finally, (92) at $t = 0$ and (6) give

$$\Pi = \int_{\Theta} [\theta - h_c(\theta)] D(\theta) dF(\theta) - U_c.$$

This is the objective in (98). Hence every feasible choice satisfying the stated continuation requirement is implementable, which proves the lower-bound and attainment conclusions of Theorem 2c. $\square$

Date zero: the local relaxation. Fix an incentive-compatible, individually rational mechanism. Theorem 1a gives $A_0 \geq 0$, hence $p_0 \geq 0$, and Remark 1 gives that D is nondecreasing. At every differentiability point of D , the GBM condition in (3) and (96) give

$$\frac{p_0(\theta)}{\gamma\theta} = Y_0(\theta) - D(\theta) \leq \theta D'(\theta),$$

which is (100). A monotone D is differentiable Lebesgue-almost everywhere and hence F -almost everywhere. The revenue identity above shows that every such mechanism is bounded by (99).

Conversely, take a maximizer satisfying the all-type conditions in Theorem 2d. Equation (91) makes $A = e^{-rt}p$ a nonnegative supermartingale. With the continuation-envelope transfers, Theorem 1a gives local incentive compatibility after each truthful initial report. The date-zero envelope and (100), equivalently the GBM line of (3), allow Theorem 1b to give local incentive compatibility at date zero. Individual rationality follows from (7), and (92) shows that the mechanism attains the local-program objective. The same feasibility and continuation identity apply to every comparator specified in the theorem, so none can yield a larger value. $\square$

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